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PHILIPS

FM93

MOBILE RADIOTELEPHONES

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Philips Radio Communication Systems (Australia)
Clarinda Road, Clayton, Victoria, Australia 3168
Telephone: 542 4500 Telex: AA20165(Auscltw +)
Fax: 543 7935

Philips Telecommunications and Data Systems Limited
(Inc. in Vic.)

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AMENDMENT LIST

Changes made to the equipment described in this manual are published as amendments which are dated and consecutively numbered. This sheet is to be used to record the incorporation of all amendments.

Reprints of the manual will incorporate all amendments to date and an entry to this effect will be recorded on the amendment list below.

Amendment No.	Subject	Date Change Completed	Signature

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Philips Radio Communication Systems Ltd.,
Publications Department,
P.O. Box 105, CLAYTON. 3168
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VEHICLE CARE

With advancing technology there is a trend by motor vehicle manufacturers to introduce the use of electronic circuitry in the control of the vehicles. Proper design, care and maintenance of these vehicle electronics will avoid the remote possibility of malfunction from radio frequency interference from adjacent or installed two way radio transmitters.

The following information is supplied to ensure that there is no radio frequency interference effect on the vehicle's electronic systems.

To prevent interference with any other electronic systems in the vehicle, e.g. ignition, anti-skid devices etc., the radiotelephone and antenna should be mounted as far away as possible from these units and their associated cables. Reference should be made to the vehicle manufacturer's handbook for the location of these items.

Finally, the following checks should be carried out if the vehicle is equipped with electronic anti-skid and/or electronic ignition systems.

NOTES:

1. The transmitter should be keyed only for the time required to make an observation.
2. An assistant will be required for the following checks.

RF COMPATIBILITY CHECKS

1. With vehicle stationary and the engine running at fast idle, key the transmitter and check that the brake lights do not illuminate, and the engine continues to run normally.
2. Operate brake pedal, key the transmitter and check that the brake lights do not extinguish.
3. Put the vehicle into motion at a moderate speed (15-20kph), key the transmitter and operate the brake pedal simultaneously. Check that the braking action is normal, and that the engine does not surge or cut out.

CAUTION

In the event of an apparent malfunction in the braking or ignition systems during the above checks, the radio installation should be rendered inoperative and the vehicle manufacturer should be contacted before any further use is made of the radio installation.

FM93 MOBILE RADIOTELEPHONE

SERVICE MANUAL

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PART 1

SERVICE MANUAL

PART 1

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SECTION 1 INTRODUCTION

1.1 GENERAL

The FM93 Series of mobile radiotelephones are frequency synthesised units, microcomputer controlled, for general local control simplex applications.

It provides a limited feature unit in the FM900 Series of mobile radios with features which include:

a **Single Digit Display** With a limit of 10 RF channels (1 through 0).

b) **Narrow Band Receiver** Switching VHF ±1%, UHF ±.5%.
Note: Transmitter switching bandwidth remains as per FM92.

c) **One Transmit Power Level** Nominally 25W.

d) **Receiver Scanning** Is available in two preprogrammable groups each with a maximum of 10 channels including a priority channel if required.

e) **Signalling** Uses the CTCSS plus Selcall Type 1 and Type 2 modules currently used in the FM91 and FM92.
Note: While the mobile-to-mobile Selcall option is still available, it is limited by the single digit display therefore providing variation of the last tone only of the 5-tone sequence.

The radio is housed in the service proven chassis, and employs the same circuit principles including double superhet receiver and microcomputer controlled single loop frequency synthesiser, that are used with all the FM900 Series radios,

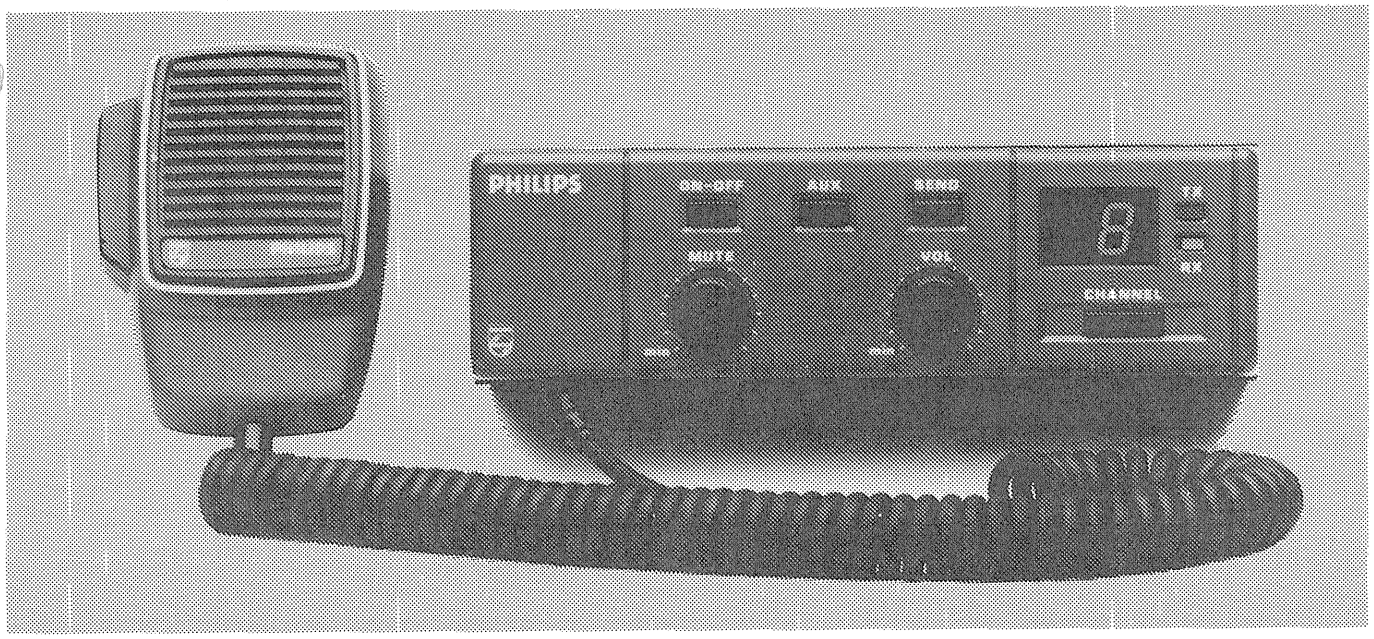


Figure 1.1 FM93 Mobile Radio

1.2.2 SOFTWARE OPTIONS CODE

The options fitted to the mobile are characterised by a 15-digit software code specified by the following table.
Certain options are mutually exclusive and the table in sub-paragraph 1.2.5 must be consulted.

	1	2	3	4	5	6	7	8	9	10	11	12/13	14/15
	CHANNEL DISPLAY	MUTE	ECON.	Tx TIMER	Tx INH.	Tx POWER	SCAN	VOTING	CTCSS	TONE PERIOD	MODE	SELCALL ENCODE	SELCALL DECODE
0			N/R	N/R	N/R	25W	N/R	N/R	N/R	N/R	SIMPLEX	SEE FOLLOWING TABLES	SEE FOLLOWING TABLES
1	CONTINUOUS	VARIABLE		1min	REQ. ALL CHANNELS		VACANT, PRE-PROG. BLOCKS		VARIABLE PER CHANNEL	100mS			
2				3min			OCCUP'D PRE-PROG BLOCKS			40mS			
3	CONT. & 8 HR TIMER			5min					VARIABLE PER CHANNEL R.T.B.	70mS			
4				10min						20mS			
5									COMMUNITY RPTR. PER CHAN.	30mS			
6													
7						1 - 25W ADJUST.			COMMUNITY RPTR. PER CHAN. & RTB				
8													

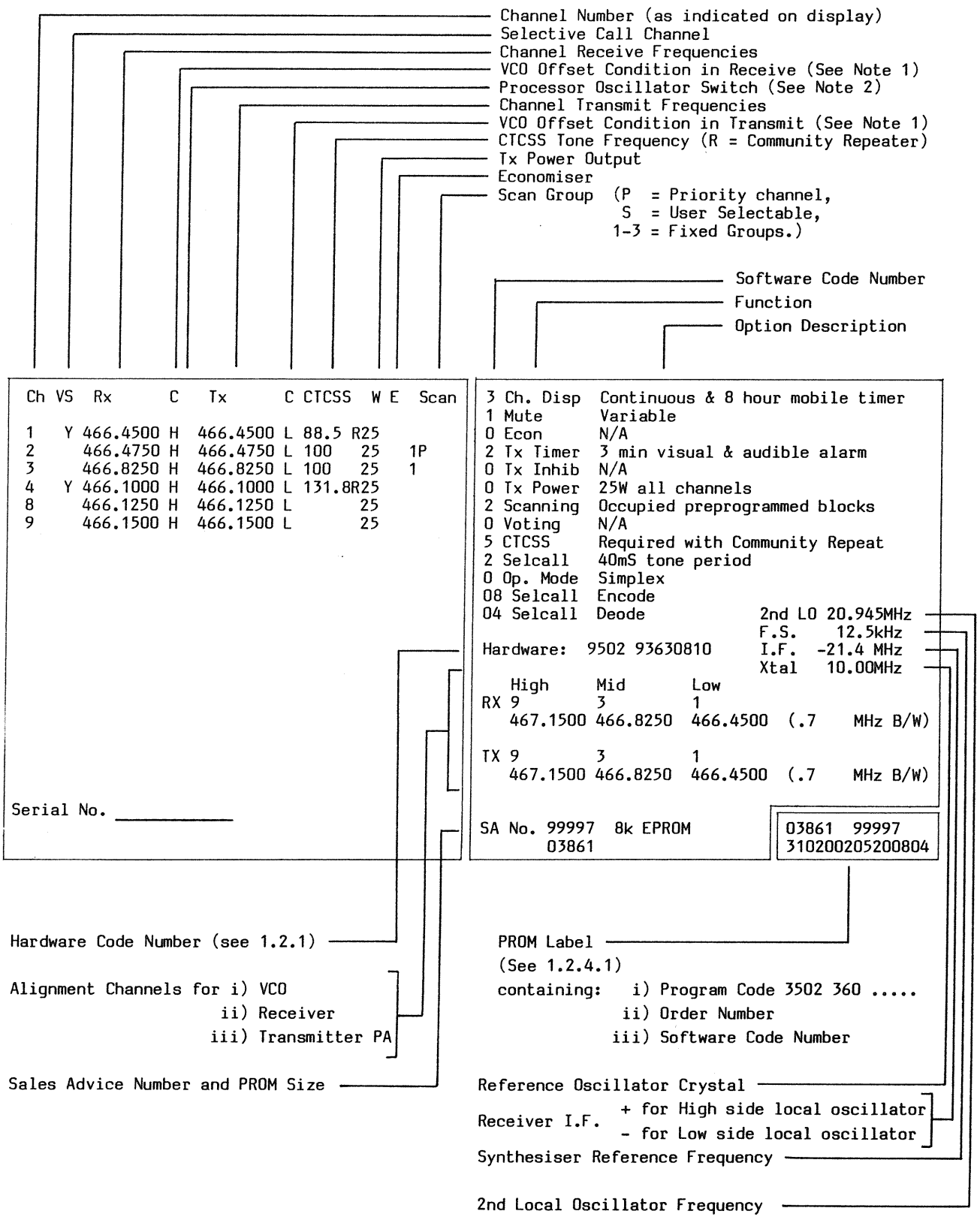
Used with Type 1 Selcall only.

1.2.2.1 Selcall Encode Tables (Col's 12/13)

TYPE 1 AND TYPE 2 SELCALL ENCODE FUNCTION (AUSTRALIA ONLY) - Col.12/13

SOFTWARE CODE NO.	STATUS	ANI CODE POSITION	SEND CODE	STATUS MEMORY	TONES PER CODE	DECODE
00	N	NA	V	NA	5	Y
01	N	S	F	NA	5	N
02	N	NA	F	NA	5	Y
03	N	S	F	NA	7	N
04	N	S	F	NA	7	Y
05	N	E	F	NA	5	N
07	N	E	F	NA	7	N
08	N	E	F	NA	7	Y
09	N	NA	F	NA	5	N
11	N	S	V	NA	7	N
12	N	S	V	NA	7	Y
15	N	E	V	NA	7	N
16	N	E	V	NA	7	Y
53	N	NA	V	NA	5	N
SC706 COMPATABILITY OPTION (* Type 1 only)						
02	N	NA	F	NA	5	Y

LEGEND:
N = Not Required
E = End of Transmission
F = Fixed Code
S = Start of Transmission
Y = Required
NA = Not Applicable
V = Variable Code



Notes:

1) The VCO offset condition refers to the voltage measured on the offset control latch output, IC405 pin 5 (H = High, L = Low).

2) The processor oscillator switch condition refers to a logic high on the processor oscillator control output.

Figure 1.2 Mobile Information Label

1.3 OPERATING INSTRUCTIONS

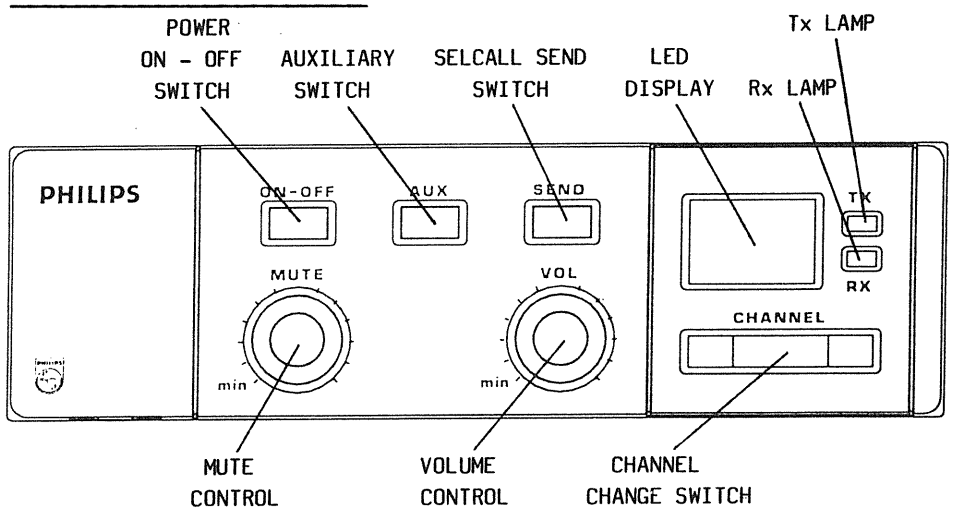


Figure 1.5 FM93 Front Panel Layout

1.3.1 CONTROLS

All controls, except the press-to-talk switch on the fist microphone, are located on the front panel of the transceiver as illustrated in Figure 1.5.

(a) **ON-OFF** Switch. Turns the transceiver on or off. Push-On, Push-Off.

(b) **MUTE** Control. This control eliminates receiver noise in the absence of a signal and also suppresses weak interfering signals.

(c) **VOLUME** Control. Adjusts the speech level at the loudspeaker.

(d) **CHANNEL CHANGE** Switch. Pressing the button enables selection of the next channel. If the button is held down the channel change will continue at a constant rate.

(e) **AUXILIARY** Switch. The function of this switch depends on the options fitted, e.g., Scanning or Car-to-Car Selcall.

(f) **SELCALL SEND** Switch. For units equipped with selective call (selcall), refer selective call operation. This switch is inoperative when selcall is not fitted.

1.3.2 INDICATOR LAMPS AND DISPLAY

(a) **LED** Display - displays the selected channel number or if enabled, the selected scan group number (flashing) or, if car-to-car selcall is fitted the last digit of the selcall code (flashing).

(b) **Tx** Lamp - a red lamp which glows when the press-to-talk switch or Selcall Send switch (if Selcall is fitted) is operated, to indicate that a signal is being transmitted. If the Selcall is queued due to the channel being busy, the red LED will flash until the Selcall is transmitted.

(c) **Rx** Lamp - an amber lamp indicates that a signal is being received. If Selcall is fitted the amber LED will flash to indicate Selcall is received.

1.3.3 SWITCHING OFF

To switch off the equipment press the ON/OFF button.The channel display will be extinguished. It is always advisable to turn the set off if the vehicle is not to be used for an extended period.

1.3.6.2 **Selcall Transmission**

Selcall transmission from a mobile radiotelephone is normally used to:

(a) Allow a base operator to receive calls from his fleet of mobiles without having to listen to the calls from other users of the same radio channel.

The mobile caller can call the base operator by pressing the SEND button. The Tx lamp will flash until the channel is free and then glow continuously while the selcall transmission is in progress.

(b) Provide the base operator with a positive identification of the caller.

In the identification mode no special action is required by the mobile operator as the identification information is automatically sent when the microphone transmit key is operated.

1.3.6.3 **Mobile-to-Mobile Calls**

In Mobile-to-mobile selcall operation a mobile may call another mobile by sending the other mobiles receive code. This is achieved in the FM93 by the following procedure.

(a) With the mobile in the receive state press the auxiliary button **AUX.**

If the mobile has been switched on for the first time after connection to the battery the channel display will flash **"0".**

(b) The flashing display **"0"** indicates that the last digit of the operator selectable selcall code is available for selection.

(c) To alter the last digit of the send code, the channel change button is used to vary the last digit between 0 and 9.

(d) When the new code is entered pressing the **"SEND"** button will cause the unit to transmit the new selcall code.

(e) When the **"SEND"** code has been successfully sent the display will revert automatically to the normal channel indication.

(f) The last manually programmed send code will be remembered automatically. If a mobile-to-mobile selective call attempt has been unsuccessful, subsequent attempts may be made by pressing the **"AUX"** button. This will cause the display to revert to the flashing mode, displaying the last entered digit. Pushing the **"SEND"** button will send this code again.

(g) The mobile-to-mobile selcall code can only be changed when the display is in the flashing mode. If the display is steady, it indicates that the channel is selected. To change to mobile-to-mobile selcall selection, press the **"AUX"** button and enter the new code as per procedure (c) above.

(h) When the display is in the flashing mode, pressing the **"AUX"** button will cause the unit to revert to normal channel indication without transmitting a selcall code. This operation may be used to check which channel the unit is on, or revert from the mobile-to-mobile selcall mode to the mobile-to-base selcall mode (See following procedure).

(i) When in the operator entry mode for selcall, the normal mobile PTT and receive function continues to operate.

1.3.6.4 **Mobile-to-Base Calls**

(a) ENSURE that the display is in the steady mode. Press the **"AUX"** button if the display is in the flashing mode. Then press the **"SEND"** button which will cause tansmission of the base address code.

SECTION 2 TECHNICAL SPECIFICATIONS

Performance data indicate typical values related to 25/30kHz unit operated at 13.8-volts dc at 25°C. ambient temperature.

2.1 GENERAL

		UHF	VHF
Operating Voltage:		13.8V dc negative earth.	
Operating Temperature:	Standard:	-10° to +60°C (Funct. -30° to +65°C)	
	Extended Temp. Range:	-20° to +60°C (Funct. -30° to +65°C)	
Current Consumption:	Receive:(Stdby)	<600mA	<600mA
	(Full audio)	<1.125A	<1.125A
	Transmit: (25W)	<6.9A	<5.3A
	(1W)	<2.0A	<1.5A
Frequency Bands:		403-440MHz (T) 440-470MHz (U) 470-500MHz (W1)	68- 88MHz (E) 132-153MHz (B) 148-174MHz (A)
Channel Spacing:		25/30kHz	25/30kHz
	(Optional)	12.5kHz	12.5kHz
Channel Capacity:		10	10
Switching Bandwidths:	Receiver: (for 1dB reduction in sensitivity)	±0.5%	±1%
	Transmitter: (for 0.5dB in output power reduction)	10MHz	6MHz (A/B Bands) 5MHz (E Band)
Frequency Stability		±5ppm	±10ppm
Duty Cycle:		1 min. Tx/ 4 mins. Rx.	1 min. Tx./ 4 mins. Rx.
E-Prom Capacity:		8k bytes	8k bytes
Dimensions:		232mm (L) 175mm (W) 52mm (H)	232mm (L) 175mm (W) 52mm (H)
Weight:		2.7kg w/o cradle 3.2kg c/w cradle	2.7kg w/o cradle 3.2kg c/w cradle
Finish:		Semi-gloss black with splatter texture paint finish.	

2.3 TRANSMITTER

	UHF	VHF
RF Power Output:	25Watts +1,-0dB	25Watts +1,-0dB
Spurious Emission: (>80dB below 25W)	<0.25uW	<0.25uW
Tx Rise Time: (to achieve 70% RF power)	<40mSec	<40mSec
Adjacent Channel Sideband Power:	>70dB	>70dB
FM Residual Noise: (@ 1kHz for 100% Deviation)	>45dB	>50dB
Audio Sensitivity: (@ 1kHz for 60% Deviation)	<40mV	<40mV
AF Distortion: (@ 1kHz for 60% Deviation)	<3%	<3%
Audio Frequency Response:	Within +1dB to -3dB of a 6db/octave pre-emphasis curve over 300 - 3000Hz relative to 1kHz modulation at 20% of maximum deviation.	

2.4 VARIATIONS OVER TEMPERATURE AND SUPPLY VOLTAGE

2.4.1 SUPPLY VOLTAGE VARIATIONS (OVER 10.8V to 16.2V RANGE)

Receiver Audio Power Output:	<±0.25dB w.r.t 1 watt at 13.8Vdc
A.F. Rated Power:	>2.5W into 4 for 10% distortion at 1kHz and 100% deviation (@ 10.8V).
Transmitter Power Output:	<+2, -3dB w.r.t output power at 13.8Vdc (25watt).

All other parameters remain within specification.

2.4.2 TEMPERATURE VARIATIONS (OVER -10°C TO +60°C RANGE)

Receiver	Sensitivity:	<±3dB for 12dB Sinad, w.r.t rated sensitivity.
	AF Power:	<±2dB w.r.t 300mW output
	Mute Sensitivity:	<±3dB RF input level variation.
Adjacent Channel Sensitivity:		<±6dB
Transmitter	Output Power:	<±0.5dB w.r.t 25W output

SECTION 3 TECHNICAL DESCRIPTION

3.1 RECEIVER

3.1.1 GENERAL

The FM93 receiver is a double conversion superhetrodyne with intermediate frequencies of 21.4MHz and 455kHz.

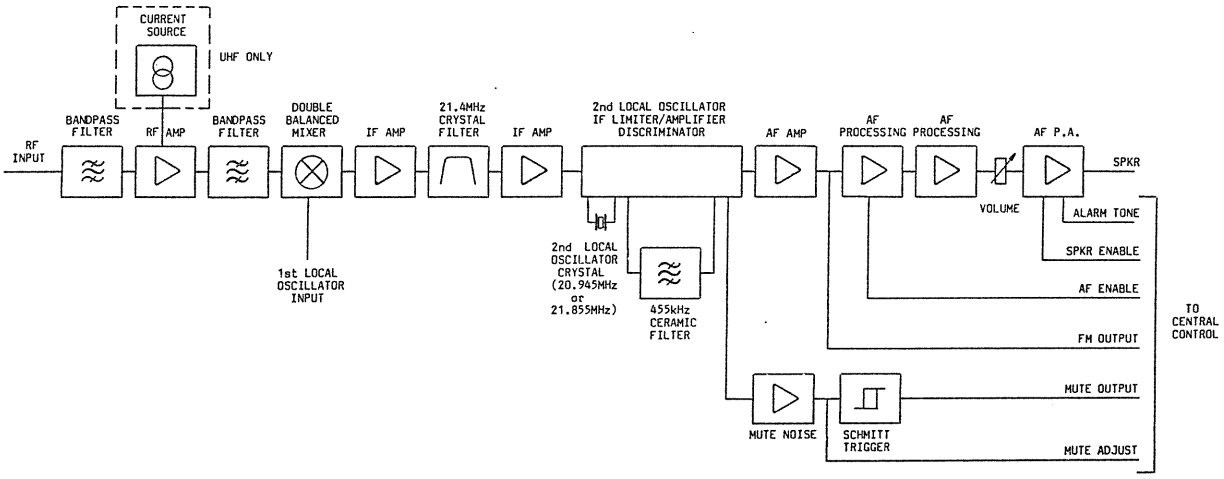


Figure 3.1 Receiver Block Diagram

3.1.2 FRONT-END

3.1.2.1 UHF Front-End

The UHF front-end consists of a discrete high performance RF amplifier (TR203) and a ring diode double balanced mixer (U202).

Two sets of 3 helical resonators (L202 and L203) provide the required RF selectivity with relatively low coupling loss.

Bias for TR203 is controlled by TR201 which is configured to provide approximately 10mA collector current for TR203.

U202 is a high dynamic range double balanced mixer. The local oscillator injection level to the mixer is approximately +10dBm.

3.1.2.2 VHF Front-End

The VHF front-end uses a common base PNP RF amplifier (TR201) and a ring diode double balanced mixer (U202).

RF selectivity is achieved with four permeability tuned circuits L210, L211, L212 and L213.

The coupling co-efficients and working Q's of the tuned circuits are controlled basically by the values of coupling capacitors except on 'A' and 'B' bands where a printed inductor (L214) is also used. This inductor is linked out on 'E' band via 2 solder bridges.

The front-end alignment requires a defined sequence of adjustment to achieve optimum performance (refer test and alignment procedure Section 4, sub-section 4.13).

3.1.3 IF SECTION

The 21.4MHz IF output from the double balanced mixer is fed to a common base amplifier TR207 which provides approximately 20dB voltage gain. Ouptut components L204, R223 and C226 provide the correct termination impedance for 4-pole crystal filter FL201.

3.1.6 POWER SUPPLY AND SWITCHING

TR210 is the main regulator set 9.8V by R234.

Other regulated lines derived from the main regulator reference are TR205 (5V6 for IC201 supply) and, in the UHF verson only, TR215 (12V P.A.). These regulators are short-circuit protected by D204 and D207 respectively.

Switched +10V Tx and +10V P.A. transistors (TR213 and TR206 respectively) are controlled by a latch in the Central Control. Timing for Tx ON and OFF sequences are contained in the software progam.

3.2 SYNTHESISER

3.2.1 GENERAL

The FM93 synthsiser is a single phase-locked-loop employing a high gain sample and hold phase comparator and a low noise VCO.

The synthesiser is based on the NJ8822, IC303, which contains two phase comparators, a reference divider and a programmable frequency divider. This IC is preceded by a divide-by-64/65 prescaler.

The main divider and the reference divider are programmed serially via three programming lines SD (Serial Data), S CLK (Serial Clock) and PE (Program Enable). A total of 28 progamming bits are required to change the synthesiser frequency (refer to Figure 3.4 for programming waveforms).

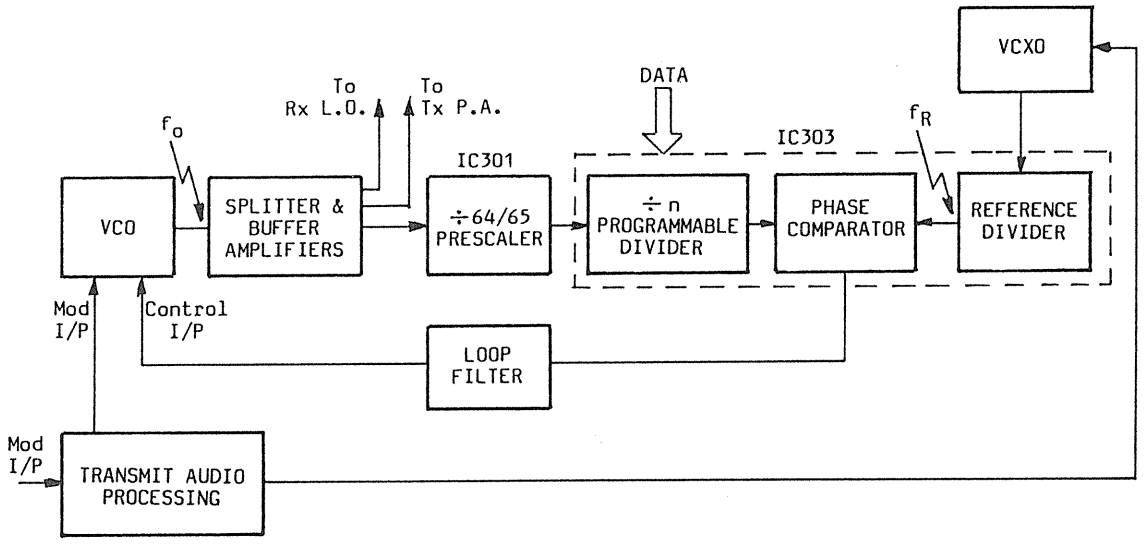


Figure 3.3 Synthesiser Simplified Block Diagram

The reference divider is programmed to provide a 5kHz, 6.25kHz or 12.5kHz reference frequency. The reference frequency is not selectable on a per-channel basis.

To align the synthesiser on transmit, the microphone cradle line should be held low; this allows continuous reprogramming of the synthesiser until the VCO is adjusted into lock. During normal operation when the microphone is off the cradle, reprogramming is limited to 16 attempts (500mS), if the loop fails to lock in this time a tone will sound while the PTT key is active. The red LED will glow only when the loop is in lock and the +10V PA line is enabled. The O.O.L. line is monitored continuously by the Central Control. In the receive mode reprogramming will continue until the loop achieves lock regardless of the cradle status.

The main logic control element, IC408, is an 8-bit CMOS microcomputer (146805E2) with external 8k program memory.

The interface to the transceiver is expanded by a single hex latch IC405 and optional RAM, I/O, Timer IC415 (NSC810).

3.3.1 MICROCOMPUTER POWER SUPPLY AND POWER FAIL RESET CIRCUITRY

3.3.1.1 Permanent 5-Volt Regulator

The permanent 5-volt regulator (IC401) is used to maintain supply (+5VP) to all critical CMOS circuit elements including the microcomputer (IC408), the on/off and power up reset circuits (IC407), peripheral control latch (IC405) and the RAM-I/O-Timer (IC415).

The permanent 5-volt regulator is connected to the unswitched 13.8-volt supply via the reverse polarity protection diode D402, and transient suppression circuit R405 and C403. Capacitors C418, C406, C408 and C427 serve to maintain supply to the above circuits whenever transients occur on the input supply, preventing loss of data and ensuring an orderly recovery from dropouts.

3.3.1.2 Switched 5-Volt Regulator

The switched 5-volt regulator (IC411) is used to supply (+5VS) to all other non-critical circuit elements.

When the set is switched off, the address strobe line (AS) will be held at logic zero. This grounds the base of the emitter follower TR404, turning off the switch transistors TR406 and TR405 and removing the input to the switched 5-volt regulator.

When the set is switched on, the address strobe pulses are applied, via TR404, to the base of the switch TR406 through the integration network R435 and C432. Following the third or fourth address strobe pulse, TR406 will be switched on, turning on TR405, and supplying approximately 12.5-volts to the input of the switched 5-volt regulator.

3.3.1.3 Power On/Off Switch

The power on/off circuit consists of IC407 configured as a monostable and schmitt trigger. When the on/off switch is depressed, pin 10 of IC407 will go to logic 1 which is applied to the PA5 input to identify the reset pulse as being due to an on/off request and not a supply transient.

When pin 10 of IC407 goes high the monostable action produces a 10mSecond logic 0 pulse at pin 11. Under normal operating conditions, pin 1 of IC407 will be at logic 1 allowing the negative pulse generated by the on/off key to be applied to the reset pin of the microcomputer.

The on/off switch is programmed as a toggle function, therefore, if the mobile was in the "on" condition, a switch off sequence will be initiated.

3.3.2 MICROCOMPUTER ADDRESS, DATA, CONTROL BUS AND CLOCK

3.3.2.1 Clock

The basic clock frequency for all microprocessor operations is 1MHz, giving an instruction cycle time of 1uS. This frequency is derived by internally dividing the signal at the oscillator input by 5.

An internal, timer-counter in the microcomputer further divides the basic clock by 416 and activates a timer interrupt. This interrupt forms the basis of the real time foreground processing section of the program.

For UHF mobiles the microprocessor oscillator signal is derived by use of a 5MHz parallel resonant crystal in conjunction with the internal clock oscillator circuit of the 146805E2.

3.3.3 RADIO INTERFACE FROM/TO MICROCOMPUTER

3.3.3.1 Microcomputer Port Lines

All interfacing to the radio system is indirectly handled by the microcomputer through the two inbuilt 8-bit bidirectional I/O ports (PA0 to PA7 and PB0 to PB7). Generally, port A lines PA0 to PA5 are used as outputs only (except during the on/off sequence when PA5 is temporarily converted to an input) and are written to with paralled data. The lines of port B all perform individual dedicated functions, and are programmed in the bit mode as follows:

PB0 - Unit latch
PB1 - Tens latch
PB2 - Mute detect
PB3 - Latch 2 strobe
PB4 - Power relay
PB5 - VHF oscillator frequency switching control
PB6 - Synthesiser program enable
PB7 - Indicator latch

3.3.3.2 Synthesiser Programming Interface

The synthesiser is programmed under control of the processor port lines PA0 (Serial Clock), PA1 (Serial Data) and PB6 (Program Enable). While the synthesiser is being programmed, PB6 will be held high, serial data will be asserted to PA1 and then clocked into the synthesiser on the negative transition of the synthesiser clock line PA0. A total of 28 bits will be clocked into the synthesiser (the first 17 bits for FREQUENCY DATA and the remaining 11 bits for the REFERENCE DIVIDER) the first bit in each case being the most significant.

3.3.3.3 TX/RX Control Lines

Transmit and receive control lines are handled primarily through latch 2 (IC405). Data is written to this latch every 833us. Lines controlled through this latch are as follows.

1. Transmit enable.
2. VCO offset.
3. Rx AF enable/Speaker enable.
4. Microphone inhibit.
5. Alarm tone.
6. P.A. enable.

3.3.4 SIGNALLING INTERFACE

Several product options require additional hardware hybrid modules (U402, U403 and U405) which are controlled by the microcomputer through memory mapped port lines of the RAM-I/O - timer (IC415). These lines are TTL level compatible and may be configured as inputs or outputs, as required by the selected option.

3.3.4.1 Selective Call

Selective calling is implemented by the hybrid circuit module U402, connected to port A of IC415. The first four lines of this port (PA0-PA3) define the current code of the tone code sequence which is being sent or received and will be configured as either an output or an input, depending on whether the module is in the Encode or Decode mode respectively. The fifth line (PA4) is always an output and is high to select encode and low for decode. Line six (PA5) is an output to the external alarm pin on the hybrid. The seventh line (PA6) is configured as an input or output in

turn adjusts the supply level to the first transistor stage of the power module and thus controls the gain of the power module. The action of the control loop is to stabilize the power input level to TR102. At low power settings of R102, stabistor D101 helps to stabilize the transmitter output power at low temperatures by providing a negative temperature coefficient on the reference voltage.

The PTC R106 senses the Tx PA internal temperature. In the event of a temperature rise above 100°C the PTC resistance will increase. This will cause a reduction of the control loop reference voltage and will therefore reduce the transmitter output power. Link LK101 is opened to disable the control loop for Tx PA tuning.

3.4.1.3 RF Power Output

RF output from TR102 is coupled to the antenna socket via the solid state antenna changeover circuit. In the transmit mode +10V Tx biasses PIN Diode D103 on and couples the output power to the antenna socket. The bias current for D103 flows into D104, D105 and D110 via a quarter wave length line. D104, D105 and D110 when biassed on, form a short circuit across the receiver input to achieve isolation from the transmitter. This short circuit is transformed to an effective open circuit at the other end of the quarter wavelength line, and therefore does not load the transmitter output. In receive mode, bias to the diodes is removed and the receiver input is coupled to the antenna socket. With D103 off the transmitter output does not load the receiver input.

An antenna filter between the antenna socket and the solid state change-over circuit rejects harmonic emmissions from the transmitter output.

3.4.2 TRANSMITTER (VHF)

The transmitter PA is based on a three stage line-up utilising discrete transistors in each stage. The first, predriver stage TR101, receives an input drive level of 25mW and delivers typically 250mW to the second stage, driver TR102. A drive level of typically 3W is then fed into the PA stage TR103, which provides a further 9.5dB gain to produce an output of 25 Watts into 50 ohms.

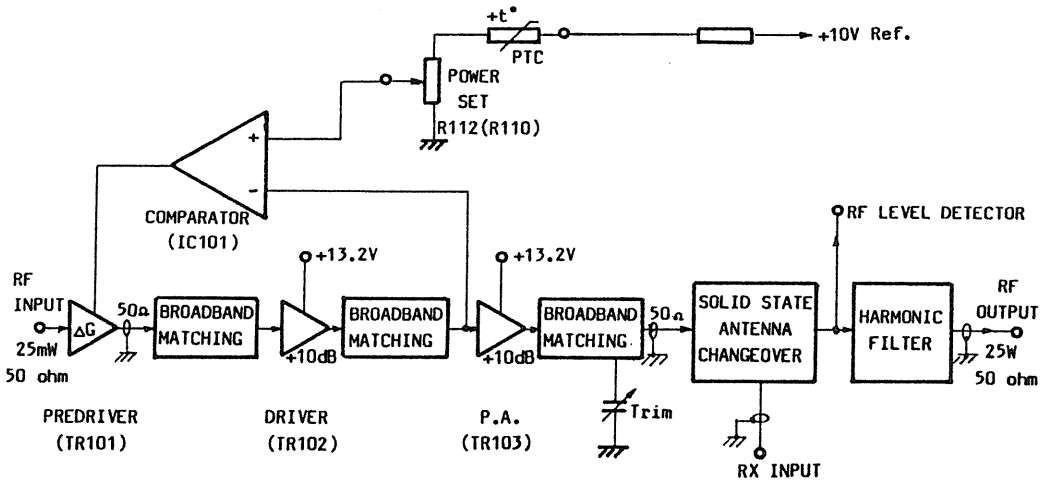


Figure 3.8 VHF Transmitter PA Block Diagram

Broadband impedance matching is employed between all stages and at the input and output to the line-up. A 50 ohm impedance point is available between the predriver and driver stages.

3.5.1 VOLUME AND MUTE CONTROL

Volume and mute controls use conventional rotary adjusting potentiometers.

The mute control R504 produces a variable voltage between 0 and 2.5 volts which is applied to the mute amplifier in the receiver. Level zero will always guarantee a mute open condition.

The volume control R505, provides an adjustable attenuation in the receiver audio between the receiver audio output and the AF power amplifier input. C504 and R502 form a high pass filter to reduce the low frequency response to the loudspeaker.

3.5.2 KEY OPERATION

Every 13.3ms the four data lines PA0 to PA3 are switched to inputs by the central control processor and tested for an active key. The keys are encoded onto the data lines as follows:

PA0	Channel +	(Active Low)
PA2	Send	(Active Low)
PA3	Auxiliary	(Active Low)

Whenever a key is depressed the corresponding input line, normally held at logic 1, will be grounded through resistors R511, R513 and R514.

If a key is pressed while the 7-segment display or the discrete LED's are being updated, resistors R511, R513 and R514 are used to buffer data on lines PA0 to PA3 from the key ground contact.

The on/off switch S505 provides a ground contact and is connected directly to the central control board reset circuitry.

3.5.3 7-SEGMENT DISPLAY CONTROL

Interface to the 7-segment display (D501) is via four parallel binary coded decimal (BCD) data lines (PA0-PA3), plus a digit strobe input.

The display D501 is updated as required by placing the appropriate BCD code on the data lines and pulsing the required digit strobe input low.

Data is effectively latched on the positive edge of the strobe pulse. This process updates the data in the appropriate display latch/decoder/driver IC501.

The outputs of the display driver IC provides a constant current source through resistors R516-R522 to D501.

3.5.4 DISCRETE LED AND DISPLAY CONTROL

The two LED indicators are the "Rx" amber LED D503 and the "Tx" red LED D504. These LED's are controlled by the 4-bit latch IC503. This latch also provides a blanking control to the display driver IC501, and a control to the decimal point input of the display.

The data input to the latch IC503 is defined as follows:

PA0	Red "Tx" LED	(Active high)
PA1	Display Blanking	(Active low)
PA2	Amber "Rx" LED	(Active high)
PA3	Decimal Point	(Active high).

The latch is strobed by a positive going pulse from the "indicator latch" input to the control unit.

The display blanking input (PA1) to the latch, is used to flash the 7-segment displays when required.

SECTION 4 TEST AND ALIGNMENT PROCEDURE

NOTES:

1. This test and alignment procedure is applicable to all versions of the FM93 Series mobile, with the exception of mobiles fitted with signalling or special options. Refer to the documentation on the appropriate option for any additional test and alignment instructions.

2. The test and alignment procedures given, follow a logical sequence and it is therefore recommended that each step be completed before proceeding to the next.

3. The Synthesiser and Receiver PC Boards are held in the servicing position by the PC Board Support Spring. This spring must be released to allow the cam on the board attachment to pass freely, or damage to the spring may result.

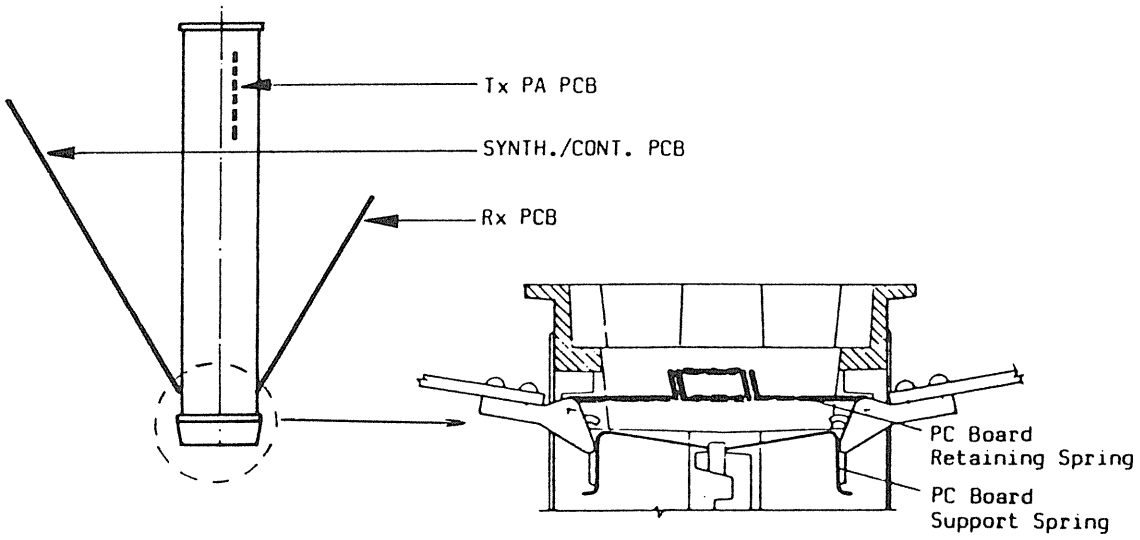


Figure 4.1 FM93 Series - Chassis with Sectional Side View showing PCB Support and Retaining Springs

4. Re-fitting of Screws:
To ensure no 'second starting' occurs the following procedure is to be followed.

a) Locate screw in hole
b) Press down lightly on screw, turning anti-clockwise until a click is felt.
c) Turn clockwise to drive screw home in normal manner.
d) Re-fitting torque for M2.5 screws - .65 to .75Nm
for M3.5 screws - 2.0 to 2.2Nm

Torque tighten VHF Stud Device nut - .40 to .50Nm.

4.4 POWER SUPPLY CHECK

STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
1	AVO 1A DC Range	In series with +ve supply lead	Activate Power On	<650mA	Channel display ON
2	DVM + 10V DC Range	Collector of TR210	R234	+9.8V ±0.1V	Checks +10V regulator output.
3	AVO +30V DC Range	Collector of TR215		+12V ±0.5V	Checks +12V regulator output. UHF VERSION ONLY
4	AVO +10V DC Range	Collector of TR213	Activate PTT	+9.7V ±0.1V	Checks +10V Tx output (VCO must be in lock)

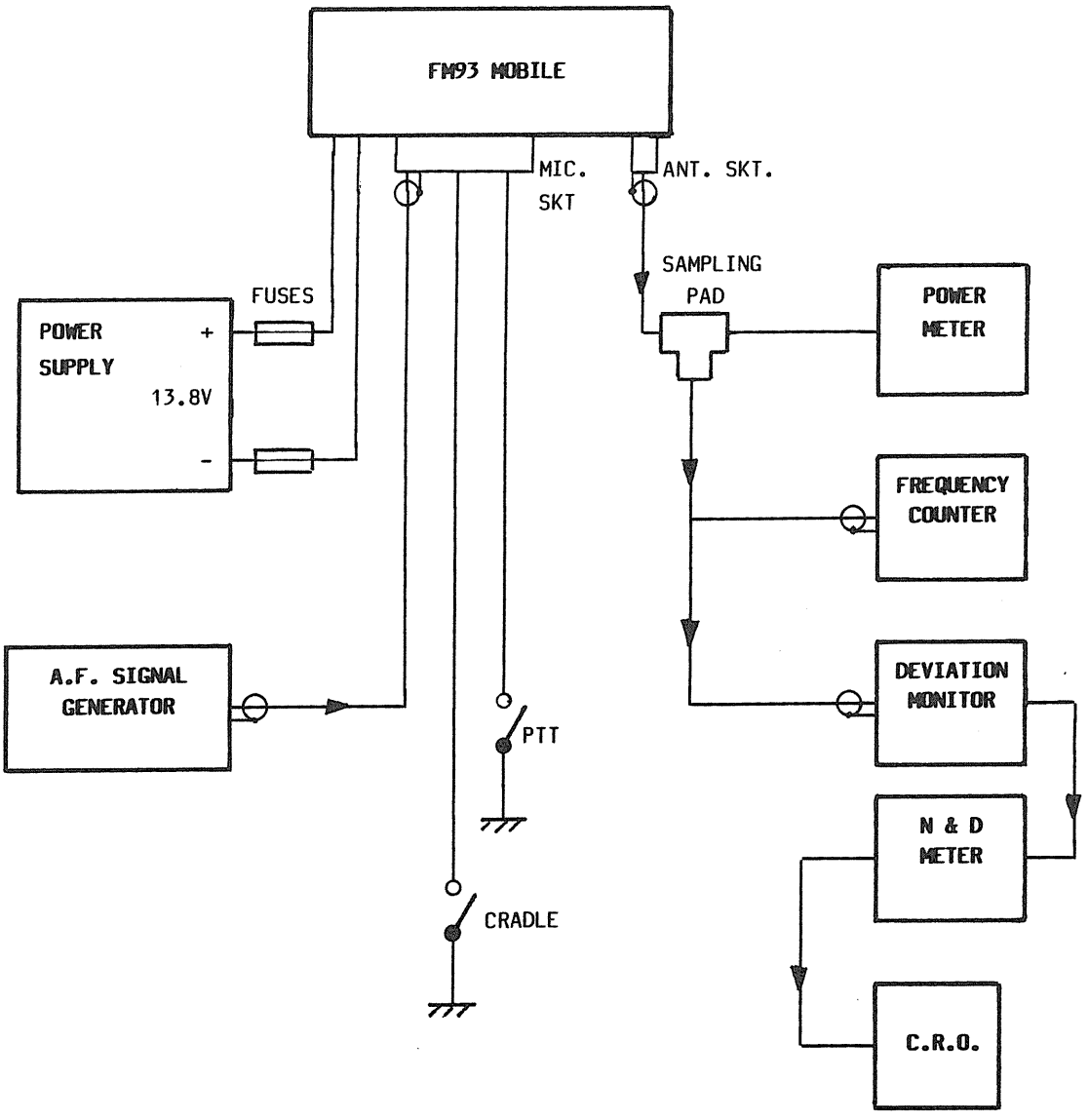


Figure 4.1 Synthesiser and Transmitter Alignment Equipment Set-up

4.6 **VCO ALIGNMENT** (Cont'd)

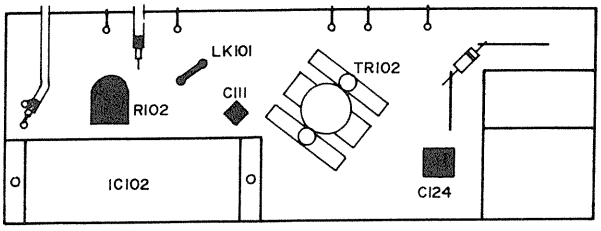
STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
6	AVO 30V DC Range	TP201	VCO offset tuning.	12V ± 0.5V	If voltage cannot be adjusted below 13V go to Special Proc. No.1 If voltage cannot be adjusted above 11V go to Special Proc. No.2
NOTE: STEPS 1 TO 6 SHOULD BE REPEATED TO ACCOUNT FOR ANY INTERACTION IN THE TUNING.					
7	AVO 30V DC Range	TP201	Select channel for lowest Receive Frequency.	>4V	See channel frequency label.
8	AVO 30V DC Range	TP201	Select channel for lowest transmit freq. & Activate PTT.	>4V	See channel frequency label.

4.6.1 **VCO SPECIAL ALIGNMENT PROCEDURE NO. 1** (Continued from step 6 of standard alignment)

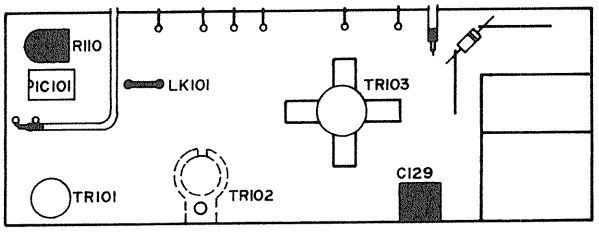
STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
7			VCO offset tuning.	Minimum capacitance or Inductance.	In VHF do not adjust slug past the top face of the VCO housing.
8	AVO 30V DC Range	TP201	VCO main tuning.	13V ±0.2V	Offset should still be on.
9	"	"	VCO offset tuning.	Minimum voltage	In VHF keep top of slug below the Top face of the VCO housing.
10	"	"	VCO main tuning.	13V ±0.2V	Accounts for interaction in tuning.
11			Select channel for highest frequency for **OFFSET OFF** condition.		
12	AVO 30V DC Range	TP201	Set mobile to **OFFSET OFF** Condition	>5V	
13	"	"	Select channel for lowest Rx fequency.	>4V	See channel freq. label.
14	"	"	Select channel for lowest Tx frequency. Activate P.T.T.	>4V	See channel freq. label.

4.7 **TX PA ALIGNMENT** (Cont'd)

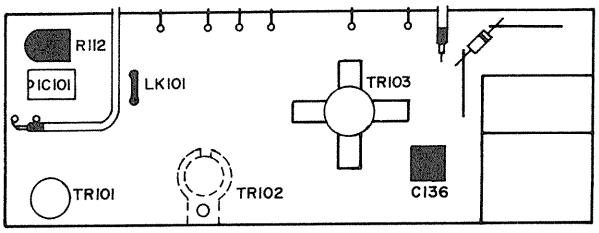
STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
5	Wattmeter (25W)	Ant.Skt.	Activate PTT	25W	
6	Wattmeter (25W)	Ant.Skt. and Tx Indicator LED	Activate PTT	OW Tx Ind.LED Extinguished	Link XL301 to ground, checks O.O.L. function.



UHF All Bands



VHF E Band



VHF A & B Bands

Figure 4.3 Tx P.A. Board Layout and Test Point Location

4.10 **TX MODULATION ALIGNMENT** (Cont'd)

STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
7	Deviation Monitor	via sampling pad at wattmeter.	R336	A ±4.8kHz B ±2.4kHz +0,-0.5dB	sets maximum dev. A = 25kHz version B = 12.5kHz version
8			AF Gen. output level.	40mV ±4mV rms	1kHz modulating frequency
9	Deviation Monitor	via sampling pad at wattmeter.	R317	A ±3kHz B ±1.5kHz ±1dB	sets standard modulation level. A = 25kHz version B = 12.5kHz version
10			AF Gen. output level.	400mV ±40mV rms	1kHz modulating frequency
11	Deviation monitor	via sampling pad at wattmeter.	AF Gen. Freq.	A ±5.0kHz B ±2.5kHz	Check for frequency range 300 to 3kHz A = 25kHz version B = 12.5kHz version
NOTE: FOR MULTI CHANNEL VERSIONS REPEAT STEP 11 FOR HIGHEST AND LOWEST Tx FREQUENCY					

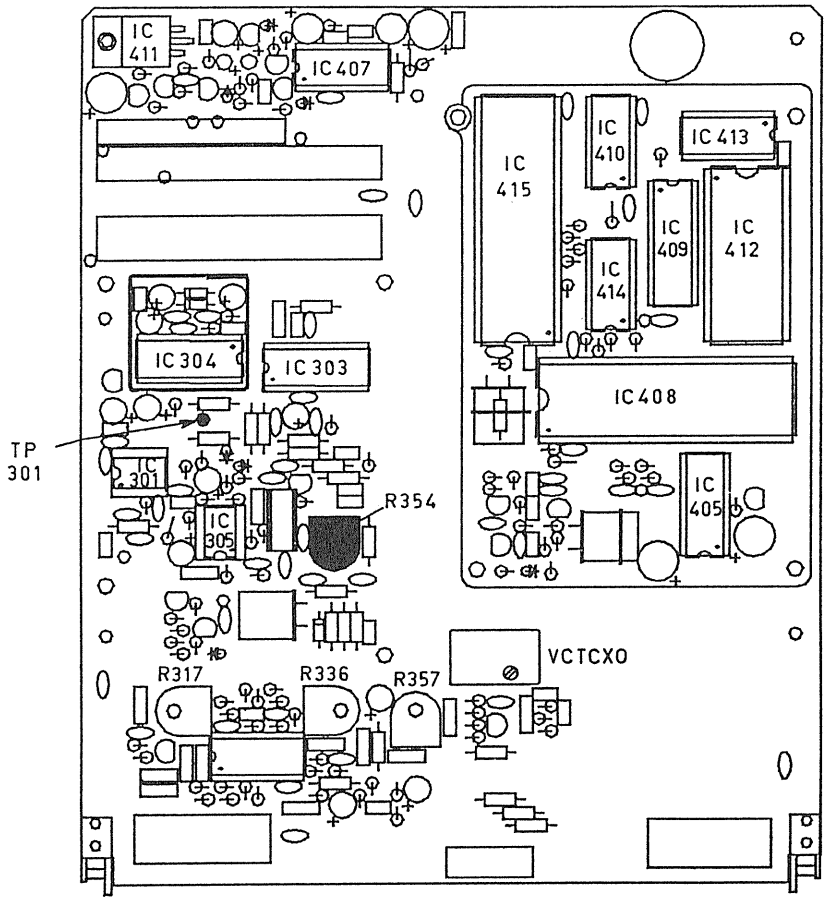


Figure 4.4 Synthesiser and Control Board Layout and Test Point Location

4.12 PRELIMINARY ADJUSTMENTS FOR RECEIVER

a) Adjust Volume Control fully counter clockwise.

b) Adjust Mute Control fully counter clockwise.

c) Adjust R251 Mute Adjust to mid position.

d) Adjust L210 to L213 cores (VHF versions) flush with top of former.

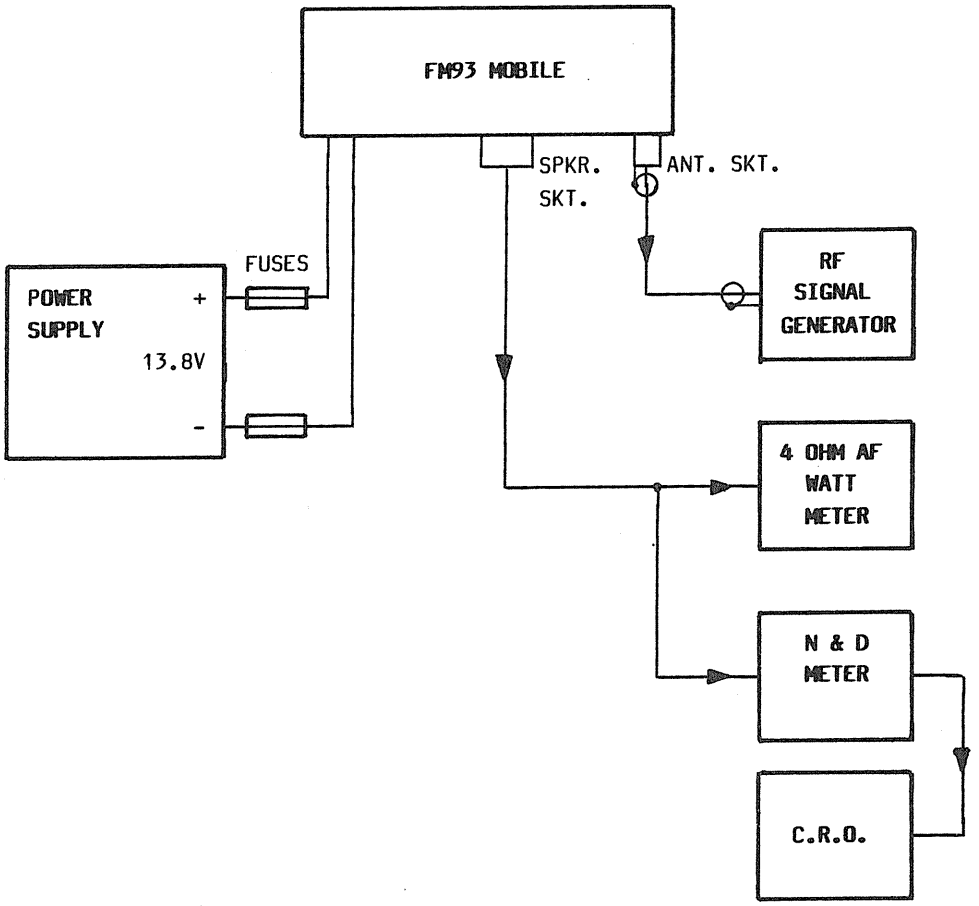


Figure 4.5 Receiver Alignment Equipment Set Up

4.13 RECEIVER ALIGNMENT

STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
1			Mobile channel for Rx **MID** Frequency.		See channel frequency label.
2			RF Sig. Gen. Frequency for Rx **MID** channel.		Modulate generator with 1KHz frequency - ±3kHz (25kHz) or ±1.5kHz (12.5kHz) deviation. Switch output off.
3	AF Watt Meter	Speaker output socket.	Volume Control	Approx. 300mW of noise output	

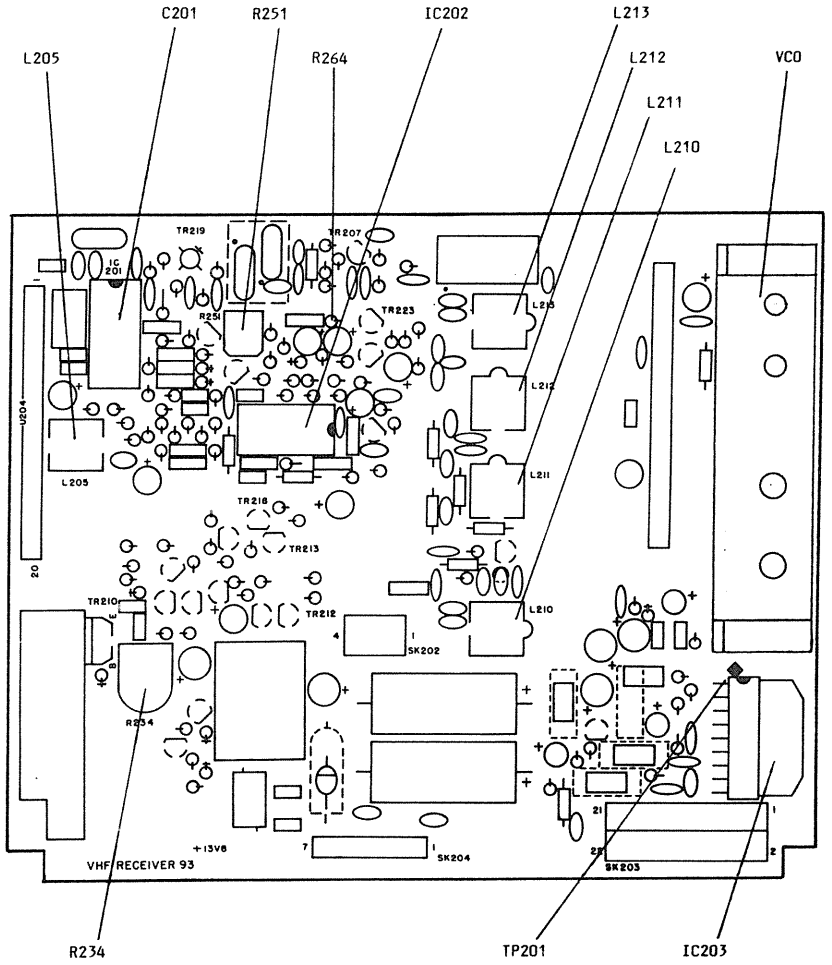


Figure 4.6 VHF Receiver Board Layout and Test Point Location

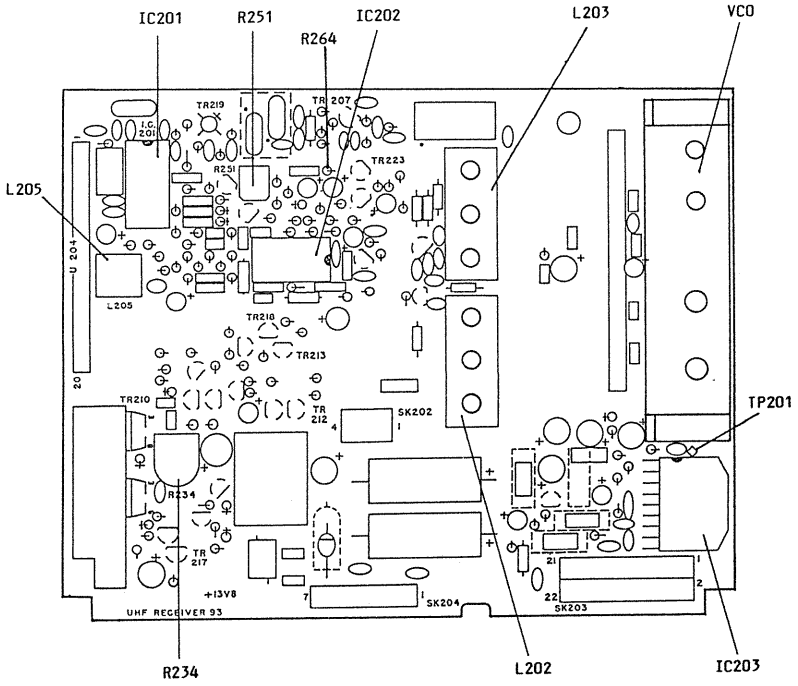


Figure 4.7 UHF Receiver Board Layout and Test Point Location

4.14 **RECEIVER MODULATION PERFORMANCE** (Cont'd)

STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
9			Switch on RF Sig. Gen. modulation		Set modulation to ±500Hz deviation
10	A.F. Wattmeter	Speaker output	Volume Control	100mW±2dB	
11	N&D meter	Speaker output	N&D meter for 0dB reference.	0dB	
12	N&D meter	Speaker output	RF Sig. Gen. modulation to 450Hz	6.9 +1dB -3dB	
13	N&D meter	Speaker output	RF Sig. Gen. modulation to: A 3kHz B 2.55kHz	A-9.5+1dB -3dB B-8.2+1dB -3dB	A = 25kHz version B = 12.5kHz version
14	N&D meter	Speaker output	RF Sig. Gen. modulation to 250Hz	<-18dB	
15			RF Sig. Gen. modulation to 1kHz		
16			Volume control to max.		
17	A.F. Wattmeter	Speaker output	RF Sig. Gen. deviation	3.5W±0.2W	Deviation should be <±2kHz/25kHz version <±1kHz/12.5kHz version

4.15 **MUTE PERFORMANCE**

STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
1			RF Sig. Gen. output level	100uVPD ±10uV	Modulate generator with 1kHz frequency and dev. ±3kHz (25kHz) or ±1.5kHz(12.5kHz)
2	A.F. Wattmeter	Speaker output	Volume control	300mW±2dB	
3			Switch off RF signal gen. output.		
4			Mute control to min.	Receiver unmuted	Yellow Rx LED should turn On.

4.17 TX LIMIT TIMER CHECK

Transmit limit options: 1, 3, 5, 10 minute. (See mobile label)
Retransmit time : 0 seconds.
Additional test equipment required: Stop watch.
Loud speaker.

NOTE: READ THIS PROCEDURE BEFORE MAKING THE TEST					
STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
1			Mobile Channel for TX Mid Frequency.		See channel frequency label.
2			Connect loudspeaker to mobile.		Loudspeaker socket located at rear of mobile
3			Reset stop watch to zero.		
4	Wattmeter 25W range.	ANT. SKT.	Activate PTT and start stop watch.	25Watt	PTT and stop watch should begin simultaneously.
5	Wattmeter	ANT. SKT.		0 Watt	At transmit time out, a continuous tone alarm should be heard from loudspeaker. Tx LED should go out.
	Stop watch.	Time out period		Limit time +5 secs.	
6			Release PTT and then reactivate PTT	25Watt	Transmitter should become active again.
7			Disable PTT		

4.18 SCANNING

4.18.1 FM93 (OCCUPIED PRE-PROGRAMMED BLOCKS)

STEP	CHECK
1	Connect mobile to test equipment set up. Ensure mic is in cradle.
2	Switch mobile on and enable the scan mode by pressing the "AUX" button. (This will select scan group 1 as shown on channel frequency label. If a second scan group is also fitted, pressing AUX button again will enable the second group. Pressing a third time will return the mobile back to its previous channel).
3	When scan group 1 is enabled the display will flash 1. Remove mic from the cradle and ensure that a channel number is displayed. This channel must be one of the scan group. If a priority channel is programmed the channel must be the priority channel. Ensure that if the mic is left out of the cradle for more than 20 secs. an alarm is emitted from the speaker. The alarm should be defeated and the scan mode re-enabled when the mic is replaced in the cradle.

4.18.2 **FM93 (VACANT PRE-PROGRAMMED BLOCKS)** (Cont'd)

STEP	CHECK
6	With mic in cradle and the scan group number displayed, switch off the marker oscillator and ensure that a channel number is displayed. This should be any one of the channels in the scan group.
7	Connect a RF signal generator into the receiver antenna socket at the frequency of the channel displayed. Set generator level to 1mV pd and ensure after approximately 2 secs. that the receiver mutes and selects another channel in the scan group. (Use a power attenuator or sampling pad between the antenna socket and the RF signal generator).
8	Activate PTT and measure transmitter frequency. Ensure that frequency corresponds to channel that is displayed in scan mode.
9	Repeat Steps 3 to Step 8 but for scan group 2 if a second scan group has been programmed.

4.19 **PTT INHIBIT**

STEP	CHECK
1	Connect mobile to test equipment set up. Ensure mic is in cradle.
2	Connect RF signal generator to receiver antenna socket input via a power attenuator (or via a sampling pad at a Wattmeter). Set generator output to 1mV PD at the receiver frequency and ensure Rx amber LED is on.
3	Activate PTT and ensure that the mobile transceiver does not transmit and that an audible alarm is sounded in the speaker for as long as the PTT is activated. The alarm should cease when the PTT is disabled.
4	Remove RF signal generator input from receiver and ensure that PTT can be activated.

4.20 **COMMUNITY REPEATER CHECK**

STEP	CHECK
1	Connect mobile to test equipment set up as shown in Figure 4.1. Ensure cradle switch is closed and PTT is disabled. Set mute to mid position.
2	Connect RF signal generator to receiver antenna socket input via a power attenuator (or via a sampling pad at wattmeter). Set generator output to 50uV PD at the receiver frequency and switch generator modulation Off.
3	Switch mobile on and ensure Rx amber LED is On. Activate PTT and ensure PTT is inhibited. An alarm tone should appear at speaker socket output. Disable PTT.
4	Modulate signal generator with 1kHz ±1.5kHz deviation and ensure no A.F. output occurs in speaker output. (Adjust Rx volume control and observe output on C.R.O. via N & D Meter).

SECTION 5 INSTALLATION

NOTE: REFER ALSO TO "VEHICLE CARE" PAGE AT FRONT OF HANDBOOK

5.1 GENERAL INFORMATION

The mobile radio unit must be fitted in a position such that the operator has easy access to the controls and microphone when wearing the seat belt. The controls must also be situated so that the driver can see them within his normal field of vision.

Fitting positions above the driver's or passenger's head must be avoided. Care should be taken to ensure that the microphone/handset lead is not installed such that the lead can interfere with the vehicle controls, and in particular the driver's feet.

The microphone/handset clip should be fitted such that the microphone/handset is easily accessible. In the case of a loudspeaker it is necessary for the loudspeaker grill to face the operator when mounted.

Caution should be exercised before drilling holes through bulkheads to ascertain if safe to do so. i.e. fuel tank, fuel line, brake line, battery, cable loom etc.

The FM93 series has been designed with safety in mind, e.g. non-reflective surface, no sharp corners or protrusions. Care must be taken that any additional metalwork necessary to fix the units into the vehicle during installation conforms to the same requirement.

Refer to Section 5.4 for mounting cradle and transceiver fitment.

5.2 VEHICLES FITTED WITH ELECTRONIC DEVICES

An additional precaution is necessary in relation to vehicles fitted with Electronic Ignition, Fuel Injection, Anti-skid Brakes or any other electronic device where temporary loss of service could be hazardous.

In theory, any of the above systems could be affected by the presence of an RF field of sufficient intensity, which when detected may cause the device to malfunction. The source of RF may be a mobile radio installed in the vehicle itself or a transmitter operating in another vehicle alongside. If interaction did occur, loss of control could result for the duration of the mobile transmission.

In the interests of safety, the user must be asked to test the vehicle when the installation is complete.

If a problem is found the owner should take the vehicle to an automobile specialist to resolve the problem. Unqualified persons should not attempt to modify these units in any way, the work should be done by a qualified automobile electrician.

5.3 SPECIALISED VEHICLES

The installation on certain specialised vehicles, such as Petrol Tankers and Fire Appliances, may be subject to safety regulations which must be closely observed.

Prior to commencing an installation on such a vehicle, be sure that any relevant safety regulations are fully understood.

SECTION 6 COMPONENT SCHEDULE

6.1 VHF PCB CODES

6.1.1 25KHZ CHANNEL SPACING

PC BOARD	E BAND 9502 932 30000	B BAND 9502 933 30000	A BAND 9502 934 30000
Tx PA PC Board	3502 349 71020	3502 349 71130	3502 349 71140
Receiver PCB	3502 349 72420	3502 349 72430	3502 349 72440
Synth/Control PC Board	3502 349 73140	3502 349 73440	3502 349 73440
Local Control Display Brd	3502 349 98860	3502 349 98860	3502 349 98860

6.1.2 12.5kHz CHANNEL SPACING

PC BOARD	E BAND 9502 932 70000	B BAND 9502 933 70000	A BAND 9502 934 70000
Tx PA PC Board	3502 349 71020	3502 349 71130	3502 349 71140
Receiver PCB	3502 349 72460	3502 349 72470	3502 349 72480
Synth/Control PC Board	3502 349 73160	3502 349 73440	3502 349 73460
Local Control Display Brd	3502 349 98860	3502 349 98860	3502 349 98860

6.2 UHF PCB CODES

6.2.1 25KHZ CHANNEL SPACING

PC BOARD	T - BAND 9502 935 30000	U - BAND 9502 936 30000	W1 - BAND 9502 937 30000
Tx PA PC Board	3502 349 81050	3502 349 81060	3502 349 81070
Receiver PCB	3502 349 82810	3502 349 82820	3502 349 82830
Synth/Control PC Board	3502 349 83120	3502 349 83120	3502 349 83120
Local Control Display Brd	3502 349 98860	3502 349 98860	3502 349 98860

6.2.2 12.5kHz CHANNEL SPACING

PC BOARD	T - BAND 9502 935 70000	U - BAND 9502 936 70000	W1 - BAND 9502 937 70000
Tx PA PC Board	3502 349 81050	3502 349 81060	3502 349 81070
Receiver PCB	3502 349 82850	3502 349 82860	3502 349 82870
Synth/Control PC Board	3502 349 83130	3502 349 83130	3502 349 83130
Local Control Display Brd	3502 349 98860	3502 349 98860	3502 349 98860

6.4 RECEIVER PC BOARDS (UHF)

6.4.1 RECEIVER BOARDS UHF 25kHz

6.4.1.1 Receiver PCB T-Band (25kHz) 3502 349 82810

	Receiver Board Basic (25kHz)	3502 349 82900			
U201	Hybrid H12	3502 358 00190	L202	Helical Filter T-Band	3502 389 90970
U203	VCO Module Assy T-Band 93	3502 357 90120	L203	Helical Filter T-Band	3502 389 90970

6.4.1.2 Receiver PCB U-Band (25kHz) 3502 349 82820

	Receiver Board Basic (25kHz)	3502 349 82900			
U201	Hybrid H12	3502 358 00190	L202	Helical Filter U-Band	3502 389 90940
U203	VCO Module Assy U-Band 93	3502 357 90390	L203	Helical Filter U-Band	3502 389 90940

6.4.1.2 Receiver PCB W1-Band (25kHz) 3502 349 82830

	Receiver Brd Basic 25kHz 93	3502 349 82900			
U201	Hybrid H12B Buffer	3502 358 00620	L202	Helical Filter W1-Band	3502 389 90930
U203	VCO Module Assy.	3502 357 90130	L203	Helical Filter W1-Band	3502 389 90930

6.4.1.2 Receiver Board Basic 25kHz 3502 349 82900

	Receiver Board UHF Basic 93	3502 349 82000	R263	Res. Carb.F. 470E, 5%, ¼W	2120 101 46471
C223	Cap. Cer.Pl. 12p, 2%, 100V NPO	2222 680 10129	R265	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153
R223	Res. Carb.F. 1k2, 5%, ¼W	2120 101 46122	R267	Therm. NTC, 220E,10%, 0.6W	2322 642 62221
R236	Res. Carb.F. 1k0, 5%, ¼W	2120 101 46102	R273	Res. Met. F. 56k, 1%, 0.6W	2322 156 15603
R250	Res. Carb.F. 12k, 5%, ¼W	2120 101 46123	R276	Res. Carb.F. 39k, 5%, ¼W	2120 101 46393
R252	Res. Carb.F. 220K, 5%, ¼W	2120 101 46224	FL201	Xtal Filter, Mnlthic, 21F15B	2722 172 90012
R257	Res. Carb.F. 68k, 5%, ¼W	2120 101 46683	FL202	Filter Cer. 445kHz, CCFW455D	2422 549 03635

6.4.2 RECEIVER BOARDS UHF 12.5kHz

6.4.2.1 Receiver PCB T-Band(12.5kHz) 3502 349 82850

	Receiver Brd Basic 12.5kHz 93	3502 349 82890			
U201	Hybrid H12	3502 358 00190	L202	Helical Filter T-Band	3502 389 90970
U203	VCO Module Assy T-Band 93	3502 357 90120	L203	Helical Filter T-Band	3502 389 90970

6.4.2.2 Receiver PCB U-Band(12.5kHz) 3502 349 82860

	Receiver Brd Basic 12.5kHz 93	3502 349 82890			
U201	Hybrid H12	3502 358 00190	L202	Helical Filter U-Band	3502 389 90940
U203	VCO Module Assy U-Band 93	3502 357 90390	L203	Helical Filter U-Band	3502 389 90940

6.4.2.3 Receiver PCB W1-Band(12.5kHz)3502 349 82870

	Receiver Brd Basic 12.5kHz 93	3502 349 82890			
U201	Hybrid H12B Buffer	3502 358 00620	L202	Helical Filter W1-Band	3502 389 90930
U203	VCO Module Assy.	3502 357 90130	L203	Helical Filter W1-Band	3502 389 90930

6.4.2.4 Receiver Board Basic 12.5kHz 3502 349 82890

	Receiver Board UHF Basic 93	3502 349 82000			
C223	Cap. Cer.Pl. 8p2,p25, 100V NPO	2222 680 09828	R265	Res. Carb.F. 18k, 5%, ¼W	2120 101 46183
R223	Res. Carb.F. 1k8, 5%, ¼W	2120 101 46182	R267	Therm. NTC, 100E,10%, 0.6W	2322 642 62101
R236	Res. Carb.F. 1k5, 5%, ¼W	2120 101 46152	R273	Res. Met. F. 82k, 1%, 0.6W	2322 156 18203
R250	Res. Carb.F. 6k8, 5%, ¼W	2120 101 46682	R276	Res. Carb.F. 18k, 5%, ¼W	2120 101 46183
R252	Res. Carb.F. 680K, 5%, ¼W	2120 101 46684	FL201	Xtal Filter, Mnlthic, 21F7.5BA	2722 172 90038
R263	Res. Carb.F. 470E, 5%, ¼W	2120 101 46471	FL202	Filter Cer. 445kHz, CCFW455F	2422 549 03637

6.4.3 RECEIVER PC BOARD (UHF) (Cont'd)

C214	Cap.Al.Elec. 4u7,-10+50%,63V	2222 035 58478
C215	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C216	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C217	Cap Al.Elec. 47u,-10+50%,25V	2222 035 56479
C218	Cap.Cer.Pl. 470p,10%,100V	2222 630 08741
C219	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C220	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C221	Cap.Cer.Pl. 18p, 2%,100V NP0	2222 680 10189
C222	Cap.Cer.Pl. 150p, 2%,100V N750	2222 680 58151
C223	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C224	Cap Al.Elec. 47u,-10+50%,25V	2222 035 56479
C225	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C226	Cap.Cer.Pl. 3p9,p25,100V NP0	2222 680 09398
C227	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C228	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C229	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C230	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C231	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C232	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C234	Cap Al.Elec. 47u,-10+50%,25V	2222 035 56479
C235	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C236	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C237	Cap.Cer.Pl. 150p, 2%,100V N750	2222 680 58151
C238	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C239	Cap Al.Elec. 47u,-10+50%,25V	2222 035 56479
C240	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C241	Cap.Cer.Pl. 22p, 2%,100V NP0	2222 680 10229
C242	Cap.Pes.Stkd 1n0, 5%,63V	2011 301 20032
C243	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C244	Cap.Cer.Pl. 47p, 2%,100V NP0	2222 680 10479
C245	Cap.Pes.Stkd 1n0, 5%,63V	2011 301 20032
C246	Cap.Pes.Stkd 1n0, 5%,63V	2011 301 20032
C247	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C248	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C249	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C250	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C251	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C252	Cap.Al.Elec. 4u7,-10+50%,63V	2222 035 58478
C253	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C254	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C255	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C256	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C257	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C258	Cap.Cer.Pl. 150p, 2%,100V N750	2222 680 58151
C259	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C260	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C261	Cap.Pes.Stkd 6n8, 5%, 63V	2011 301 20029
C262	Cap.Pes.Stkd 6n8, 5%, 63V	2011 301 20029
C263	Cap.Al.Elec. 47u,-10+50%,25V	2222 035 56479
C264	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C265	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C266	Cap.Pes.Stkd 6n8, 5%, 63V	2011 301 20029
C267	Cap.Al.Elec. 1u,-10+50%,10V	2222 035 58108
C268	Cap.Cer.Pl. 2n2,10%,100V	2222 630 08222
C269	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C270	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C271	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C272	Cap.Pes.Stkd100n, 5%,63V	2011 301 20031
C273	Cap.Pes.Stkd100n, 5%,63V	2011 301 20031
C274	Cap.Al.Elec. 47u,-10+50%,25V	2222 035 56479
C275	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C276	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C277	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C278	Cap.Pes.Stkd100n, 5%,63V	2011 301 20031
C279	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C280	Cap.Pes.Stkd 1n0, 5%,100V	2011 301 20032
C281	Cap.Al.Elec. 47u,-10+50%,25V	2222 035 56479
C282	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C283	Cap.Al.Elec.470u,-10+50%,16V	2222 031 35471
C284	Cap.Al.Elec.470u,-10+50%,16V	2222 031 35471
C285	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C286	Cap.Al.Elec. 4u7,-10+50%,63V	2222 035 58478
C287	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
L201	Choke Wide Band RF	4312 020 36640
L204	Choke 6.8uH,10%, AXL 70-23	2422 535 97559
L205	Coil, Discr. Toko RMC-2A6597HM	2422 549 18111
RL201	Relay, Min 10Amp 12V	2402 132 00015
XL201	Crystal 20.945Hz, 2nd L.Osc	3502 413 48410
IC201	Intgrd.Cct. MC3361P	9337 941 60682
IC202	Intgrd.Cct. LM348M	9333 705 60682
IC203	Intgrd.Cct. TDA1010	9333 849 50112
TR201	Transistor BC558	9331 977 30112
TR202	Transistor BC548C	9331 976 70112
TR203	Transistor 2SC3355	9337 925 10682
TR204	Transistor BC337	9331 492 00112
TR205	Transistor BC548C	9331 976 70112
TR206	Transistor BC328	9331 491 90112
TR207	Transistor 2SC3355	9337 925 10682
TR208	Transistor BC548C	9331 976 70112
TR209	Transistor BC548C	9331 976 70112
TR210	Transistor BD946	9334 481 30112
TR211	Transistor BC548C	9331 976 70112
TR212	Transistor BC548C	9331 976 70112
TR213	Transistor BC328	9331 491 90112
TR214	Transistor BC548C	9331 976 70112
TR215	Transistor BD946	9334 481 30112
TR216	Transistor BC548C	9331 976 70112
TR217	Transistor BC548C	9331 976 70112
TR218	Transistor BC548C	9331 976 70112
TR219	Transistor BF960	9335 105 20112
TR220	Transistor BC548C	9331 976 70112
TR221	Transistor BC548C	9331 976 70112
TR222	Transistor BC548C	9331 976 70112
TR223	Transistor BC548C	9331 976 70112
TR224	Transistor BC548C	9331 976 70112
TR225	Transistor BC548C	9331 976 70112
D201	Diode 1N4148	9330 839 90112
D202	Diode 1N4148	9330 839 90112
D203	Diode BZT03-C20	9336 015 80112
D204	Diode 1N4148	9330 839 90112
D205	Diode BZX79-C6V2	9331 177 40112
D206	Diode 1N4148	9330 839 90112
D207	Diode 1N4148	9330 839 90112
D208	Diode 1N4148	9330 839 90112
D209	Diode 1N4148	9330 839 90112
D210	Diode 1N4148	9330 839 90112
U202	Freq. Mixer Dbl.Bal.	2722 162 91963

6.5.2 RECEIVER BOARDS VHF 12.5kHz

6.5.2.1 Receiver PCB E-Band(12.5kHz) 3502 349 72460

	Rx Brd Basic 12.5kHz 93	3502 349 72450	C20P	Cap. Cer.Pl. 33p, 2%,100V NPO	2222 680 10339
	Core Screw Frte M4.5x10mm F29	2413 545 00494	C20R	Cap. Cer.Pl. 6p8,p25,100V NPO	2222 680 09688
R20P	Res. Carb.F. 2k2, 5%, ¼W	2120 101 46222	C20S	Cap. Cer.Pl. 15p 2%,100V NPO	2222 680 10159
C20A	Cap. Cer.Pl. 18p, 2%,100V NPO	2222 680 10189	C20T	Cap. Cer.Pl.100p, 2%,100V NPO	2222 680 10101
C20B	Cap. Cer.Pl. 56p, 2%,100V NPO	2222 680 10569	C213	Cap. Cer.Pl.470p,10%,100V	2222 630 08471
C20C	Cap. Cer.Pl. 6p8,p25,100V NPO	2222 680 09688	U201	Hybrid H9	3502 358 00140
C20D	Cap. Cer.Pl. 6p8,p25,100V NPO	2222 680 09688	U203	VCO Module Asy. "E" (21.4)MHz	3502 357 90180
C20H	Cap. Cer.Pl. 18p, 2%,100V NPO	2222 680 10189	L210	Coil Front-End FM93 E-Bnd	3502 389 90960
C20J	Cap. Cer.Pl.270p,2%,100V N750	2222 680 58271	L211	Coil Front-End FM93 E-Bnd	3502 389 90960
C20K	Cap. Cer.Pl. 15p, 2%,100V NPO	2222 680 10159	L212	Coil Front-End FM93 E-Bnd	3502 389 90960
C20N	Cap. Cer.Pl. 4p7,p25,100V NPO	2222 680 09478	L213	Coil Front-End FM93 E-Bnd	3502 389 90960

6.5.2.2 Receiver PCB B-Band(12.5kHz) 3502 349 72470

	Rx Brd Basic 12.5kHz 93	3502 349 72450	C20L	Cap. Cer.Pl. 56p, 2%,100V NPO	2222 680 10569
	Core Screw Frte M4.5x10mm F29	2413 545 00494	C20S	Cap. Cer.Pl. 8p2,p25,100V NPO	2222 680 09828
R20P	Res. Carb.F. 2k2, 5%, ¼W	2120 101 46222	C20T	Cap. Cer.Pl. 68p, 2%,100V NPO	2222 680 10689
C20A	Cap. Cer.Pl. 5p6,p25,100V NPO	2222 680 09568	C213	Cap. Cer.Pl. 33p, 2%,100V NPO	2222 680 10339
C20B	Cap. Cer.Pl. 27p, 2%,100V NPO	2222 680 10279	U201	Hybrid H10E	3502 358 00670
C20C	Cap. Cer.Pl. 2p7,p25,100V NPO	2222 680 09278	U203	VCO Module Asy. "B" (21.4)MHz	3502 357 90350
C20D	Cap. Cer.Pl. 3p9,p25,100V NPO	2222 680 09398	L210	Coil Front-End FM93 A-B Bnd	3502 389 90950
C20H	Cap. Cer.Pl. 4p7,p25,100V NPO	2222 680 09478	L211	Coil Front-End FM93 A-B Bnd	3502 389 90950
C20J	Cap. Cer.Pl. 68p, 2%,100V NPO	2222 680 10689	L212	Coil Front-End FM93 A-B Bnd	3502 389 90950
C20K	Cap. Cer.Pl. 8p2,p25,100V NPO	2222 680 09828	L213	Coil Front-End FM93 A-B Bnd	3502 389 90950

6.5.2.3 Receiver PCB A-Band(12.5kHz) 3502 349 72480

	Rx Brd Basic 12.5kHz 93	3502 349 72450			
	Core Screw Frte M4.5x10mm F29	2413 545 00494			
R20P	Res. Carb.F. 3k9, 5%, ¼W	2120 101 46392	C20S	Cap. Cer.Pl. 6p8,p25,100V NPO	2222 680 09688
C20A	Cap. Cer.Pl. 4p7,p25,100V NPO	2222 680 09478	C20T	Cap. Cer.Pl. 56p, 2%,100V NPO	2222 680 10569
C20B	Cap. Cer.Pl. 22p, 2%,100V NPO	2222 680 10229	C213	Cap. Cer.Pl. 33p, 2%,100V NPO	2222 680 10339
C20C	Cap. Cer.Pl. 2p2,p25,100V NPO	2222 680 09228	U201	Hybrid H10A	3502 358 00150
C20D	Cap. Cer.Pl. 3p9,p25,100V NPO	2222 680 09398	U203	VCO Module Asy. "A" (21.4)MHz	3502 357 90190
C20H	Cap. Cer.Pl. 3p9,p25,100V NPO	2222 680 09398	L210	Coil Front-End FM93 A-B Bnd	3502 389 90950
C20J	Cap. Cer.Pl. 68p, 2%,100V NPO	2222 680 10689	L211	Coil Front-End FM93 A-B Bnd	3502 389 90950
C20K	Cap. Cer.Pl. 6p8,p25,100V NPO	2222 680 09688	L212	Coil Front-End FM93 A-B Bnd	3502 389 90950
C20L	Cap. Cer.Pl. 56p, 2%,100V NPO	2222 680 10569	L213	Coil Front-End FM93 A-B Bnd	3502 389 90950

6.5.2.4 Receiver Basic 12.5kHz VHF 3502 349 72450

	Receiver Board VHF Basic 93	3502 349 72090			
C233	Cap. Cer.Pl. 8p2,p25,100V NPO	2222 680 09828	R265	Res. Carb.F. 18k, 5%, ¼W	2120 101 46183
R223	Res. Carb.F. 1k8, 5%, ¼W	2120 101 46182	R267	Therm. NTC, 100E,10%, 0.6W	2322 642 62101
R236	Res. Carb.F. 1k5, 5%, ¼W	2120 101 46152	R273	Res. Met. F. 82k, 1%, 0.6W	2322 156 18203
R250	Res. Carb.F. 6k8, 5%, ¼W	2120 101 46682	R276	Res. Carb.F. 18k, 5%, ¼W	2120 101 46183
R252	Res. Carb.F. 680k, 5%, ¼W	2120 101 46684	FL201	Xtal Filter, Mnlthic,21F7.5BA	2722 172 90038
R263	Res. Carb.F. 10E, 5%, ¼W	2120 101 46109	FL202	Filter Cer. 445kHz, CCFW455F	2422 549 03637

6.5.3 **RECEIVER BOARD VHF** (Cont'd)

C212	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C214	Cap.Al.Elec. 4u7,-10+50%,63V	2222 035 58478
C215	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C216	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C217	Cap Al.Elec. 47u,-10+50%,25V	2222 035 56479
C219	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C220	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C221	Cap.Cer.Pl. 18p, 2%,100V NPO	2222 680 10189
C222	Cap.Cer.Pl. 150p, 2%,100V N750	2222 680 58151
C224	Cap Al.Elec. 47u,-10+50%,25V	2222 035 56479
C225	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C226	Cap.Cer.Pl. 3p9,p25,100V NPO	2222 680 09398
C227	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C228	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C229	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C231	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C232	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C234	Cap Al.Elec. 47u,-10+50%,25V	2222 035 56479
C236	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C237	Cap.Cer.Pl. 150p, 2%,100V N750	2222 680 58151
C238	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C240	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C241	Cap.Cer.Pl. 22p, 2%,100V NPO	2222 680 10229
C242	Cap.Pes.Stkd 1n0, 5%,63V	2011 301 20032
C243	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C244	Cap.Cer.Pl. 47p, 2%,100V NPO	2222 680 10479
C245	Cap.Pes.Stkd 1n0, 5%,63V	2011 301 20032
C246	Cap.Pes.Stkd 1n0, 5%,63V	2011 301 20032
C247	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C248	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C249	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C250	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C251	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C252	Cap.Al.Elec. 4u7,-10+50%,63V	2222 035 58478
C253	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C254	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C255	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C256	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C257	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C258	Cap.Cer.Pl. 150p, 2%,100V N750	2222 680 58151
C259	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C260	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C261	Cap.Pes.Stkd 6n8, 5%, 63V	2011 301 20029
C262	Cap.Pes.Stkd 6n8, 5%, 63V	2011 301 20029
C263	Cap.Al.Elec. 47u,-10+50%,25V	2222 035 56479
C264	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C265	Cap.Al.Elec. 10u,-10+50%,50V	2222 035 90008
C266	Cap.Pes.Stkd 6n8, 5%, 63V	2011 301 20029
C267	Cap.Al.Elec. 1u,-10+50%,10V	2222 035 58108
C268	Cap.Cer.Pl. 2n2,10%,100V	2222 630 08222
C269	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C270	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C271	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C272	Cap.Pes.Stkd100n, 5%,63V	2011 301 20031
C273	Cap.Pes.Stkd100n, 5%,63V	2011 301 20031
C274	Cap.Al.Elec. 47u,-10+50%,25V	2222 035 56479
C275	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C276	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C277	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C278	Cap.Pes.Stkd100n, 5%,63V	2011 301 20031
C279	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C280	Cap.Pes.Stkd 1n0, 5%,100V	2011 301 20032
C281	Cap.Al.Elec. 47u,-10+50%,25V	2222 035 56479
C282	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C283	Cap.Al.Elec.470u,-10+50%,16V	2222 031 35471
C284	Cap.Al.Elec.470u,-10+50%,16V	2222 031 35471
C285	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251
C286	Cap.Al.Elec. 4u7,-10+50%,63V	2222 035 58478
C287	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
L202	Choke Wide Band RF	4312 020 36640
L204	Choke 6.8uH,10%, AXL 70-23	2422 535 97559
L205	Coil, Discr. Toko RMC-2A6597HM	2422 549 18111
RL201	Relay, Min 10Amp 12V	2402 132 00015
XL201	Crystal 20.945MHz 2nd L.Osc	3502 413 48410
IC201	Intgrd.Cct. MC3361P	9337 941 60682
IC202	Intgrd.Cct. LM348M	9333 705 60682
IC203	Intgrd.Cct. TDA1010	9333 849 50112
TR201	Transistor BF324	9331 677 30112
TR202	Transistor BC548C	9331 976 70112
TR204	Transistor BC337	9331 492 00112
TR205	Transistor BC548C	9331 976 70112
TR206	Transistor BC328	9331 491 90112
TR207	Transistor 2SC3355	9337 925 10682
TR208	Transistor BC548C	9331 976 70112
TR209	Transistor BC548C	9331 976 70112
TR210	Transistor BD946	9334 481 30112
TR211	Transistor BC548C	9331 976 70112
TR212	Transistor BC548C	9331 976 70112
TR213	Transistor BC328	9331 491 90112
TR214	Transistor BC548C	9331 976 70112
TR218	Transistor BC548C	9331 976 70112
TR219	Transistor BF960	9335 105 20112
TR220	Transistor BC548C	9331 976 70112
TR221	Transistor BC548C	9331 976 70112
TR222	Transistor BC548C	9331 976 70112
TR223	Transistor BC548C	9331 976 70112
TR224	Transistor BC548C	9331 976 70112
D201	Diode 1N4148	9330 839 90112
D202	Diode 1N4148	9330 839 90112
D203	Diode BZT03-C20	9336 015 80112
D204	Diode 1N4148	9330 839 90112
D205	Diode BZX79-C6V2	9331 177 40112
D206	Diode 1N4148	9330 839 90112
D208	Diode 1N4148	9330 839 90112
D209	Diode 1N4148	9330 839 90112
D210	Diode 1N4148	9330 839 90112
U202	Freq. Mixer Dbl.Bal. CBL-1X	2722 162 90163

6.6.4 TX PA PC BOARD UHF BASIC (3502 349 81000)

	PCB Tx PA 25W Basic	3502 309 81000
	Shield Aerial Filter	3502 319 91100
	Partition Aerial Filter	3502 319 91110
	Bead Ferrite	4332 020 34400
R101	Res Carb F 2k7 5% ¼W	2120 101 46272
R102	Pot Cermet 10k 10% 1W	2113 391 00507
R103	Res Carb F 1kO 5% ¼W	2120 101 46102
R104	Res Carb F 470k 5% ¼W	2120 101 46474
R105	Res Carb F 100E 5% ¼W	2120 101 46101
R106	Thermistor PTC 90C with Leads	2322 672 91004
R108	Res Carb F 10E 5% ¼W	2120 101 46109
R109	Res Carb F 10k 5% ¼W	2120 101 46103
R110	Res Carb F 10k 5% ¼W	2120 101 46103
R111	Res Carb F 33k 5% ¼W	2120 101 46333
R112	Res Carb F 4E7 5% ½W	2322 186 13478
R115	Res Carb F 10k 5% ¼W	2120 101 46103
R117	Res Carb F 68k 5% ¼W	2120 101 46683
R118	Res Carb F 10E 5% ¼W	2120 101 46109
R119	Res Carb F 3k3 5% ¼W	2120 101 46332
C101	Cap Cer D/M 100n 20% 50V X7R	2022 552 01751
C102	Cap Cer Pl 470p 10% 100V	2222 630 08471
C103	Cap Cer Pl 4n7 10% 100V	2222 630 08472
C104	Cap Elec S/Al 3u3 -20+40% 16V	2222 122 55338
C105	Cap Cer D/M 100n 20% 50V X7R	2022 552 01751
C106	Cap Elec S/AL 3u3 -20+40% 16V	2222 122 55338
C107	Cap Cer D/M 100n 20% 50V X7R	2022 552 01751
C108	Cap Cer Pl 220p 10% 100V	2222 630 08221
C109	Cap Elec S/AL 3u3 -20+40% 16V	2222 122 55338
C110	Cap Cer Pl 47p 2% 100V NPO	2222 632 10479
C111	Cap Trim Film H/T 1p8-10p 300V	2222 809 05002
C112	Cap Cer P1 220p 10% 100V	2222 630 08221
C113	Cap Cer Pl 2p2 p25 100V NPO	2222 632 09228
C116	Cap Elec S/AL 4u7 -20+40% 25V	2222 122 56478
C117	Cap Cer Pl 120p 2% 500V N750	2222 650 58121
C120	Cer D/M 100n 20% 50V X7R	2022 552 01751
C121	Cap Cer Pl 2n2 10% 100V	2222 630 08222
C124	Cap Trim Film 3H/T 2p-9p 300V	2222 809 09002
C126	Cap Cer Pl 120p 2% 500V N750	2222 650 58121
C128	Cap Cer Pl 120p 2% 500V N750	2222 650 58121
C129	Cap Cer Pl 1n0 10% 100V	2222 630 08102
C130	Cap Cer Pl 4p7 p25 500V NPO	2222 650 09478
C131	Cap Cer Pl 10p 2% 500V NPO	2222 650 10109
C132	Cap Cer Pl 1p8 p25 500V NPO	2222 650 09188
C133	Cap Cer Pl 10p 2% 500V NPO	2222 650 10109
C134	Cap Mica 5p6 5% 250V	2022 900 00094
C135	Cap Cer Pl 2p2 p25 100V NPO	2222 632 09228
IC101	Intgrd Circt LM358N	9333 935 10112
TR101	Transistor BD945	9334 481 20112
TR102	Transistor MRF646	9333 928 70112
D101	Diode BZV46-2V0	9334 339 40112
D102	Diode 1N4148	9330 839 90112
D106	Diode 1N4148	9330 839 90112
L101	Choke RF 2.5turn 4B	4312 020 36700
L103	Coil A/W Tx PA UHF	3502 382 90000
L104	Choke RF 2.5turn 4B	4312 020 36700
L110	Coil A/W Tx PA UHF	3502 382 90040
L111	Coil A/W Tx PA UHF	3502 382 90050
L112	Coil A/W Tx PA UHF	3502 382 90060

6.7.2 TX PA PC BOARD (B-BAND) (3502 349 71130)

	PCB Tx PA (Basic)	3502 349 71100	C131	Cap Mica 390p 10% 350V	2022 751 00037
R116	Res Carb F 100k 5% ¼W	2120 101 46104	C134	Cap Mica 150p 10% 350V	2022 751 00038
R119	Res Carb F 470E 5% ½W	2322 186 13471	C135	Cap Mica 47p 5% 250V	2022 751 00017
C102	Cap Cer Pl 47p 2% 100V NPO	2222 680 10479	C137	Cap Cer Pl 33p 2% 500V NPO	2222 650 10339
C107	Cap Cer Pl 33p 2% 100V NPO	2222 680 10339	C143	Cap Cer Pl 22p 2% 100V NPO	2222 680 10229
C108	Cap Cer Pl 33p 2% 100V NPO	2222 680 10339	C144	Cap Cer Pl 22p 2% 500V NPO	2222 650 10229
C112	Cap Cer Pl 47p 2% 100V NPO	2222 680 10479	C147	Cap Cer Pl 18p 2% 500V NPO	2222 650 10189
C113	Cap Cer Pl 100p 2% 100V NPO	2222 680 10101	C152	Cap Cer Pl 18p 2% 500V NPO	2222 650 10189
C118	Cap Cer Pl 33p 2% 100V NPO	2222 680 10339	L104	Coil A/W Tx PA 'A'	3502 382 90280
C119	Cap Cer Pl 220p 10% 100V	2222 630 08221	L114	Coil A/W 900	3502 382 90590
C120	Cap Cer Pl 33p 2% 100V NPO	2222 680 10339	L120	Coil A/W 900	3502 382 90580
C122	Cap Mica 130p 5% 350V	2022 751 00025	L121	Coil A/W 900	3502 382 90600
C124	Cap Cer Pl 100p 2% 100V NPO	2222 680 10101	L122	Coil A/W 900	3502 382 90580
C128	Cap Mica 390p 10% 350V	2022 751 00037			

6.7.3 TX PA PC BOARD (A-BAND) (3502 349 71140)

	PCB Tx PA (Basic)	3502 349 71100	C131	Cap Sl/Mic 350p 10% 350V	2022 751 00036
R116	Res Carb F 39k 5% ¼W	2120 101 46393	C134	Cap Mica 130p 5% 350V	2022 751 00025
R119	Res Carb F 220E 5% ½W	2322 186 13221	C135	Cap Mica 22pF 5% 250V	2022 751 00018
C102	Cap Cer Pl 39p 2% 100V NPO	2222 680 10399	C137	Cap Cer Pl 27p 2% 500V NPO	2222 650 10279
C107	Cap Cer Pl 27p 2% 100V NPO	2222 680 10279	C143	Cap Cer Pl 18p 2% 100V NPO	2222 680 10189
C108	Cap Cer Pl 39p 2% 100V NPO	2222 680 10399	C144	Cap Cer Pl 18p 2% 500V NPO	2222 650 10189
C112	Cap Cer Pl 27p 2% 100V NPO	2222 680 10279	C147	Cap Cer Pl 15p 2% 500V NPO	2222 650 10159
C113	Cap Cer Pl 120p 2% 100V NPO	2222 680 10121	C152	Cap Cer Pl 15p 2% 500V NPO	2222 650 10159
C118	Cap Cer Pl 22p 2% 100V NPO	2222 680 10229	L104	Coil A/W Tx PA 'A'	3502 382 90520
C119	Cap Cer Pl 180p 10% 100V	2222 630 08181	L114	Coil A/W Tx PA 'A'	3502 382 90280
C120	Cap Cer Pl 27p 2% 100V NPO	2222 680 10279	L120	Coil A/W Tx PA 'A'	3502 382 90330
C122	Cap Sl/Mic 100p 5% 350V	2022 751 00033	L121	Coil A/W Tx PA 'A'	3502 382 90340
C124	Cap Sl/Mic 80p 10% 350V	2022 751 00035	L122	Coil A/W Tx PA 'A'	3502 382 90330
C128	Cap Sl/Mic 350p 10% 350V	2022 751 00036			

6.7.4 TX PA PC BOARD (BASIC) (3502 349 71100)

	PCB Tx PA (A Band)	3502 309 71100	R127	Res Carb F 100E 5% ¼W	2120 101 46101
	Heatsink PA Diode (x2)	3502 319 91070	R128	Res Carb F 10k 5% ¼W	2120 101 46103
	Shield Aerial Filter	3502 319 91100	R129	Res Carb F 68k 5% ¼W	2120 101 46683
	Partition Aerial Filter	3502 319 91110	R130	Res Carb F 10E 5% ½W	2322 101 16109
	Spacer PCB Support	3502 319 91860	R131	Res Carb F 68k 5% ¼W	2120 101 46683
R102	Res Carb F 10E 5% ¼W	2120 101 46109	C101	Cap Cer Pl 68p 2% 100V NPO	2222 680 10689
R103	Res Carb F 1k5 5% ¼W	2120 101 46152	C103	Cap Cer Pl 4n7 10% 100V	2222 630 08472
R104	Res Carb F 470E 5% ¼W	2120 101 46471	C104	Cap Cer D/M 100n 20% 50V X7R	2022 552 01751
R105	Res Carb F 330k 5% ¼W	2120 101 46334	C105	Cap Cer Pl 4n7 10% 100V	2222 630 08472
R106	Res Carb F 10E 5% ½W	2322 212 16109	C106	Cap Cer Pl 1n0 10% 100V	2222 630 08102
R107	Res Carb F 180E 5% ¼W	2120 101 46181	C109	Cap Cer Pl 1n0 10% 100V	2222 630 08102
R108	Res Carb F 2E2 5% ¼W	2120 101 46228	C111	Cap Cer Pl 1n0 10% 100V	2222 630 08102
R109	Res Carb F 3k3 5% ¼W	2120 101 46332	C114	Cap Elec S/AL 3u3 -20+40% 63V	2222 122 56338
R110	Res Carb F 470E 5% ¼W	2120 101 46471	C115	Cap Cer D/M 100n 20% 50V X7R	2022 552 01751
R111	Thermistor PTC 90C with leads	2322 672 91004	C116	Cap Cer D/M 100n 20% 50V X7R	2022 552 01751
R112	Pot Cermet 2k 10% 1W	2113 391 00505	C117	Cap Cer Pl 180p 10% 100V	2222 630 08181
R113	Res Carb F 3k3 5% ¼W	2120 101 46332	C121	Cap Cer Pl 39p 2% 100V NPO	2222 680 10399
R114	Res Carb F 22E 5% ¼W	2120 101 46229	C123	Cap Cer Pl 1n0 10% 100V	2222 630 08102
R117	Res Carb F 10E 5% ¼W	2120 101 46109	C125	Cap Cer Pl 6p8 p25, 100V NPO	2022 680 09688
R118	Res Carb F 10E 5% ½W	2322 101 16109	C129	Cap Cer Pl 220p 10% 100V	2222 630 08221
R123	Res Carb F 10k 5% ¼W	2120 101 46103	C130	Cap Cer Pl 180p 2% 100V N750	2222 680 58181
R124	Res Carb F 10E 5% ½W	2120 101 16109	C132	Cap Elec S/AL 4u7 -20+40% 25V	2222 122 56478
R125	Res Carb F 10E 5% ½W	2322 101 16109	C133	Cap Cer D/M 100n 20% 50V X7R	2022 552 01751
R126	Res Carb F 47E 5% ¼W	2120 101 46479	C136	Cap Trim Flm 3H/T 2P-18P 300V	2222 809 09003

6.8 SYNTHESISER/CONTROL PC BOARDS

6.8.1 SYNTHESISER/CONTROL BOARDS UHF

6.8.1.1 Synth/Cont PCB 25kHz UHF 3502 349 83120

	Synth/Cont PC Brd (Basic)93	3502 349 73040					
R335	Res. Carb.F. 1kO, 5%, ¼W	2120 101 46102		C325	Cap.Cer.Pl. 27p, 2%,100V N750	2222 680 58279	
R337	Res. Carb.F. 27k, 5%, ¼W	2120 101 46273		C327	Cap.Pes.Stkd 47n, 5%,63V	2011 301 20501	
R340	Res. Met. F. 220k, 1%, 0.6W	2322 156 12204		C330	Cap.Cer.Pl. 39p, 2%,100V N750	2222 680 58399	
R343	Res. Met. F. 220k, 1%, 0.6W	2322 156 12204		C344	Cap.Cer.Pl. 68p, 2%,100V NPO	2222 680 10689	
R344	Res. Met. F. 120k, 1%, 0.6W	2322 156 11204		C347	Cap.Pes.Stkd 15n, 5%,63V	2011 301 20503	
R350	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153		C349	Cap.Pes.Stkd 22n, 5%,63V	2011 301 20502	
R351	Res. Carb.F. 1MO, 5%, ¼W	2120 101 46105		C424	Cap.Cer.Pl. 18p, 2%,100V NPO	2222 680 10189	
R366	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153		C425	Cap.Cer.Pl. 18p, 2%,100V NPO	2222 680 10189	
R372	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153		XL301	Crystal 10000kHz, Holder	3502 414 98930	
R430	Res. Carb.F. 10M,10%, 0.33W	2322 211 12106		XL401	Crystal 5MHz Daiwa KDS	2422 543 00051	

6.8.1.2 Synth/Cont PCB 12.5kHz UHF 3502 349 83130

	Synth/Cont PC Brd (Basic)93	3502 349 73040					
R335	Res. Carb.F. 3k9, 5%, ¼W	2120 101 46392		C325	Cap.Cer.Pl. 27p, 2%,100V N1500	2222 680 70279	
R337	Res. Carb.F. 27k, 5%, ¼W	2120 101 46273		C327	Cap.Pes.Stkd 15n, 5%,63V	2011 301 20503	
R340	Res. Met. F. 270k, 1%, 0.6W	2322 156 12704		C330	Cap.Cer.Pl. 39p, 2%,100V N1500	2222 680 70399	
R343	Res. Met. F. 270k, 1%, 0.6W	2322 156 12704		C344	Cap.Cer.Pl. 82p, 2%,100V NPO	2222 680 10829	
R344	Res. Met. F. 150k, 1%, 0.6W	2322 156 11504		C347	Cap.Pes.Stkd 15n, 5%,63V	2011 301 20503	
R350	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153		C349	Cap.Pes.Stkd 22n, 5%,63V	2011 301 20502	
R351	Res. Carb.F. 1MO, 5%, ¼W	2120 101 46105		C424	Cap.Cer.Pl. 18p, 2%,100V NPO	2222 680 10189	
R366	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153		C425	Cap.Cer.Pl. 18p, 2%,100V NPO	2222 680 10189	
R372	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153		XL301	Crystal 10000kHz, Holder	3502 414 98930	
R430	Res. Carb.F. 10M,10%, .33W	2322 211 12106		XL401	Crystal 5MHz Daiwa KDS	2422 543 00051	

6.8.2 SYNTHESISER/CONTROL BOARDS VHF

6.8.2.1 Synth/Cont 25kHz E Band 3502 349 73140

	Synth/Cont PC Brd (Basic)93	3502 349 73070		R366	Res. Carb.F. 12k, 5%, ¼W	2120 101 46123	
R335	Res. Carb.F. 1kO, 5%, ¼W	2120 101 46102		R372	Res. Carb.F. 2k7, 5%, ¼W	2120 101 46272	
R337	Res. Carb.F. 33k, 5%, ¼W	2120 101 46333		C327	Cap.Pes.Stkd 47n, 5%,63V	2011 301 20501	
R340	Res. Met. F. 220k, 1%, 0.6W	2322 156 12204		C339	Cap.Cer.Pl. 47p, 2%,100V NPO	2222 680 10479	
R343	Res. Met. F. 220k, 1%, 0.6W	2322 156 12204		C344	Cap.Cer.Pl. 68p, 2%,100V NPO	2222 680 10689	
R344	Res. Met. F. 120k, 1%, 0.6W	2322 156 11204		C347	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251	
R350	Res. Carb.F. 120E, 5%, ¼W	2120 101 46121		C349	Cap.Con/Ml Cer. 220n, 10%,50V	2022 552 02379	
R351	Res. Carb.F. 820k, 5%, ¼W	2120 101 46824		XL301	Crystal 10000kHz, Holder	3502 414 98990	

6.8.2.2 Synth/Cont 12.5kHz E Band 3502 349 73160

	Synth/Cont PC Brd (Basic)93	3502 349 73070		R366	Res. Carb.F. 12k, 5%, ¼W	2120 101 46123	
R335	Res. Carb.F. 3k9, 5%, ¼W	2120 101 46392		R372	Res. Carb.F. 2k7, 5%, ¼W	2120 101 46272	
R337	Res. Carb.F. 33k, 5%, ¼W	2120 101 46333		C327	Cap.Pes.Stkd 15n, 5%,63V	2011 301 20503	
R340	Res. Met. F. 270k, 1%, 0.6W	2322 156 12704		C339	Cap.Cer.Pl. 47p, 2%,100V NPO	2222 680 10479	
R343	Res. Met. F. 270k, 1%, 0.6W	2322 156 12704		C344	Cap.Cer.Pl. 82p, 2%,100V NPO	2222 680 10829	
R344	Res. Met. F. 150k, 1%, 0.6W	2322 156 11504		C347	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251	
R350	Res. Carb.F. 120E, 5%, ¼W	2120 101 46121		C349	Cap.Con/Ml Cer. 220n, 10%,50V	2022 552 02379	
R351	Res. Carb.F. 820k, 5%, ¼W	2120 101 46824		XL301	Crystal 10000kHz, Holder	3502 414 98930	

6.8.2.3 Synth/Cont 25kHz A Band 3502 349 73440

	Synth/Cont PC Brd (Basic)93	3502 349 73070		R366	Res. Carb.F. 12k, 5%, ¼W	2120 101 46123	
R335	Res. Carb.F. 1kO, 5%, ¼W	2120 101 46102		R372	Res. Carb.F. 2k7, 5%, ¼W	2120 101 46272	
R337	Res. Carb.F. 27k, 5%, ¼W	2120 101 46273		C327	Cap.Pes.Stkd 47n, 5%,63V	2011 301 20501	
R340	Res. Met. F. 220k, 1%, 0.6W	2322 156 12204		C339	Cap.Cer.Pl. 47p, 2%,100V NPO	2222 680 10479	
R343	Res. Met. F. 220k, 1%, 0.6W	2322 156 12204		C344	Cap.Cer.Pl. 68p, 2%,100V NPO	2222 680 10689	
R344	Res. Met. F. 120k, 1%, 0.6W	2322 156 11204		C347	Cap.Pes.Stkd 10n, 5%,100V	2011 301 40251	
R350	Res. Carb.F. 2k7, 5%, ¼W	2120 101 46272		C349	Cap.Pes.Stkd 22n, 5%,63V	2011 301 20502	
R351	Res. Carb.F. 1MO, 5%, ¼W	2120 101 46105		XL301	Crystal 10000kHz, Holder	3502 414 98990	

6.8.3 SYNTH/CONT PCB BASIC FM93 (Cont'd)

R401	Res. Carb.F. 2k2, 5%, ¼W	2120 101 46222
R402	Res. Carb.F. 1E0, 5%, ¼W	2120 101 46108
R403	Res. Carb.F.470E, 5%, ¼W	2120 101 46471
R404	Res. Carb.F.560E, 5%, ¼W	2120 101 46561
R405	Res. Carb.F. 10E, 5%, ¼W	2120 101 46109
R406	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153
R407	Res. Carb.F. 47k, 5%, ¼W	2120 101 46473
R408	Res. Carb.F. 10E, 5%, ¼W	2120 101 46109
R409	Res. Carb.F.560k, 5%, ¼W	2120 101 46564
R410	Res. Carb.F.100k, 5%, ¼W	2120 101 46104
R411	Res. Carb.F. 22k, 5%, ¼W	2120 101 46223
R412	Res. Carb.F. 47k, 5%, ¼W	2120 101 46473
R413	Res. Carb.F.220k, 5%, ¼W	2120 101 46224
R414	Res. Carb.F. 4k7, 5%, ¼W	2120 101 46472
R415	Res. Carb.F.100k, 5%, ¼W	2120 101 46104
R416	Res. Carb.F. 10k, 5%, ¼W	2120 101 46103
R421	Res. Carb.F.100k, 5%, ¼W	2120 101 46104
R422	Res. Carb.F.100k, 5%, ¼W	2120 101 46104
R423	Res. Carb.F. 47k, 5%, ¼W	2120 101 46473
R424	Res. Carb.F. 10k, 5%, ¼W	2120 101 46103
R425	Res. Carb.F. 4k7, 5%, ¼W	2120 101 46472
R426	Res. Carb.F. 4k7, 5%, ¼W	2120 101 46472
R427	Res. Carb.F. 47k, 5%, ¼W	2120 101 46473
R428	Res. Carb.F. 47k, 5%, ¼W	2120 101 46473
R429	Res. Carb.F. 10k, 5%, ¼W	2120 101 46103
R432	Res. Carb.F. 10E, 5%, ¼W	2120 101 46109
R434	Res. Carb.F. 10k, 5%, ¼W	2120 101 46103
R435	Res. Carb.F. 4k7, 5%, ¼W	2120 101 46472
R436	Res. Carb.F. 2k2, 5%, ¼W	2120 101 46222
R444	Res. Carb.F. 47k, 5%, ¼W	2120 101 46473
R447	Res. Carb.F. 4k7, 5%, ¼W	2120 101 46472
R448	Res. Carb.F. 4k7, 5%, ¼W	2120 101 46472
R449	Res. Carb.F. 4k7, 5%, ¼W	2120 101 46472
R450	Res. Carb.F. 4k7, 5%, ¼W	2120 101 46472
R451	Res. Carb.F. 1k0, 5%, ¼W	2120 101 46102
R452	Res. Carb.F. 10k, 5%, ¼W	2120 101 46103
C301	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C302	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C303	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C304	Cap.Pes.Stkd 22n, 5%,63V	2011 301 20502
C305	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C306	Cap.Pes.Stkd 22n, 5%,63V	2011 301 20502
C307	Cap.Pes.Stkd 22n, 5%,63V	2011 301 20502
C308	Cap.Cer.Pl. 330p,10%,100V	2222 630 08331
C309	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C310	Cap Al.Elec. 1u0,-10+50%,63V	2222 035 58108
C311	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C312	Cap.Pes.Stkd 1n0, 5%,63V	2011 301 20032
C313	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08742
C314	Cap.Cer.Pl. 47p, 2%,100V NPO	2222 680 10479
C315	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C316	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C317	Cap.Cer.Pl. 33p,10%,100V NPO	2222 680 10339
C318	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C319	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C320	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C321	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C322	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C323	Cap.Cer.Pl. 47p, 2%,100V NPO	2222 680 10479
C324	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C325	Cap.Cer.Pl. 27p, 2%,100V N750	2222 680 58279
C326	Cap.Cer.Pl. 18p, 2%,100V NPO	2222 680 10189
C328	Cap.Cer.Pl. 1n0,10%,100V NPO	2222 630 08102
C329	Cap.Pes.Stkd 47n, 5%,63V	2011 301 20501
C330	Cap.Cer.Pl. 39p, 2%,100V N750	2222 680 58399
C331	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C332	Cap.Cer.Pl. 6p8,p25,100V N750	2222 680 57688
C333	Cap.Cer.Pl. 270p, 2%,100V N750	2222 680 58271
C335	Cap Al.Elec. 4u7,-10+50%,63V	2222 035 58478
C336	Cap Al.Elec. 1u0,-10+50%,63V	2222 035 58108
C337	Cap.Cer.DP-Ml.1n0, 5%,50V,N750	2011 557 25581
C338	Cap.Pes.Stkd 1n5, 5%,63V	2011 301 20028
C342	Cap.Cer.Pl. 270p, 2%,100V N750	2222 680 58271
C343	Cap.Pes.Stkd 47n, 5%,63V	2011 301 20501
C345	Cap.Cer.Con/Ml. 220n,10%,50V	2022 552 02379
C346	Cap.Cer.Pl. 470p,10%,100V	2222 630 08471
C348	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C352	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C353	Cap.Pcarb. 150n,10%,100V	2222 344 21154
C354	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C355	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C356	Cap.Cer.Con/Ml. 220n,10%,50V	2022 552 02379
C357	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C358	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C359	Cap.Cer.Con/Ml. 220n,10%,50V	2022 552 02379
C360	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C361	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C362	Cap. Tant. 22u,20%,16V	2020 004 90041
C363	Cap Al.Elec. 10u,-10+50%,50V	2222 035 90008
C364	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C365	Cap. Tant. 15u,20%,25V	2020 004 90062
C366	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C393	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C403	Cap Al.Elec. 22u,-10+50%,50V	2222 035 90003
C404	Cap. Tant. 2u2,20%,35V	2020 004 90027
C405	Cap.Cer.Pl. 4n7,10%,100V	2222 630 08472
C406	Cap. Tant. 22u,20%,25V	2020 004 90041
C407	Cap.Cer.Con/Ml. 220n,10%,50V	2022 552 02379
C408	Cap Al.Elec. 47u,-10+50%,25V	2222 035 56479
C409	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C415	Cap. Tant. 4u7,20%,35V	2020 004 90001
C416	Cap.Cer.D/M 100n,20%,50V,X7R	2022 552 01751
C417	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C418	Cap Al.Elec.100u,-10+50%,25V	2222 035 56101
C419	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C420	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C421	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
C422	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C423	Cap.Cer.Pl. 47p, 2%,100V NPO	2222 680 10479
C426	Cap.Cer.Pl. 39p, 2%,500V NPO	2222 650 10399
C427	Cap.Elect 47000u,+80-20%, NEC	2020 900 00002
C428	Cap Al.Elec.100u,-10+50%,25V	2222 035 56101
C430	Cap.Cer.D/M 100n,20%,50V,Z5U	2022 552 02482
C432	Cap.Cer.Pl. 220p,10%,100V	2222 630 08221

SECTION 7 CIRCUIT DIAGRAMS

Figure 7.1 FM93 INTERCONNECTION

Figure 7.2 UHF Receiver Board

Figure 7.3 VHF Receiver Board

Figure 7.4 UHF Transmitter P.A. Board

Figure 7.5 VHF Transmitter P.A. Board

Figure 7.6A Synthesiser/Control Board (Synthesiser)
Figure 7.6A Synthesiser/Control Board (Control)

Figure 7.7 Local Control Board

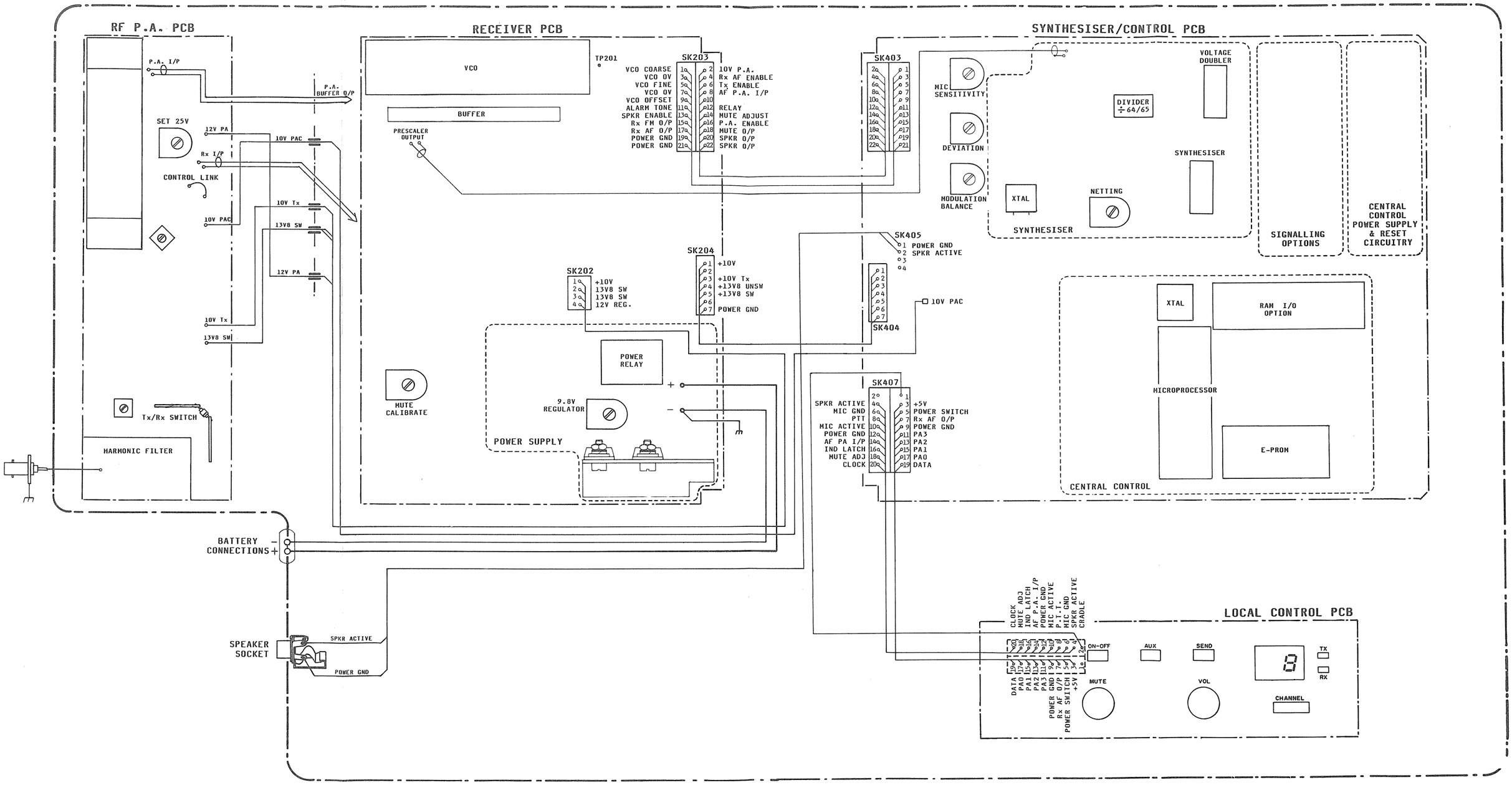
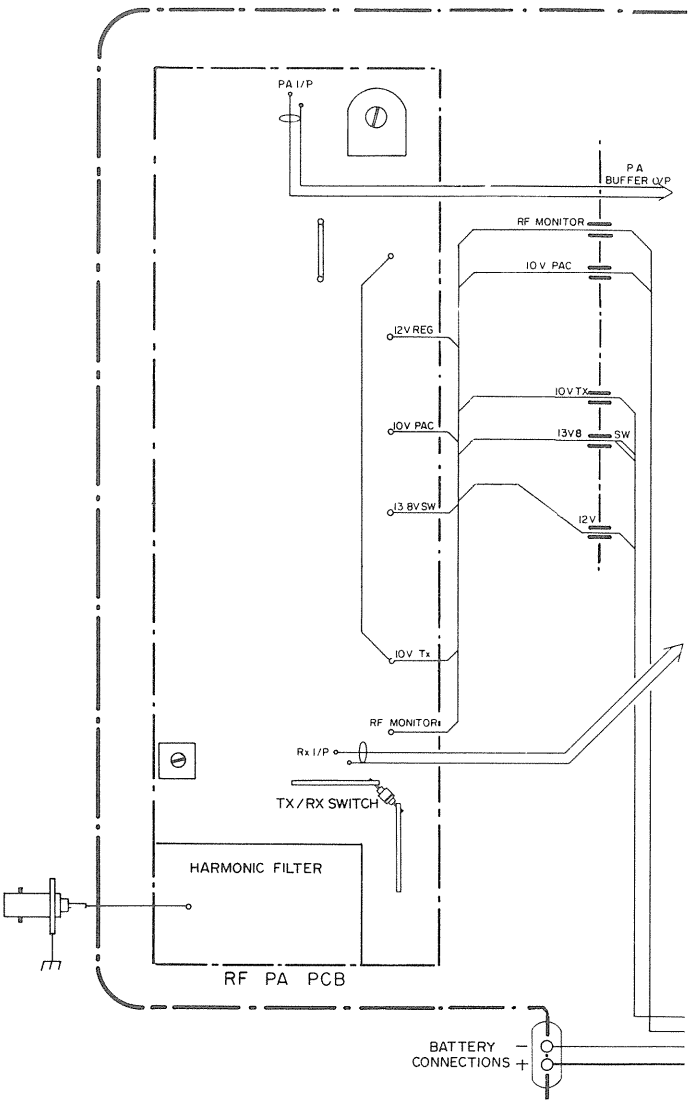
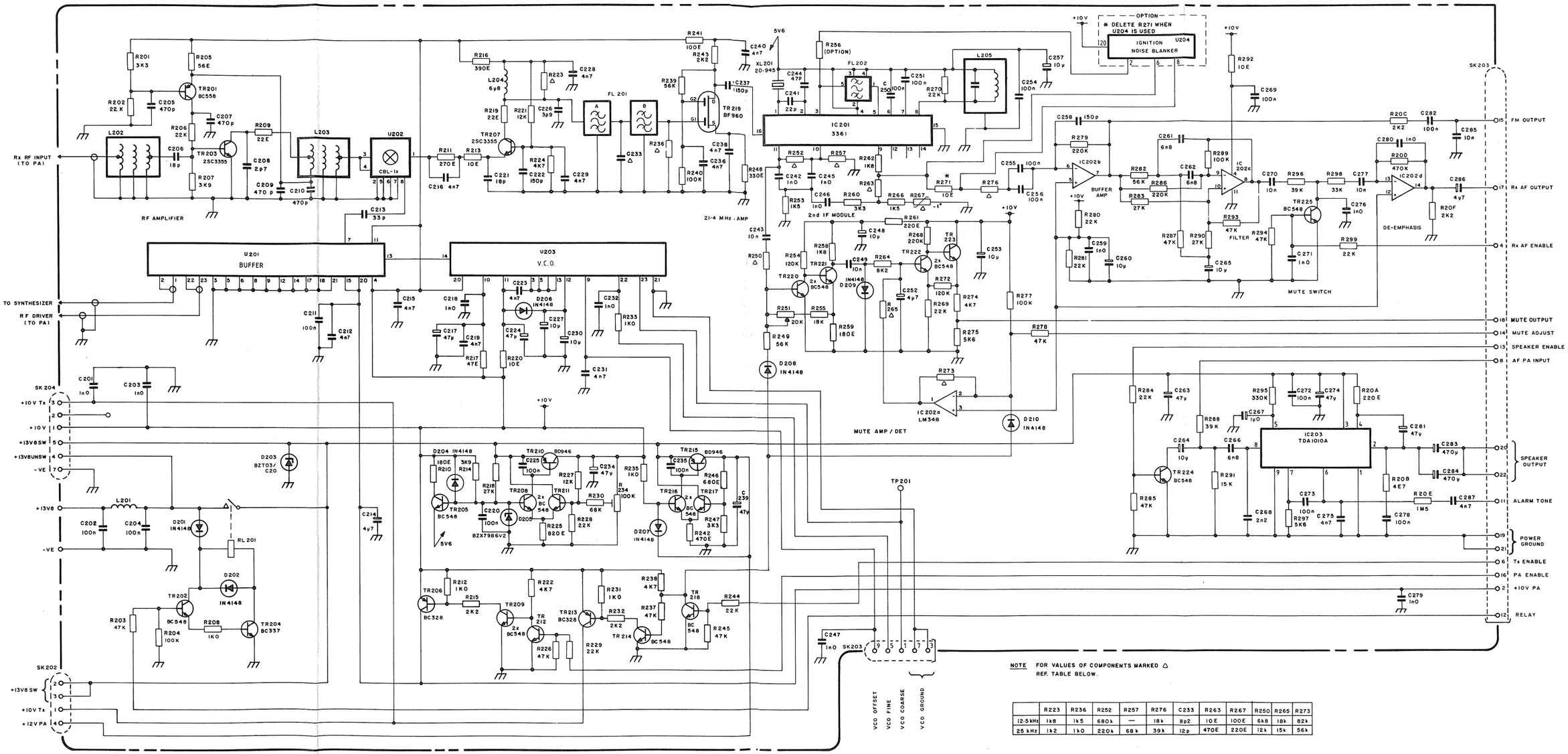


FIGURE 7.1 FM93 INTERCONNECTION DIAGRAM
UHF TRANSCEIVER

(WITH VHF TX PA VARIATION)



VHF VARIATION

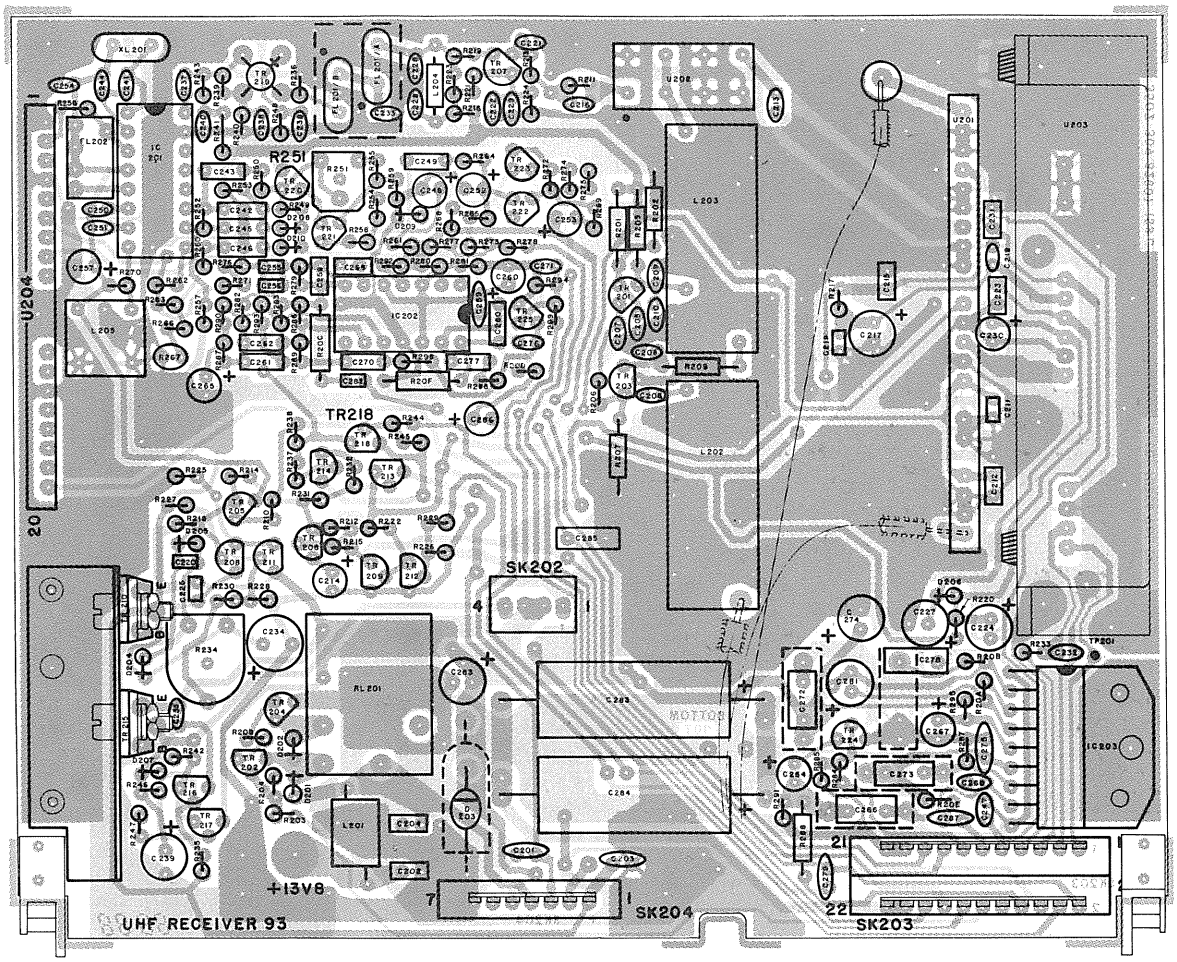


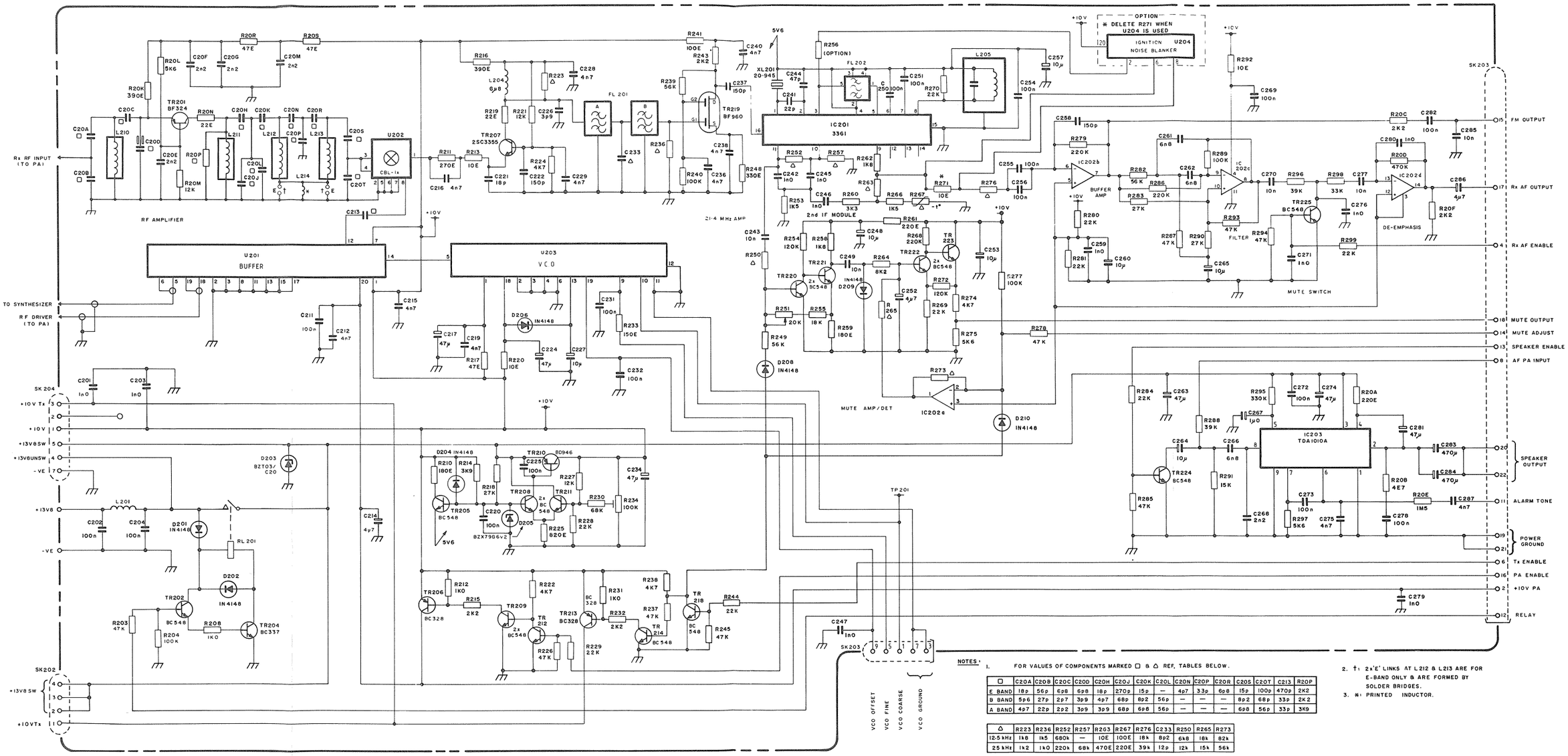
NOTE FOR VALUES OF COMPONENTS MARKED △ REF. TABLE BELOW.

	R223	R236	R252	R257	R276	C233	R263	R267	R250	R265	R273
12·5 kHz	1k8	1k5	680k	—	18k	8p2	10E	100E	6k8	18k	82k
25 kHz	1k2	1k0	220k	68k	39k	12p	47OE	220E	12k	15k	56k

FIGURE 7.2 UHF RECEIVER P.C. BOARD

CIRCUIT AND COMPONENT LAYOUT


UHF RECEIVER 93
+13V8
SK202
SK204
SK203
TR218
R251
L203
L202
RL201
IC202
IC203
U204
U203
U201
U208
20
21
22
4
7
1



NOTES:

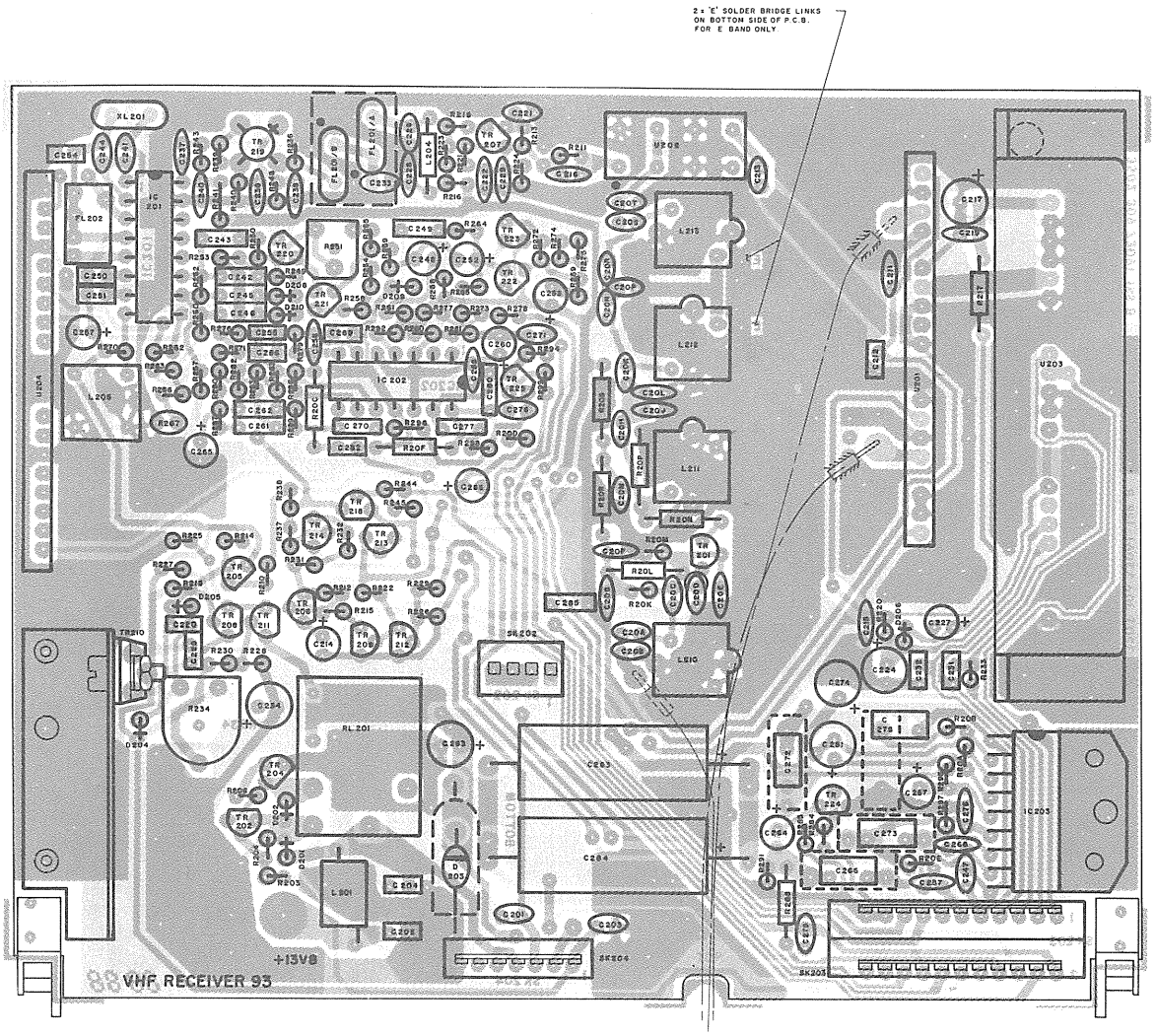
1. FOR VALUES OF COMPONENTS MARKED □ & △ REF, TABLES BELOW.

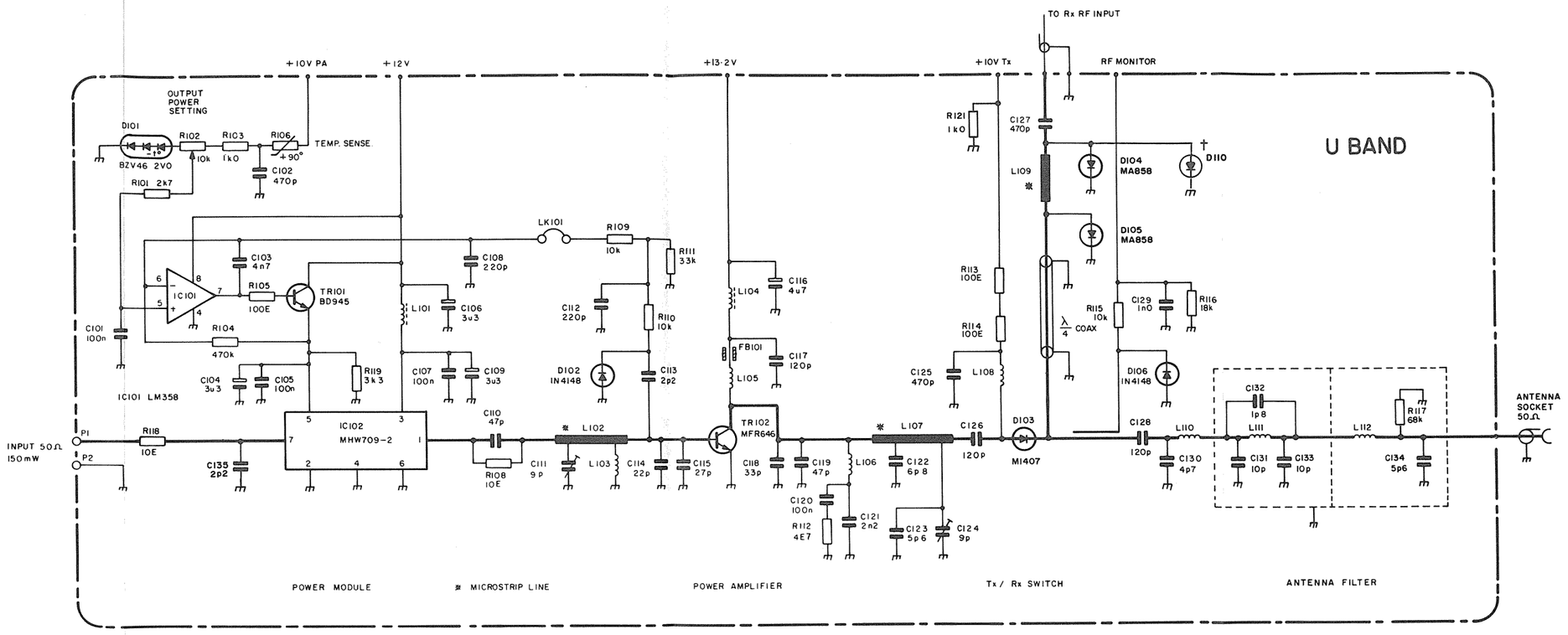
□	C20A	C20B	C20C	C20D	C20H	C20J	C20K	C20L	C20N	C20P	C20R	C20S	C20T	C213	R20P
E BAND	18p	56p	6p8	6p8	18p	270p	15p	—	4p7	33p	6p8	15p	100p	470p	2K2
B BAND	5p6	27p	2p7	3p9	4p7	68p	8p2	56p	—	—	—	8p2	68p	33p	2K2
A BAND	4p7	22p	2p2	3p9	3p9	68p	6p8	56p	—	—	—	6p8	56p	33p	3K9

△	R223	R236	R252	R257	R263	R267	R276	C233	R250	R265	R273
12.5 kHz	1k8	1k5	680k	—	10E	100E	18k	8p2	6k8	18k	82k
25 kHz	1k2	1k0	220k	68k	470E	220E	39k	12p	12k	15k	56k

2. †: 2x'E' LINKS AT L212 & L213 ARE FOR E-BAND ONLY & ARE FORMED BY SOLDER BRIDGES.
3. *: PRINTED INDUCTOR.

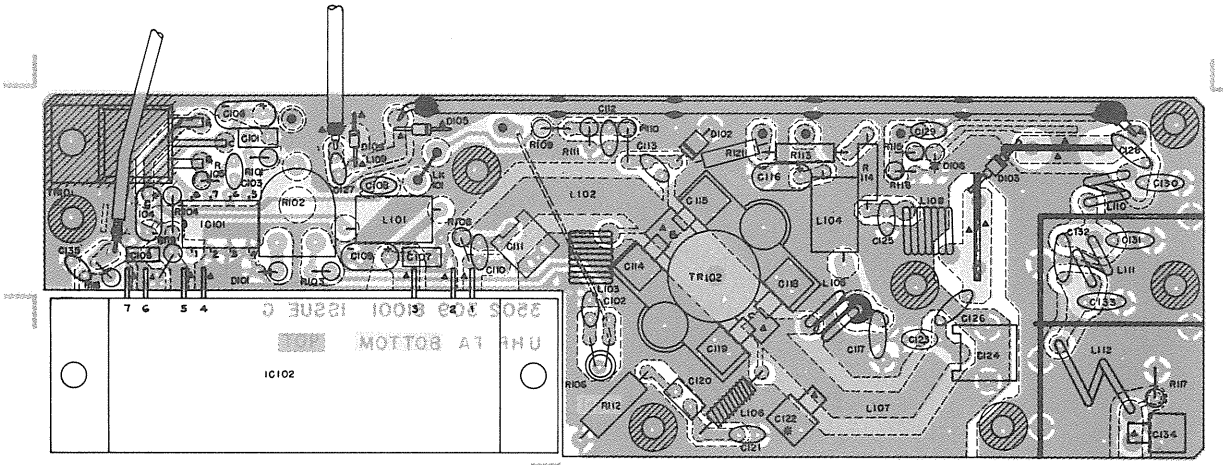
FIGURE 7.3 VHF RECEIVER P.C. BOARD
CIRCUIT AND COMPONENT LAYOUT

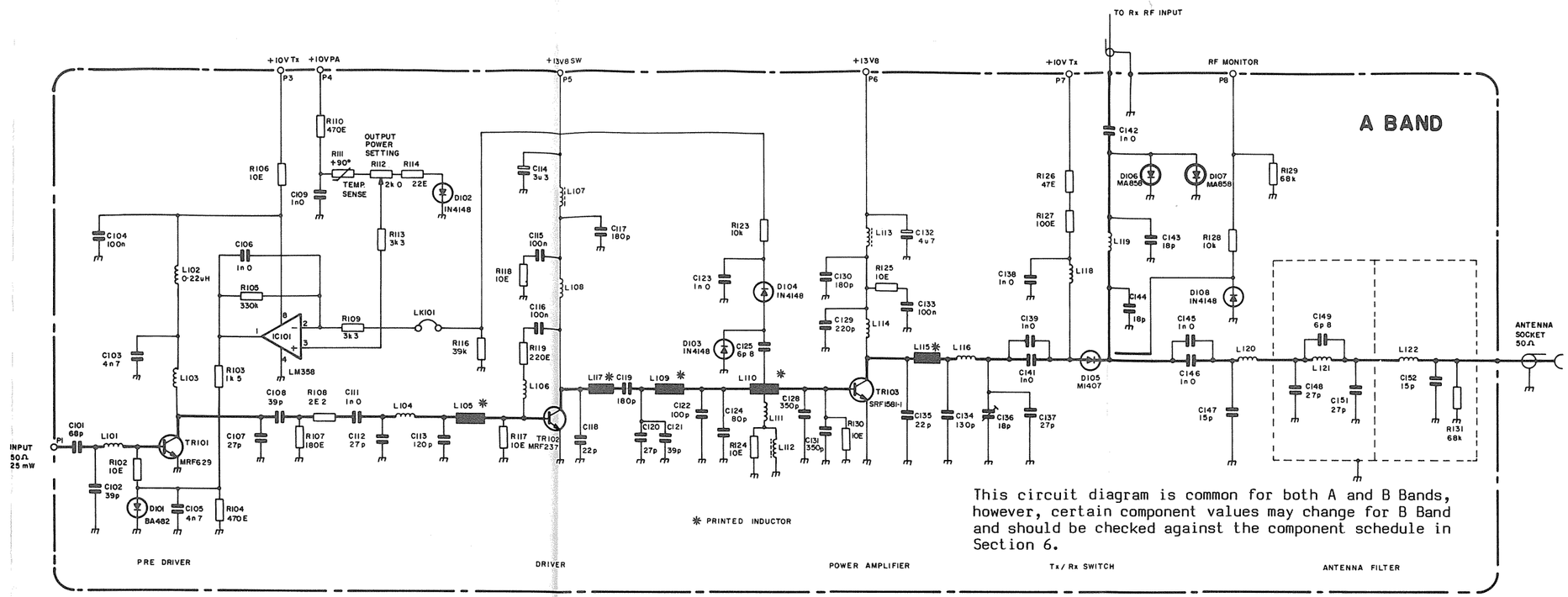

2 x 'E' SOLDER BRIDGE LINKS
ON BOTTOM SIDE OF P.C.B.
FOR E BAND ONLY
VHF RECEIVER 93
+13V8
BOTTOM



This circuit diagram is common for all UHF bands, however, certain component values will change with each band and should be checked against the component schedule in Section 6.

FIGURE 7.4 UHF TRANSMITTER PA P.C. BOARD
CIRCUIT AND COMPONENT LAYOUT


IC102
TR102
L102
L104
L107
L108
L110
L111
L112
C102
C103
C105
C106
C107
C108
C109
C110
C111
C112
C113
C114
C115
C116
C117
C118
C119
C120
C121
C122
C123
C124
C125
C126
C127
C128
C129
C130
C131
C132
C133
C134
C135
R101
R102
R103
R104
R105
R106
R107
R108
R109
R110
R111
R112
R113
R114
R115
R116
R117
R118
D101
D102
D103
D104
D105
D106
7 6 5 4



This circuit diagram is common for both A and B Bands, however, certain component values may change for B Band and should be checked against the component schedule in Section 6.

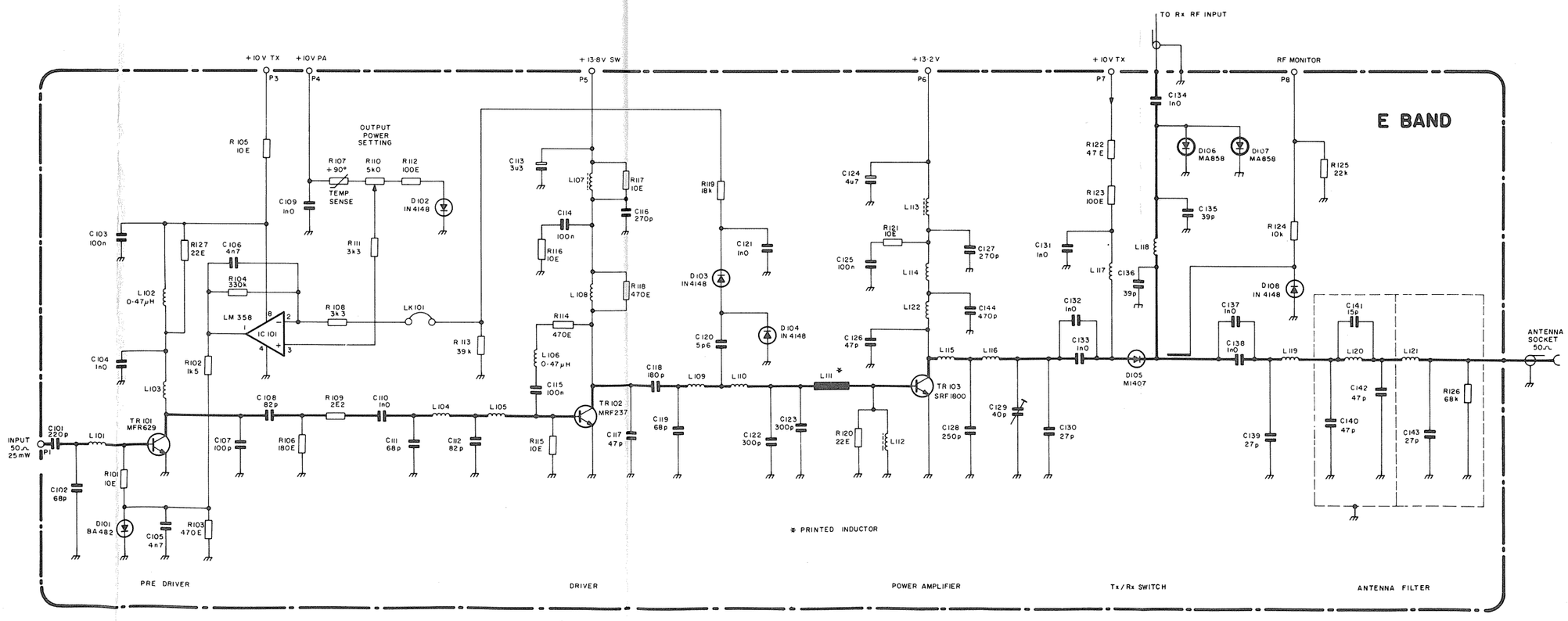
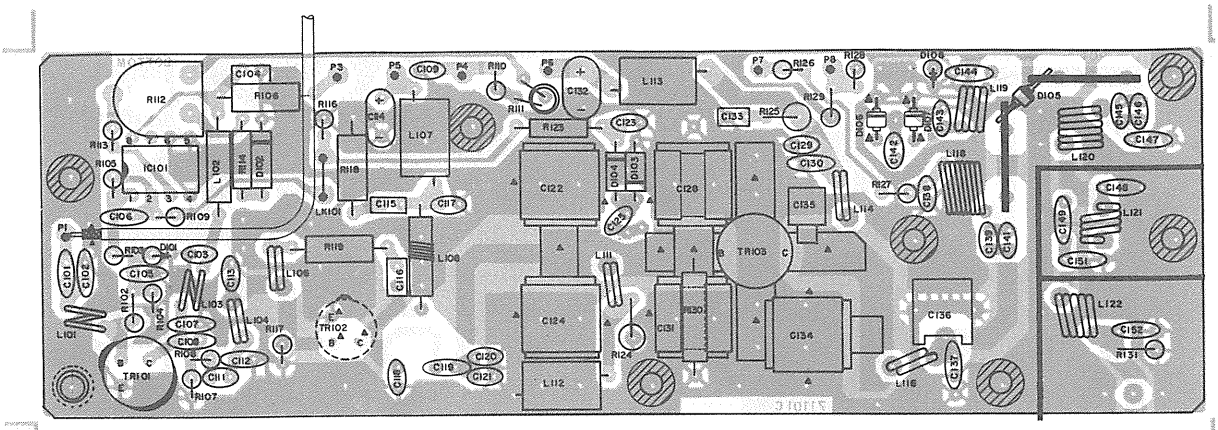
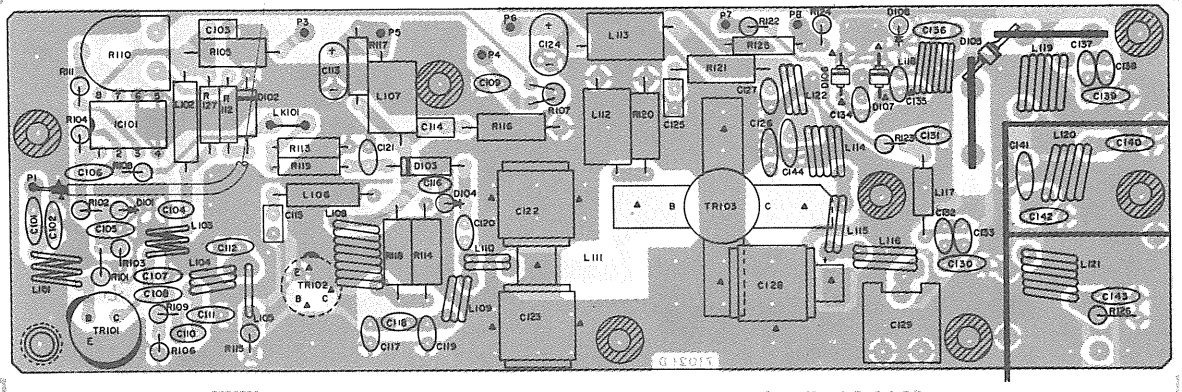


FIGURE 7.5 VHF TRANSMITTER PA P.C. BOARD
CIRCUIT AND COMPONENT LAYOUT



A & B BAND



E BAND

	FREQ BAND			CHAN. SPACING	
	E	A/B	UHF	12·5kHz	25kHz
R337	33k	27k	27k		
R335				3k9	1kO
R351	820k	1MO	1MO		
R366	12k	12k	15k		
R372	2k7	2k7	15k		
R340				270k	220k
R343				270k	220k
R344				150k	120k
R350	120E	2k7	15k		
C347	10n	10n	15n		
C327				15n	47n
C339	47p	47p		82p	68p
C344					
C349	220n	22n	22n		

CODE Nº	FREQ. BAND	CHANNEL SPACING kHz	
3502 349 73140	E	25/30	
3502 349 73160	E	12·5	
3502 349 73440	A&B	25/30	
3502 349 73460	A&B	12·5	
3502 349 83120	T,U,W1,W2	25	
3502 349 83130	T,U,W1,W2	12·5	

NOTES:-

1. FOR VALUES OF COMPONENTS MARKED * SEE TABLE.
2. COMPONENTS MARKED ● ONLY FITTED FOR "LOW TEMP XTAL HEATER" OPTION
3. COMPONENTS MARKED □ ONLY FITTED FOR "TCXO" OPTION.
4. COMPONENTS MARKED # ARE NOT FITTED WHEN "TCXO" OPTION IS INSTALLED.
5. EXTERNAL CONNECTIONS MARKED —▸ ARE CONTINUED ON SHEET 2 of 2

FIGURE 7.6A SYNTHESISER/CONTROL P.C. BOARD (SYNTHESISER SECTION)

CIRCUIT AND COMPONENT LAYOUT

EXT I/P
IC412 (REF)
IC415
IC408
IC409
IC410
IC413
IC414
IC405
IC407
IC411
IC301
IC302
IC303
IC304
XL301
XL302
XL401
XL402
TP301
TR354
TR336
TR357
TR317
BOTTOM

NOTES

SIGNALLING OPTIONS

1. COMPONENTS MARKED THUS □ ARE SIGNALLING OPTIONS AND DETAILS ON THESE MAY BE FOUND IN THE HANDBOOK.

GENERAL

2. COMPONENTS MARKED THUS :
   ■ ARE FITTED FOR UHF ONLY.
   ✳ ARE FITTED FOR VHF ONLY.
3. CONNECTIONS MARKED ⟶ ON PERIMETER OF THIS CIRCUIT INTERFACE WITH CIRCUITRY ON SHEET 1 OF THIS DRAWING.

FIGURE 7.6B SYNTHESISER/CONTROL P.C. BOARD (CONTROL SECTION)

CIRCUIT AND COMPONENT LAYOUT

IC411
IC407
IC415
IC410
IC413
IC409
IC412 (REF)
IC414
IC408
IC405
IC303
IC304
IC305
IC301
IC302
EXT I/P
R317
R336
R357
R354
XL301
XL302
XL401
XL402
TP301
TR380
TR390
TR409
TR410
TR411
TR403
D405
D304
D303
D380
C427
C428
C418
C440
C426
C401
C402
C445

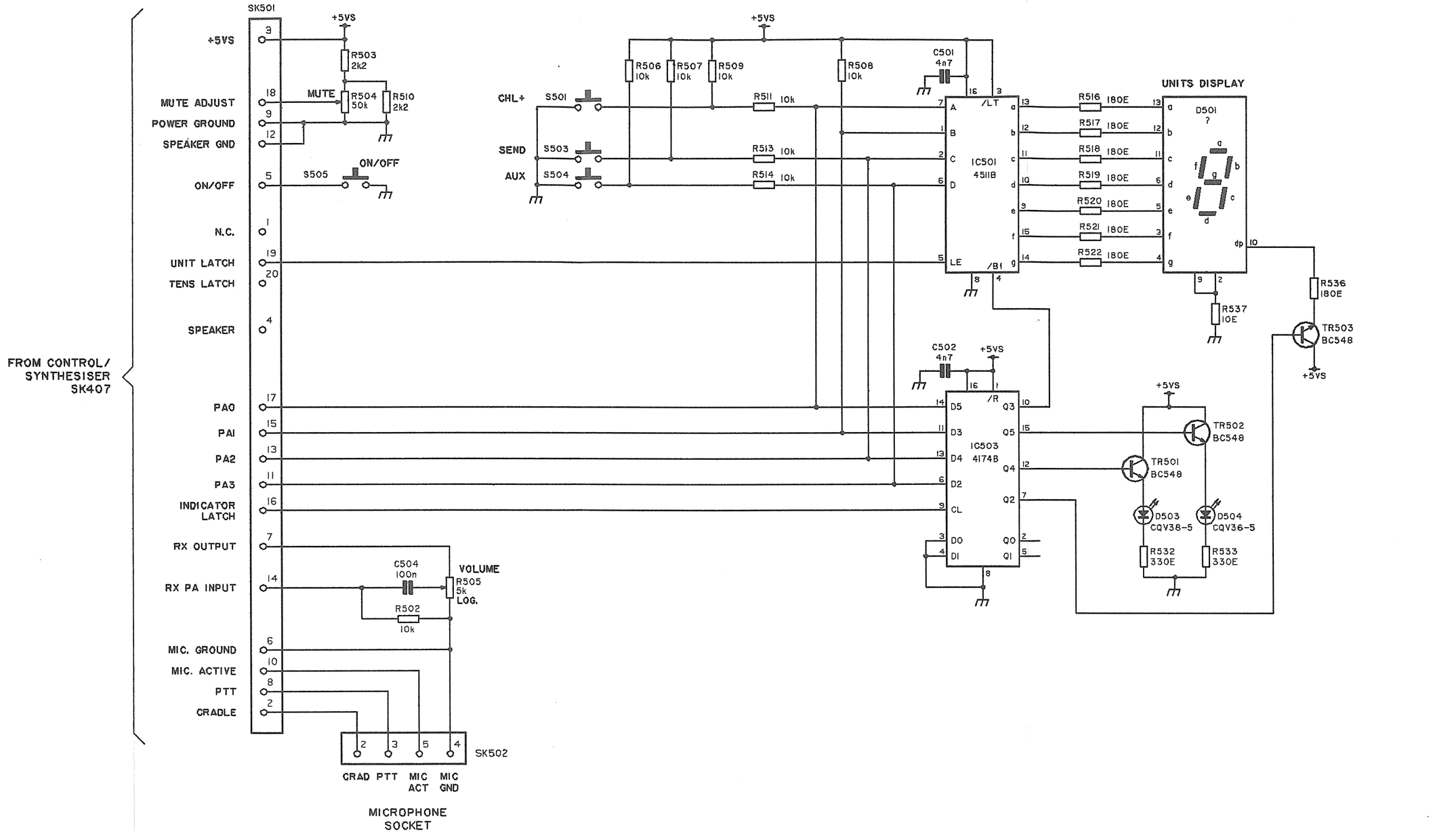
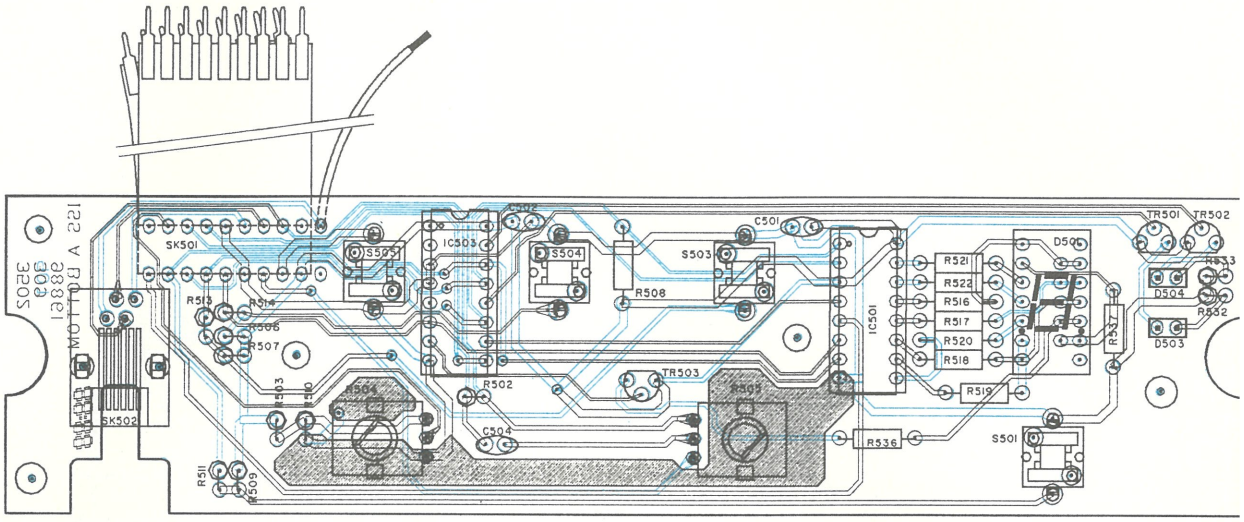
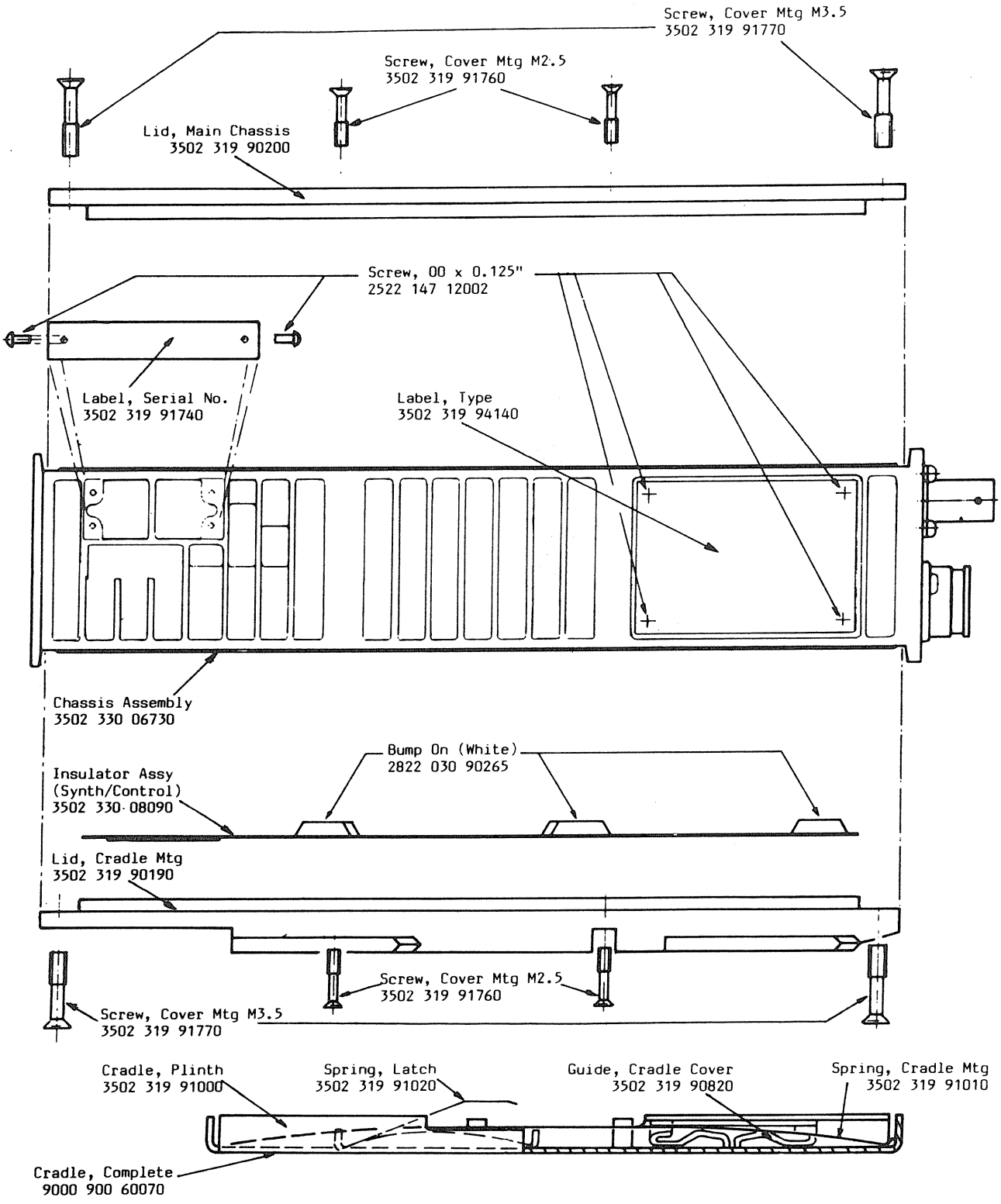


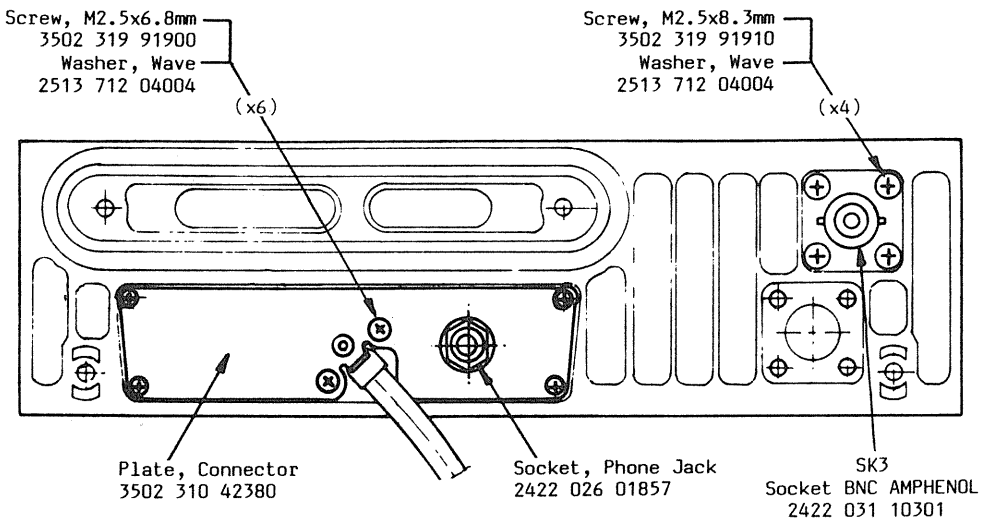
FIGURE 7.7 LOCAL CONTROL P.C. BOARD
CIRCUIT AND COMPONENT LAYOUT

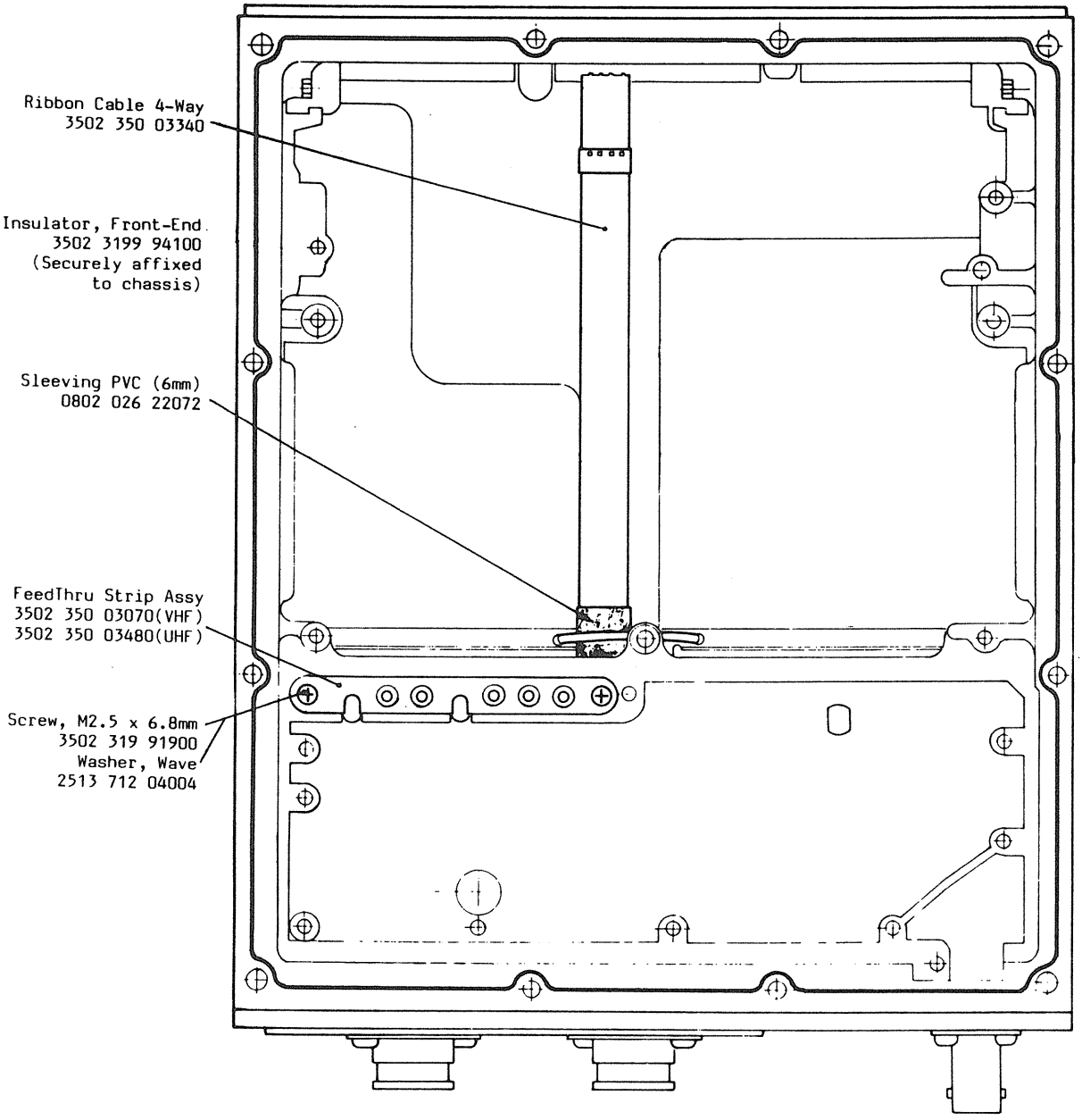

SK501
S505
IC503
S504
C502
C501
S503
R508
TR501
TR502
D504
D503
R532
R533
R521
R522
R516
R517
R520
R518
R519
R537
IC501
D501
R513
R514
R506
R507
R503
R510
R502
TR503
R505
C504
R536
S501
SK502
R511
R509

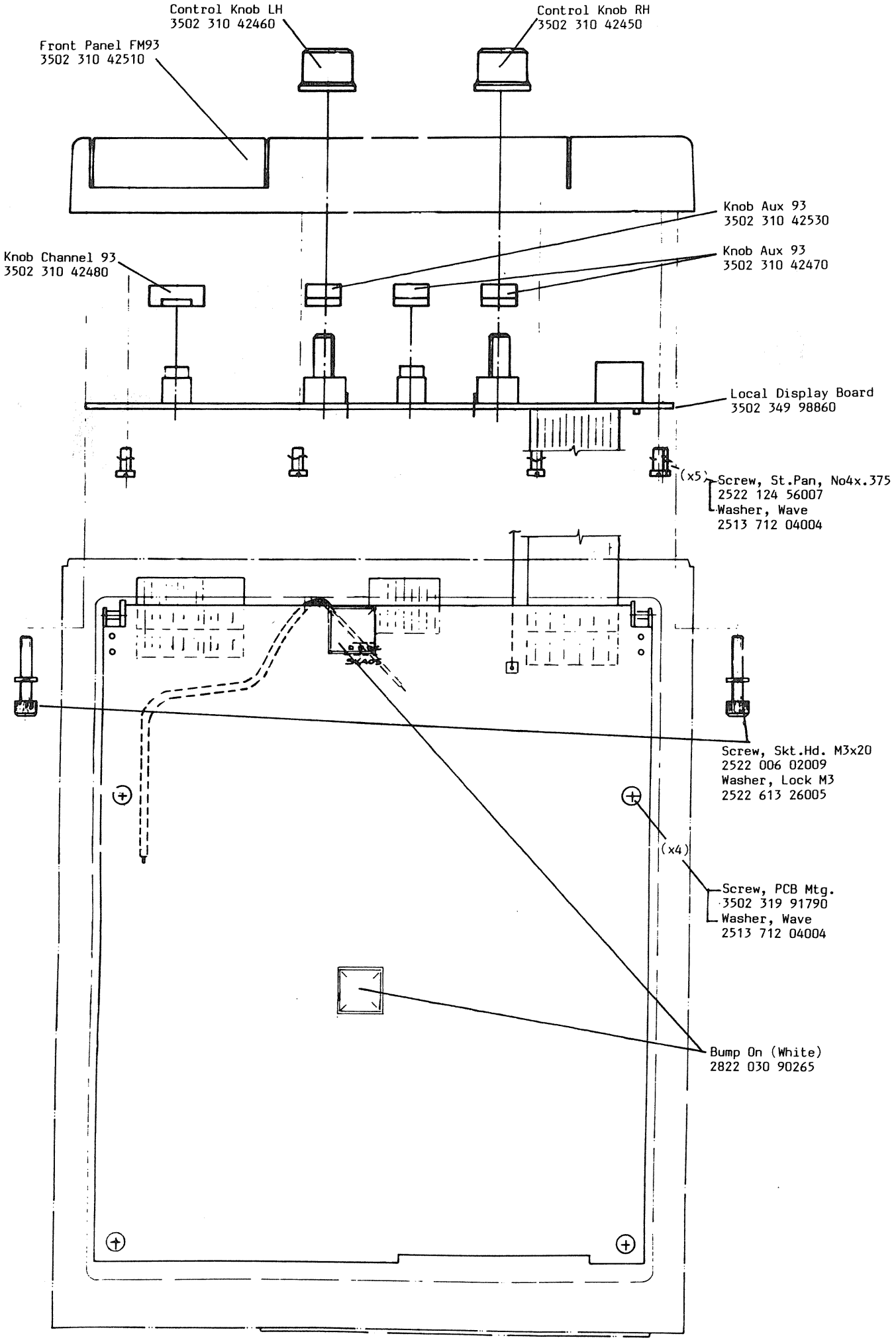
SECTION 8 MECHANICAL QUICK REFERENCE

8.1 COVERS




Screw, M2.5x6.8mm
3502 319 91900
Washer, Wave
2513 712 04004
(x6)
Screw, M2.5x8.3mm
3502 319 91910
Washer, Wave
2513 712 04004
(x4)
Plate, Connector
3502 310 42380
Socket, Phone Jack
2422 026 01857
SK3
Socket BNC AMPHENOL
2422 031 10301


Ribbon Cable 4-Way
3502 350 03340
Insulator, Front-End
3502 3199 94100
(Securely affixed
to chassis)
Sleeving PVC (6mm)
0802 026 22072
FeedThru Strip Assy
3502 350 03070(VHF)
3502 350 03480(UHF)
Screw, M2.5 x 6.8mm
3502 319 91900
Washer, Wave
2513 712 04004


Control Knob LH
3502 310 42460
Control Knob RH
3502 310 42450
Front Panel FM93
3502 310 42510
Knob Aux 93
3502 310 42530
Knob Channel 93
3502 310 42480
Knob Aux 93
3502 310 42470
Local Display Board
3502 349 98860
(x5)
Screw, St.Pan, No4x.375
2522 124 56007
Washer, Wave
2513 712 04004
Screw, Skt.Hd. M3x20
2522 006 02009
Washer, Lock M3
2522 613 26005
(x4)
Screw, PCB Mtg.
3502 319 91790
Washer, Wave
2513 712 04004
Bump On (White)
2822 030 90265

PART 2

OPTIONS

PART 2 OPTIONS

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SECTION 1 GENERAL

1.1 INTRODUCTION

The FM93 is designed to enable a range of software and hardware options to be incorporated to suit customer requirements as shown in Part 1, sub-section 1.4 of this manual. This part (Part 2) of the manual provides the details of the available hardware options.

1.2 AVAILABLE OPTIONS

1.2.1 HARDWARE OPTIONS

a) Selective Calling
   i) Selcall Type 1
   ii) Selcall Type 2
   - CCIR Encoder/Decoder
   - CCIR Encoder Only
   - ZVEI Encoder/Decoder
   - EEA Encoder/Decoder

b) Continuous Tone Controlled Squelch System
   i) CTCSS Encoder Only (9585 900 61020)
   ii) CTCSS Encoder/Decoder/RTB (9585 900 61130)

c) Ignition Noise Blanker

d) 300Hz to 3000Hz Receiver AF Response

e) Crystal Heater

f) VCTCXO (Voltage Controlled Temperature Compensated Xtal Oscillator)

1.2.2 SOFTWARE OPTIONS

The following software options are available, descriptions of which are contained in Part 1, Section 1.4.2, with checking procedures detailed in Part 1, Section 4, Sub-section 4.19 to 4.23.

a) Tx PTT Inhibit - Not available on a "per channel" basis.

b) Scanning

c) Economiser

d) Community Repeater

NOTE: Some options are mutually exclusive, refer to Part 1, Section 1, Sub-section 1.2.5 for table of valid combinations.

SECTION 2 - SELECTIVE CALLING

2.1 SELCALL TYPE 1

2.1.1 GENERAL DESCRIPTION

The Type 1 Selcall signalling module is designed for operation with the FM93 series mobile and fits into the existing U402 module slot. It offers high stability encoding/decoding of one of 8 defined international tonesets. Tonesets include CCIR, EEA, EIA, ZVEI 1, ZVEI 2, ZVEI 3, NATEL and CCIRH.

When decoding, the module is capable of continuously monitoring a signal input terminal for a pre-specified tone sequence and then producing audible and visual alert for the operator.

When encoding, a synthesized tone sequence output is produced.

All interface to the central control section is at standard TTL levels with inverted logic.

2.1.1.1 Toneset Selection

A total of 8 variations of toneset can be selected by solder bridging the appropriate copper lands TS2, 1, 0 situated on the modem module.

TABLE 2.1 - Sequential Signalling Tonesets

LINK SETTING	CCIR	EEA	EIA	ZVEI-1	ZVEI-2	ZVEI-3	NATEL	CCIRH	TONE ADDRESS
TS2	H	H	H	H	L	L	L	L	
TS1	H	H	L	L	H	H	L	L	
TS0	H	L	H	L	H	L	H	L	
TONE									$\overline{D3}$ $\overline{2}$ $\overline{1}$ $\overline{0}$
0	1981	1981	600	2400	2400	2200	1633	1981	H H H H
1	1124	1124	741	1060	1060	970	631	1124	H H H L
2	1197	1197	882	1160	1160	1060	697	1197	H H L H
3	1275	1275	1023	1270	1270	1160	770	1275	H H L L
4	1358	1358	1164	1400	1400	1270	852	1358	H L H H
5	1446	1446	1305	1530	1530	1400	941	1446	H L H L
6	1540	1540	1446	1670	1670	1530	1040	1540	H L L H
7	1640	1640	1587	1830	1830	1670	1209	1640	H L L L
8	1747	1747	1728	2000	2000	1830	1336	1747	L H H H
9	1860	1860	1869	2200	2200	2000	1477	1860	L H H L
A	2110	2110	459	2600	970	2400	1805	2400	L H L H
B	2400	1055	2151	2800	885	885	1995	930	L H L L
C	(1055)	(2400)	(2600)	(741)	(741)	(741)	(1300)	2247	L L H H
D	(2247)	(2247)	(2010)	(970)	(2600)	(2600)	(1700)	991	L L H L
E	(930)	(930)	(2433)	(810)	(2800)	(2800)	(2175)	2110	L L L H
F	(991)	(991)	(2292)	(886)	(600)	(600)	(2937)	(1055)	L L L L

Tones indicated in brackets are not specifically defined in the respective International Toneset.

2.1.1.2 Tone Number Selection

Selection of the appropriate tone number from the toneset table above is achieved by placing a static (logically inverted) binary value on the $\overline{D0}$ to $\overline{D3}$ inputs of the module. See Section 2.1.1.3 and 2.1.1.4 for timing details.

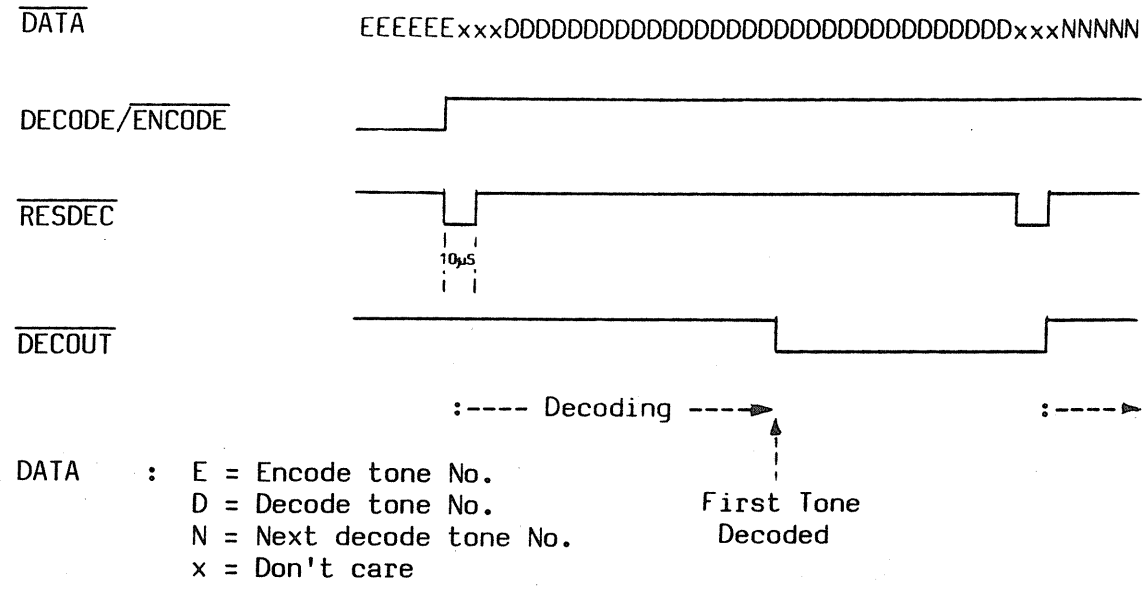


Figure 2.2 - Decoder Timing Diagram

2.1.1.4 **Encoder Operation**

To begin encoding, $\overline{\text{RESDEC}}$ is taken LO for 10uS, and upon returning HI. DEC/ENC input is taken LO. This remains LO permanently during the encoding process.

a) **Lead In Delay**

The lead in delay is set at factory standard 600mS.

b) **Encoding A Tone**

To encode a tone within the selected toneset (hardware selected), a binary (logically inverted) value is placed on D0 to D3.

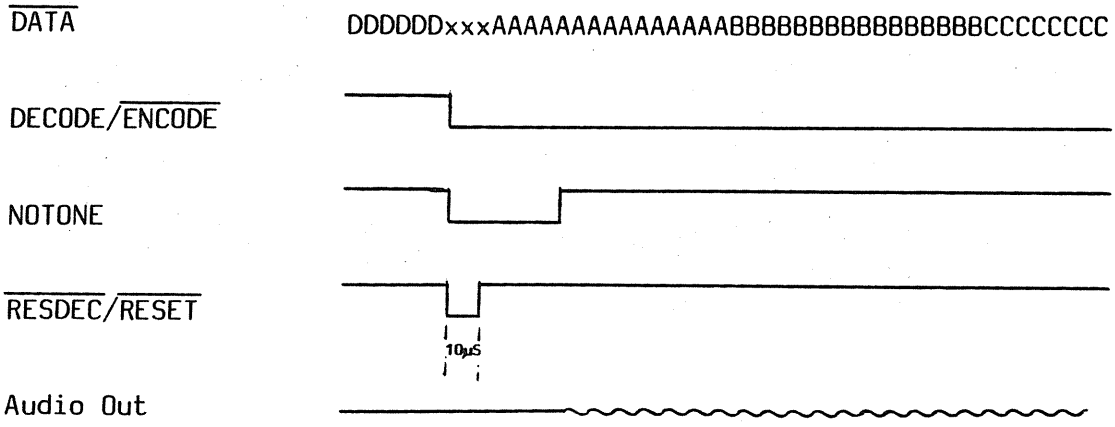


Figure 2.3 - Encoder Timing Diagram

2.1.3 SPECIFICATIONS

2.1.3.1 Electrical

Supply Voltage:	5V ±10%
Maximum Rating On Any Pin:	-1.0V, +7.0V
Current Consumption:	55mA
Internal Clock Frequency:	4MHz
Tonesets Available:	CCIR, EEA, EIA, ZVEI-1,ZVEI-2,ZVEI-3, NATEL, CCRIH.
Individual Toneset Frequencies:	See Table 2.1.
Encode AF Output Level:	750mV ±150mV RMS
Encode AF Output Change vs Frequency:	±2dB wrt 750mV RMS
Encode AF Output Distortion:	< 10% THD
Encode AF Output Muting:	> 50dB
Encode Output Impedance:	< 5k ac coupled
Frequency Stability:	±0.01%
Frequency Accuracy:	±0.2%
Decode Sensitivity:	< 40mV RMS
Decode AF Input Dynamic Range:	40mV to 2V RMS
Decode AF Input Impedance:	> 100k Ohm
Decode Bandwidth: (See Note 1)	±1.5% Pass ±2.0% Stop (CCIR)
Decode Response Time: (See Note 2)	9mS for tone 5 CCIR @ 20dB SINAD
False Decode Rate (See Note 3): (Gaussian noise 6kHz BW, 5V p-p)	Zero
Alarm Output Sink Current: (Alarm transistor not normally fitted)	> 30mA Sink @ 0.2V
Logic Levels: (Internal 6k pullups on all I/O)	'1' = 2.0V to 5.8V '0' = -0.3V to +0.8V
Operating Temperature Range:	-10° to +80°C (-40° to +90°C Available)
Storage Temperature Range:	-55° to +125°C

2.1.3.2 **Mechanical**

Size:	72mm x 19mm x 12mm (xtal extra)
Construction:	Double Sided PCB
Mounting:	The module is designed to fit in the existing U402 slot. The area allocated for U403 will also be required.
Vibration:	Capable of withstanding vibration conditions encountered in mobile radio environment.
Moisture:	Protected from condensation (Not hermetically sealed).

2.1.4 **TEST AND ALIGNMENT**

This test procedure covers the testing of all FM93 mobiles fitted with "Type 1" Selcall systems. The procedure covers CCIR, ZVEI and EEA toneset systems and tone periods of 20, 40, 70 and 100mSecs. Normal transmitter and receiver alignment must be completed before Selcall testing is performed. As this is a generalised test procedure which covers a variety of Selcall systems, it will be necessary to consult the mobile's programming chart and the software code charts (refer Part 1, Section 1.4) to determine which facilities are to be tested in a particular mobile.

2.1.4.1 **Test Equipment**

a) Power Supply. PS775 or similar (13.8V @ 8A)

b) Rohde & Schwarz test unit model SMFP.

c) Selective call test unit:

i) Option-2 fitted to SMFP suitable for testing:
CCIR, ZVEI, EEA, 5 and 7-tone systems.
20, 40, 70, 100mS tone period.

ii) Schlumberger Selective Call test unit model 4920, suitable for testing:
CCIR, ZVEI, EEA, 5-tone systems.
20, 40, 70 100mS tone period.

d) Noise and distortion meter, HP331A or similar.

e) CRO BWD Model 539D or similar.

2.1.4.2.2 **Encoder Test, 7-Tone Systems (SEPAC Compatible)**

a) Select channel on mobile for which Selcall is programmed.

b) Close "ENCODE TEST" Switch.

c) Adjust SMFP to measure FM93 mobile's transmit deviation.

d) Press Selcall "Send" button on FM93 mobile and check transmit deviation of all Selcall tones. **(NOTE:** Tones will be extended in length).

Ensure deviation is between ±3kHz (±1.5kHz) and ±4.9kHz (±2.4kHz).

Check the tone distortion for one of the tones (preferably the lowest frequency tone). Distortion should be less than 10% THD.

e) Open "encode test" switch.

f) Adjust Selcall test unit to the toneset used by the mobile's Selcall system (i.e. CCIR, ZVEI or EEA as specified in the mobile's hardware code).

Press mobile's "SEND" button and record code as decoded by test unit. Ensure that this code has the correct 5-tone address plus the "NULL status" tone.

g) Variable Send Function (if applicable)

Reprogram the mobile's "Send" code to any arbitrary code. Press the "SEND" button and verify that the manually programmed code is correctly sent.

h) ANI Function (if applicable)

Press PTT switch and release. Verify that the mobile's ident is transmitted. Verify that the ident code is correctly positioned at either the start or end of transmission in accordance with the mobile's software code.

i) Variable Status (if applicable)

Reprogram the status digit to any arbitrary code. Press the PTT switch momentarily and verify that the mobile's ident code is transmitted with the new status digit appended.

d) Decode Sensitivity Check

i) Establish a 4dB SINAD condition on receiver for 1kHz/±3kHz with 300mW into 4 ohm load. Modulate signal generator with Selcall test unit and set tone deviation to ±4kHz.

ii) Send 5-tone code to mobile and ensure mobile responds. Repeat sending Selcall code several times and ensure that the mobile responds at least 9 out of 10 times.

TABLE 2.1 - TYPE 1 Selcall Decode Response

	AUDIBLE ALARM	DISPLAY	RECEIVER MUTE	TIMED ALARM	AUTOMATIC ACKNOWLEDGE
STANDARD CALL	SHORT BEEP	"BUSY" LED FLASHING	OPENS CONTINUOUS	N/A	YES
STANDARD CALL WITH URGENT BEEP	CONTINUOUS BEEPING			N/A	YES
STANDARD CALL WITH ALARM TIMED	SHORT BEEP			SET FOR 10 SECONDS	YES
STANDARD CALL WITH URGENT BEEP AND ALARM TIMED	CONTINUOUS BEEPING			SET FOR 10 SECONDS	YES
ANY OF ABOVE WITH GROUP CODE	AS PER ABOVE			AS FOR ABOVE	NO
STANDARD CALL WITH SC706 COMPATIBILITY	SHORT BEEP	N/A	OPENS FOR 7 SECONDS	N/A	NO
STANDARD CALL WITH URGENT BEEP AND SC706 COMPATIBILITY	2 SECONDS OF BEEPING	N/A		SET FOR 10 SECONDS	NO

N/A = Not Applicable.

2.2 SELCALL TYPE 2

2.2.1 GENERAL DESCRIPTION

The Selective Call encoder/decoder signalling option module, is constructed on a 60mm x 16.9mm x 0.635mm hybrid substrate. The hybrid is fitted in the U402 position on the synthesiser/control board of the FM93 series mobile. The circuit design utilizes very large scale integration CMOS technology.

The hybrid has four versions to cover CCIR, ZVEI, and EEA tone frequency standards, in both encoder/decoder and encoder only variants.

The hybrid contains a 17-tone programmable encoder, producing frequencies which conform to the appropriate standard, and a universal decoder which responds to 15 frequencies in the tone set. Both the encoder and decoder operate from a common 560kHz reference frequency which is generated by a ceramic resonator. A trimmer capacitor is provided to fine tune this clock frequency. Programming of the encoder frequency is under microcomputer control and is achieved via four bi-directional logic control inputs (D0 to D3 at Pins, 5, 2, 3 and 4 respectively) which also act as data outputs when in decode. The encoder only variant has no decode circuitry assembled on the hybrid.

An input at Pin 12 sets either the encode or decode mode and is at logic '1' for encode. A bi-directional input at Pin 14 (Data change) is used to latch-in the new encode data when encoding, and produces a positive going pulse output in decode, each time the output data has changed. A dual function input at Pin 6 is used in encode to select two alternative modes of the encoder, i.e. logic '1' sets a hexadecimal tone set format and '0' sets the quadradecimal tone set format. In decode this input is used in conjunction with the microcomputer control to hold the state of the output data for reading, this effectively establishes handshake routines for selcall decoding.

AF input to the selcall decoder is derived from the receiver unde-emphasised output and is connected to Pin 16 of the hybrid.

AF output for selcall encode is at Pin 22 and couples the selcall tones to the transmitter audio processing hybrid, U301.

Pin 18 of the selcall hybrid is internally linked to Pin 17 and provides a connection to the mobile microcomputer. The output at this pin is provided as a logic control for selcall alarm decode options.

The selcall hybrid operates in tone sequential signalling systems under microcomputer control in the mobile. The microcomputer establishes all timing and sequencing of code patterns.

2.2.3 SPECIFICATIONS

2.2.3.1 Encoder

a)	Tone Deviation	±3.5kHz to ±4.9kHz deviation. Nominal ±4kHz. (For 25/30kHz channel spacing).
b)	Tone Distortion	> 10% THD
c)	Frequency Standards	CCIR, ZVEI, EEA. (See Note 1)
d)	Frequency Stability	±0.3% (Over temperature range -30°C to +60°C Ambient).
e)	Tone Period Timing	40, 70, 100mSec (See Note 2) (EEA),(ZVEI),(CCIR)
f)	Tone Period Tolerance	40 ±2mSec, 70 ±2mSec, 100 ±4mSec.
g)	Lead In Delay	Adjustable, 10mSec to 2.5 Sec. in 10mSec steps. Factory preset at 600mSec.
h)	Lead Out Delay	Adjustable, 10mSec to 2.5 Sec. in 10mSec steps. Factory preset to 100mSec.
i)	Send Repeat Rate	Delay between successive calls is fixed at 10 Secs.
j)	Code Formats	i) 5-Tone consecutive address code. OR ii) 7-Tone (5-Tone consecutive address code followed by a two tone period gap followed by a two tone period "Status" tone).

NOTE: 1. A different hybrid is required for each of the tone sets.

2. The tone period for CCIR may be either 40mSec or 100mSec on request.

2.1.3.2 Decoder

a)	SINAD Sensitivity (90% decode success rate of 5-tone sequence).	⩽3dB SINAD (measured at loud speaker for receiver reference input of 1kHz ±3kHz deviation).
b)	Deviation Sensitivity (Min.deviation for decode, with receiver in full quieting).	⩽±300Hz

2.2.3.3 **U402 Module Pin Connections Selcall Hybrid**

PIN NO.	CONNECTION
1, 10, 19, 20	0V
8	+5V
5	D0 Data Input/Output
2	D1 Data Input/Output
3	D2 Data Input/Output
4	D3 Data Input/Output
14	Data Change Input/Output
6	SEE NOTE
12	Encode/$\overline{\text{Decode}}$
16	AF Input
22	AF Output
17	Connected to Pin 18
11, 13, 15, 21, 23	Not Connected
7, 9	Not Fitted

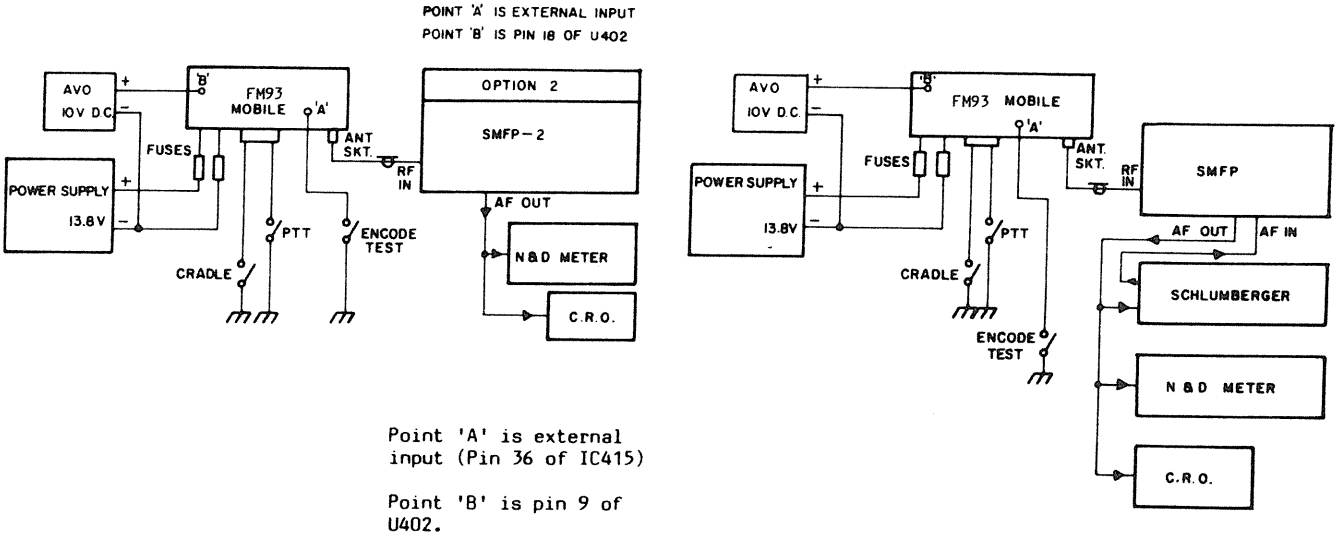
NOTE: Pin 6 is $\overline{\text{hold}}$ on Decode and XTC/$\overline{\text{QTC}}$ on Encode.

2.2.4 TEST & ALIGNMENT

This test and alignment procedure assumes that the normal transmitter and receiver alignment is already completed.

2.2.4.1 Test Equipment & Test Set-Up

a) Power supply, PS775 or similar. (13.8V @ 8a)
b) Rohde and Schwarz test unit model SMFP. (with Option 2)
c) Schlumberger Selective Call test unit model 4920.
This may be used with SMFP if option 2 not fitted, but will only be suitable for 5 tone system testing).
d) Noise and Distortion meter. HP331A or similar.
e) C.R.O. BWD Model 539D.
f) Frequency Counter Philips PM6668.
g) High impedance probe.



(a) Equipment Set Up Suitable For All Selcall Systems

(b) Alternative Equipment Set-up. Suitable for 5-Tone Selcall Systems Only.

NOTE: "Encode Test" switch when closed extends the tone period on selcall encode, if held closed when mobile switched on.

Figure 2.5 - Equipment Set-Up for Selcall Test

2.2.4.1.4 **Encoder Test 7-Tone Systems (Sepac)**

e) Adjust Selcall test unit to decode the mobile code sent. Press mobile's "Send" button and record code as decoded by test unit. Ensure this code has correct 5-tone address plus the "null status" tone. For Selcall plus ANI system ensure the 5-tone address corresponds to the "base address" code for the customer. For ANI only system ensure the 5-tone address corresponds to the "mobile ident" code for the customer.

f) For mobiles programmed with ident at start of transmission, activate PTT and ensure "mobile ident" code is sent at start of transmission. If mobile is programmed with ident at end of transmission, ensure correct code is sent when PTT is released. This code should have the 5-tone address plus the "null status" tone.

g) For FM93 mobiles with variable encode, programme a 5-tone code (arbitrary code) and press "SEND" button. Verify that manually programmed code is correctly sent.

2.2.4.1.5 **Decode Test**

a) Ensure cradle switch is closed and "Encode Test" and PTT switches are open. In this condition verify that the receiver mute is closed.

b) Select channel on mobile for which Selcall decode is programmed.

c) Adjust SMFP signal generator to mobile receive frequency and set output level to 100uv pd. Ensure the amber Rx LED is on but receiver mute is closed.

Programme SMFP to have fast response time on the transmit mode, for demodulated audio output. (See SMFP handbook for set up procedure.)

d) Standard Decode

i) Modulate SMFP with mobile's 5-tone standard decode address. (Set modulation at ±4kHz deviation and ensure code is sent at correct tone period length and initially sent at ±0% frequency error).

ii) Check that mobile responds in the following way:

(a) When mobile decodes correctly an audible "beep" is produced in the loudspeaker and the 'BUSY' LED will flash.

(b) PTT is activated and Tx Red LED is active during transmit. Selcall code is sent. This is automatic acknowledge, measure this selcall transmission and ensure code corresponds to the mobile ident code.

(c) Receiver mute should open.

iii) Check that the flashing 'BUSY' LED is cleared when the cradle switch is opened, and that the receiver is muted when switch is closed again.

h) Alarm ON/OFF Decode

i) If mobile is programmed for alarm on/off decode response, connect an AVO positive lead to Pin 18 of U402 module and negative lead to chassis. Set AVO +10Vdc range. Modulate SMFP with mobile's alarm on address code.

ii) Check that mobile has the following response:

When mobile decodes correctly, the AVO reads between 2.4V & 5V. No other response should occur.

iii) Modulate SMFP with mobile's alarm off address code and ensure that mobile responds in the following way:

The AVO reads between 0V & 0.4V.
No other response should occur.

j) Decode Bandwidth Test

i) Modulate SMFP with mobile's 5-tone standard decode address. Set modulation to ±4kHz deviation and offset tone frequencies by +3.5%.

ii) Ensure mobile does not respond to Selcall code.

iii) Offset tone frequencies by -3.5% and again modulate SMFP with Selcall code.

iv) Ensure mobile does not respond to Selcall code.

v) OPTIONAL - If facilities are available on Selcall test unit, offset Selcall tone frequencies by +1.5% and -1.5% and repeat decode test.

vi) Ensure mobile does respond to Selcall code.

k) Decode Sensitivity Check

i) Establish a 4dB SINAD condition on receiver for 1kHz ±3kHz with 300mW into 4ohm load. Change SMFP modulation to Selcall code for mobile standard decode address. Modulate at ±4kHz deviation.

ii) Ensure mobile successfully decodes. Repeat sending Selcall to mobile several times and ensure that the mobile responds at least 9 out of 10 times.

SECTION 3 CTCSS
(Continuous-Tone-Controlled-Squelch-System)

3.1 GENERAL DESCRIPTION

3.1.1 CTCSS ENCODER ONLY

The CTCSS encoder only signalling option module is a discrete design constructed on a 60mm x 17mm x 0.8mm printed circuit board. The module is fitted in the U405 position on the synthesiser/control board of the FM93 series mobile. The module synthesises a low distortion tone from an external 1MHz clock input. Frequency programming control inputs to the module (D0 to D5 at Pins 7, 6, 4, 13, 22 and 14 respectively) enable selection of tone frequencies in the range 67Hz to 250Hz.

The frequencies selected conform to the EIA standard tone set. A control input to the module at Pin 15 is provided for gating of the output tone. The module control inputs interface to the mobile's central control microcomputer via the NSC810 I/O expander. The AF output of the module at Pins 20 and 23 interfaces directly to the transmitter audio processing circuit via an A.I.T. resistor (nominally 39k) fitted in place of C336.

3.1.2 CTCSS ENCODER/DECODER (+ RTB or ENCODER ONLY + RTB)

The CTCSS encoder/decoder signalling option module, is a thick film hybrid constructed on a 60mm x 16.9mm x 0.635mm ceramic substrate. The hybrid is fitted in the U405 position on the synthesiser/control board.

The hybrid is controlled by the central control and can be configured as encoder/decoder with RTB or encoder only with RTB.

In the encode mode the encoder/decoder circuit synthesises a low distortion tone from its 5MHz reference crystal oscillator. Frequency programming control inputs to the module (D0 to D5 at Pins 7, 6, 4, 13, 22 and 14 respectively) enable selection of tone frequencies in the range 67Hz to 250Hz. The frequencies selected conform to the EIA standard tone set. The encode/decode changeover control input at Pin 15 of the module disables the AF output tone when in the decode state. The encode AF output at Pins 20 and 23 interfaces directly to the transmitter audio processing circuit via C336.

In the decode mode unde-emphasised receiver AF output ("FM Output") is connected to Pin 2 and processed in the hybrid for detection of the sub-audible tone. The decoder centre frequency is programmed the same way as the encoder. A successful tone decode causes a control output at Pin 16 to go to a logic '0' state. Control inputs and outputs interface to the mobile's central control microcomputer via the NSC810 I/O expander IC415.

The audio processing filters on the receiver p.c.board are designed to suppress CTCSS tones up to 250Hz (response 450Hz to 3000Hz).
Where full 300Hz to 3000Hz AF response is required CTCSS tones below 171.3Hz should be used. (refer Section 5).

3.1.3 REVERSE TONE BURST

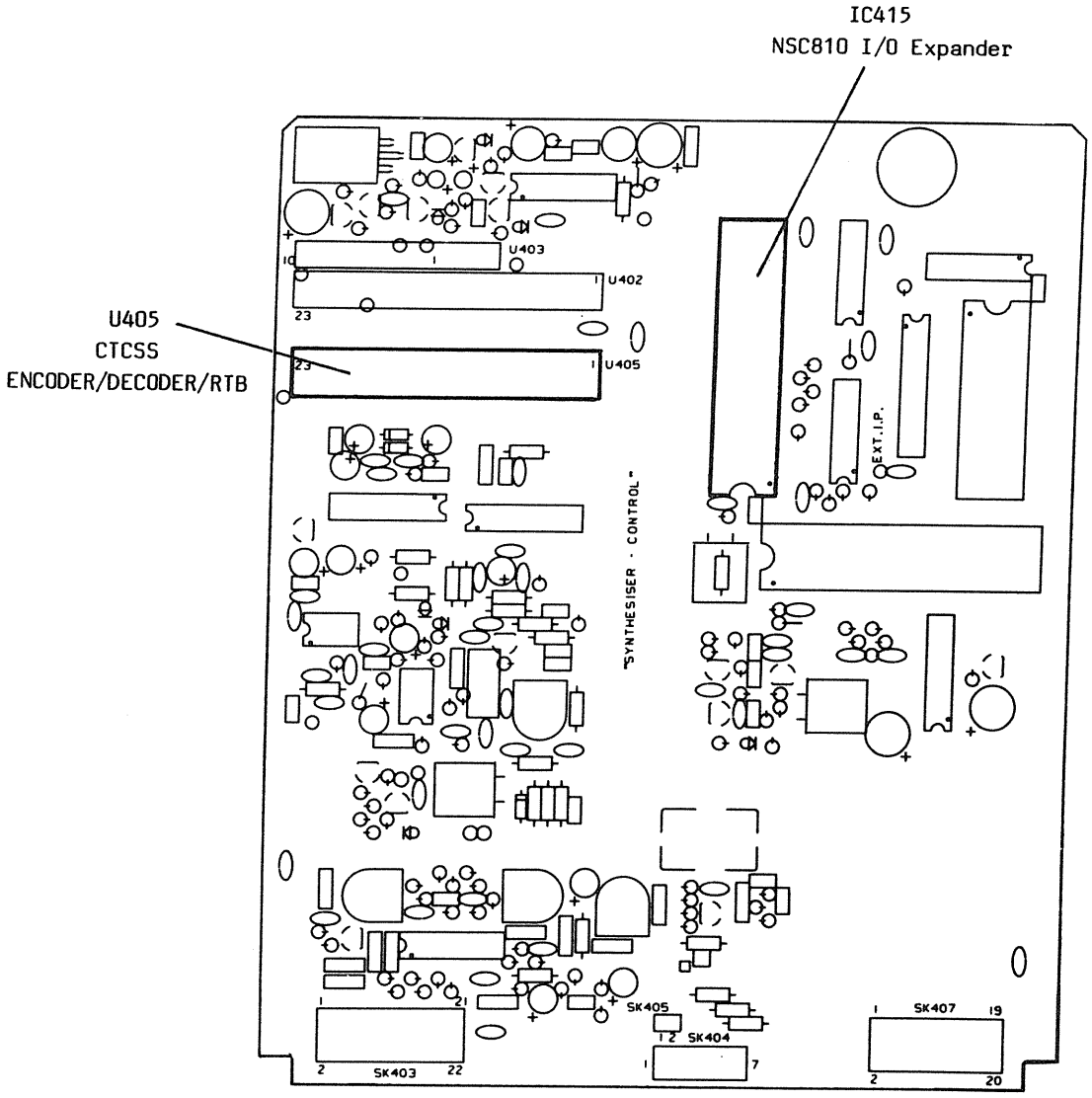
The Reverse-Tone-Burst option is enabled by the control software.

The purpose of the RTB is to change the phase of the sub-audible tone at the end of the transmission for rapid quenching of vibrating reed decoders.

The module produces a 120° phase lag for a period of approximately 147mS.

The RTB enable to the module is active low via pin 17.

3.2.2 CTCSS ENCODER/DECODER/RTB



COMPONENTS

a)	U405	CTCSS Enc/Dec/RTB Module	9585 900 61130
b)	IC415	Int.Cct. NSC810AN	9337 507 50682

3.3.2.2 **Decoder**

a)	Sensitivity (mute opening Sinad)	3dB typical (6dB max.). See Note 2. (for ±500Hz receive deviation)
b)	Bandwidth	±1.0%
c)	Response Time	f_o < 100Hz: 290mS max.(<250mS typ.) f_o 100-200Hz: 190mS max.(<150mS typ.) f_o > 200Hz: 100mS max.(<90mS typ.)
d)	Frequency Range	As per Encoder
e)	Number of Tones	As per Encoder
f)	Centre Frequency Stability	As per Encoder
g)	Absolute Centre Frequency Accuracy	As per Encoder
h)	Speech Deviation for loss of Decode	> ±5kHz Peak to Peak 350Hz to 3kHz
i)	False Decode Rate (no carrier input)	1 per hour max. (see note 4)
j)	Receiver Hum & Noise Ratio Ref. 1kHz at 3kHz deviation 300mW into 4ohm loudspeaker.	> 32dB (see note 3)
k)	Receiver Audio Response	Within +1, -3dB of 6dB/octave de-emphasis over a frequency range of 450Hz to 3kHz standard. Optional response +1, -3dB of a 6dB/octave de-emphasis over a response of 300Hz to 3000Hz can be used with CTCSS tones up to 167.9Hz while maintaining Hum and Noise figure greater than 32dB.
l)	Reverse Tone Burst	Phase shift 120° ±5° lagging

NOTES:

1. CTCSS encode tone output is disabled when in decode (receive) mode.

2. Mute opening threshold for carrier controlled mute at minimum. CTCSS mute and carrier controlled mute normally wired as an "AND" function.

3. This hum and noise ratio is the worse case figure with 250Hz tone. The CTCSS tone is normally of low distortion therefore the degradation to receiver hum and noise ratio is normally not noticed.

4. This false decode typically consists of a pulse 200mSec in duration and because the tone mute is wired as an "AND" function with the carrier mute, the false decode is not noticed. When a carrier is received no false decodes occur.

3.4 TEST AND ALIGNMENT

3.4.1 CTCSS ENCODER ONLY

3.4.1.1 Test Equipment and Test Set-Up

a) Noise and distortion meter. HP339A or similar.
b) Power supply PS775 or similar.
c) Frequency Counter.
d) R.F. Power Attenuator/Power Meter 25W Bird 8321
e) Sampling Pad (-40dB)
f) Deviation Monitor. Marconi TF2300B.
g) Audio Signal Generator HP208A.
h) C.R.O. BWD 539D.

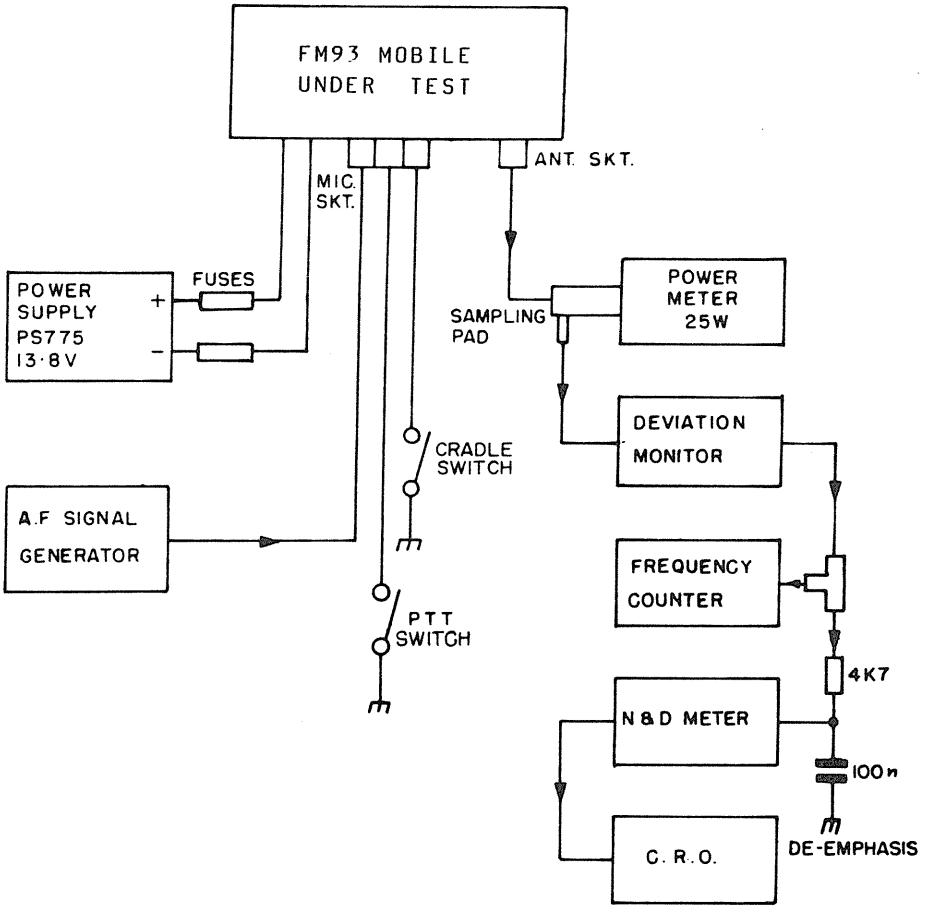


Figure 3.1. Equipment Set Up.

3.4.1.2 Preliminary Adjustments

a) Disable CTCSS tone output by closing Cradle Switch.

b) Perform normal mobile transmitter and receiver test and alignment procedure.

c) Enable CTCSS Encoder output by opening Cradle Switch.

d) Deviation settings bracketed are for 12.5kHz mobiles.

3.4.2.2 **Preliminary Operations**

a) Disable encoder output. (Close Cradle Switch to disable encoder output. Open Cradle Switch to disable decoder receiver mute).

b) Perform normal mobile transmitter and receiver test and alignment procedure.

c) Open Cradle Switch and re-enable CTCSS encoder output.

d) Connect mobile under test to test equipment as in Figure 3.2a.

e) Deviation settings in parenthesis are for 12.5kHz mobiles.

3.4.2.3 **Encoder Test**

a) Select transmit channel for which CTCSS encode is programmed.

b) Activate PTT and monitor deviation of transmitter via a sampling pad at the Wattmeter. Ensure Cradle Switch is open.

c) The deviation of the sub-audible tone should be in the range ±560Hz to ±700Hz (±280Hz to ±350Hz). Observe waveform on C.R.O. Adjust module preset level control pot to achieve the required result.

d) Measure frequency of tone. Frequency should be within ±0.2Hz of EIA tone frequency programmed.

e) Use N & D Meter to measure SINAD (with de-emphasis) of CTCSS tone. SINAD should be better than 27dB.

f) Connect AF generator to mic input skt and set to 1kHz/400mV rms.

g) Measure transmitter deviation and if deviation is greater than ±5kHz (±2.5kHz) slightly re-adjust deviation set pot R336 for ±4.9kHz (±2.4kHz).

h) For mobiles fitted with multi-tone access option, check encode tone deviation and frequency for other channels.

j) Disable PTT.

3.4.2.4 **Decoder**

a) Connect mobile to test equipment as shown in Figure 3.2(b). Disable receiver mute.

b) Select channel for which CTCSS decode is fitted.

c) Set RF generator to mobile frequency, and modulate with 1kHz ±3kHz (1kHz ±1.5kHz).

d) Adjust RF generator level to give 6dB receiver SINAD. Switch off RF generator modulation. Do not alter RF level.

e) Adjust receiver carrier mute to minimum. Check that the cradle switch is closed. Carrier received LED should be on but receiver AF output should be muted.

f) Modulate RF generator with correct CTCSS tone (for mobile decoder) at ±500Hz (±250Hz) deviation. Rx AF output should now be enabled.

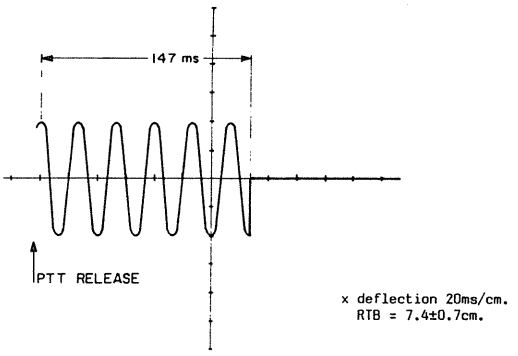
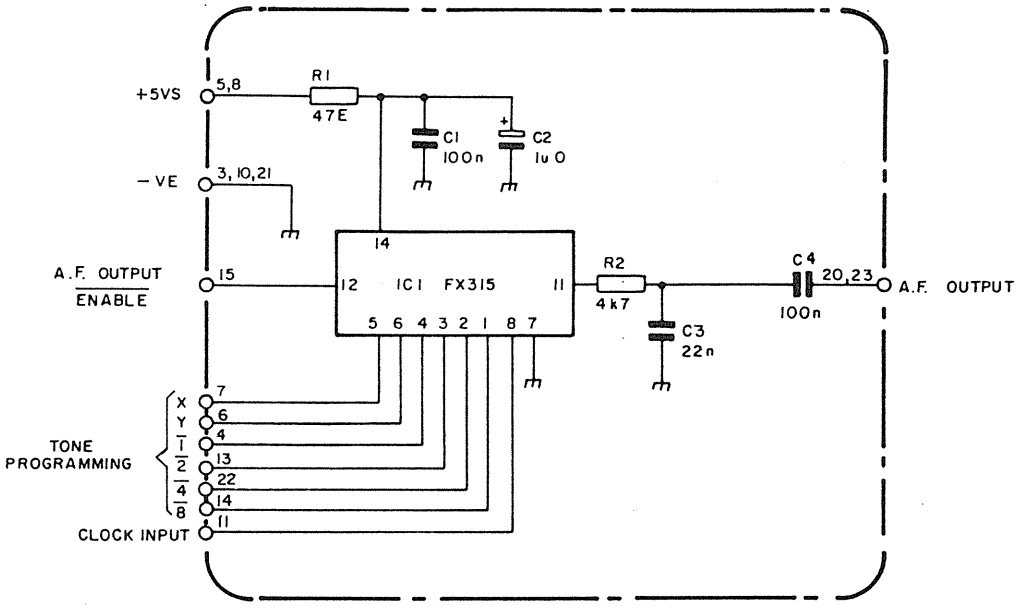


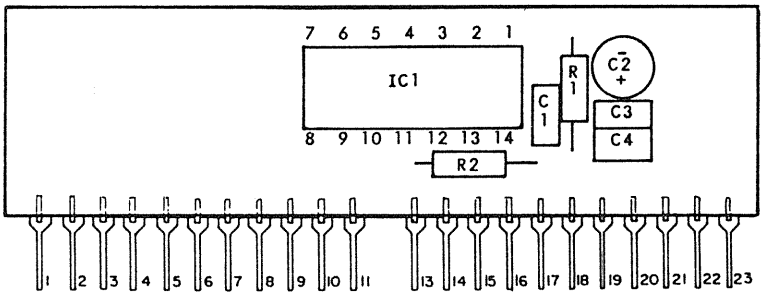
Figure 3.4 Reverse-Tone-Burst Waveform

3.5 CTCSS PC BOARDS

3.5.1 CTCSS ENCODER PC BOARD (3502 349 96430)

	PCB CTCSS Encoder	3502 329 96430	C1	100n,20%,50V, Cer D/M	2022 552 02482
	47E, 5%, ¼W, Carb.F.	2120 101 46479	C2	1u0, 20%,35V, Tant.	2020 004 90037
R2	4k7, 5%, ¼W, Carb.F.	2120 101 46472	C3	22n, 20%,50V, Cer D/M	2022 552 02019
			C4	100n,20%,50V, Cer D/M	2022 552 02482
IC1	Intgrd Circt FX315	9336 526 30682			





SECTION 4 IGNITION NOISE BLANKER

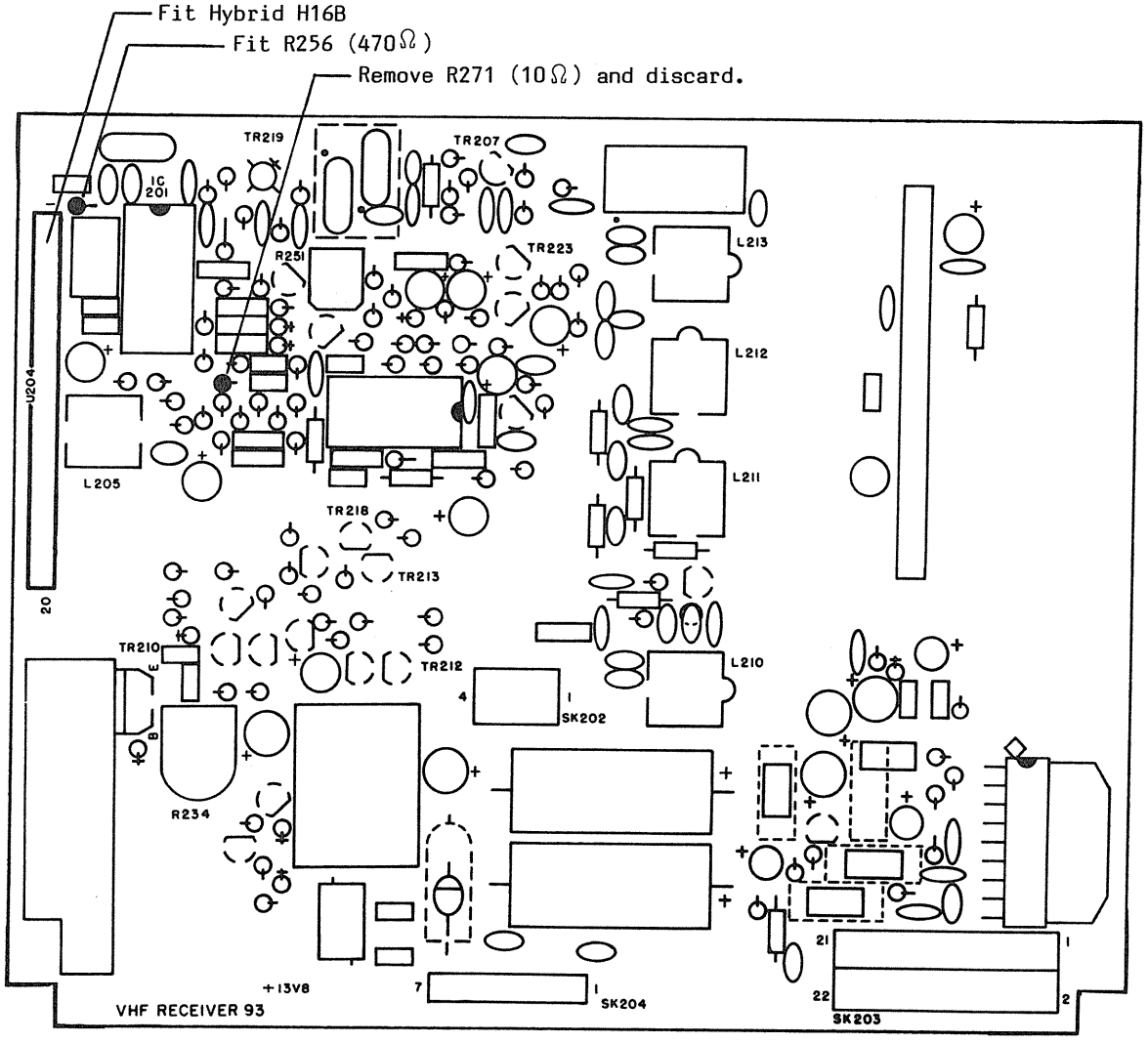
4.1 GENERAL DESCRIPTION

The module is fitted in the U204 position of the receiver pcb of the FM93.

The module contains a buffer and IF amplifier stage which samples the IF signal, including noise impulses, at the input of FL202. Pulse detector, pulse amplifier and blanking pulse generator stages then control an audio gate. The audio gate interrupts the AF output from IC201 for 1.2mSec when a high level noise impulse is received.

Blanking effectively stops the high level noise pulses from reaching the filters and de-emphasis stages. With the ignition pulses removed at this point a significant improvement is achieved in message intelligibility in the presence of high level ignition interference.

4.2 IGNITION NOISE BLANKER KIT



COMPONENTS

U204	Hybrid H16B	3502 358 00760
R256	Res.Carb F., 2k2, 5%, ¼W	2120 101 46222

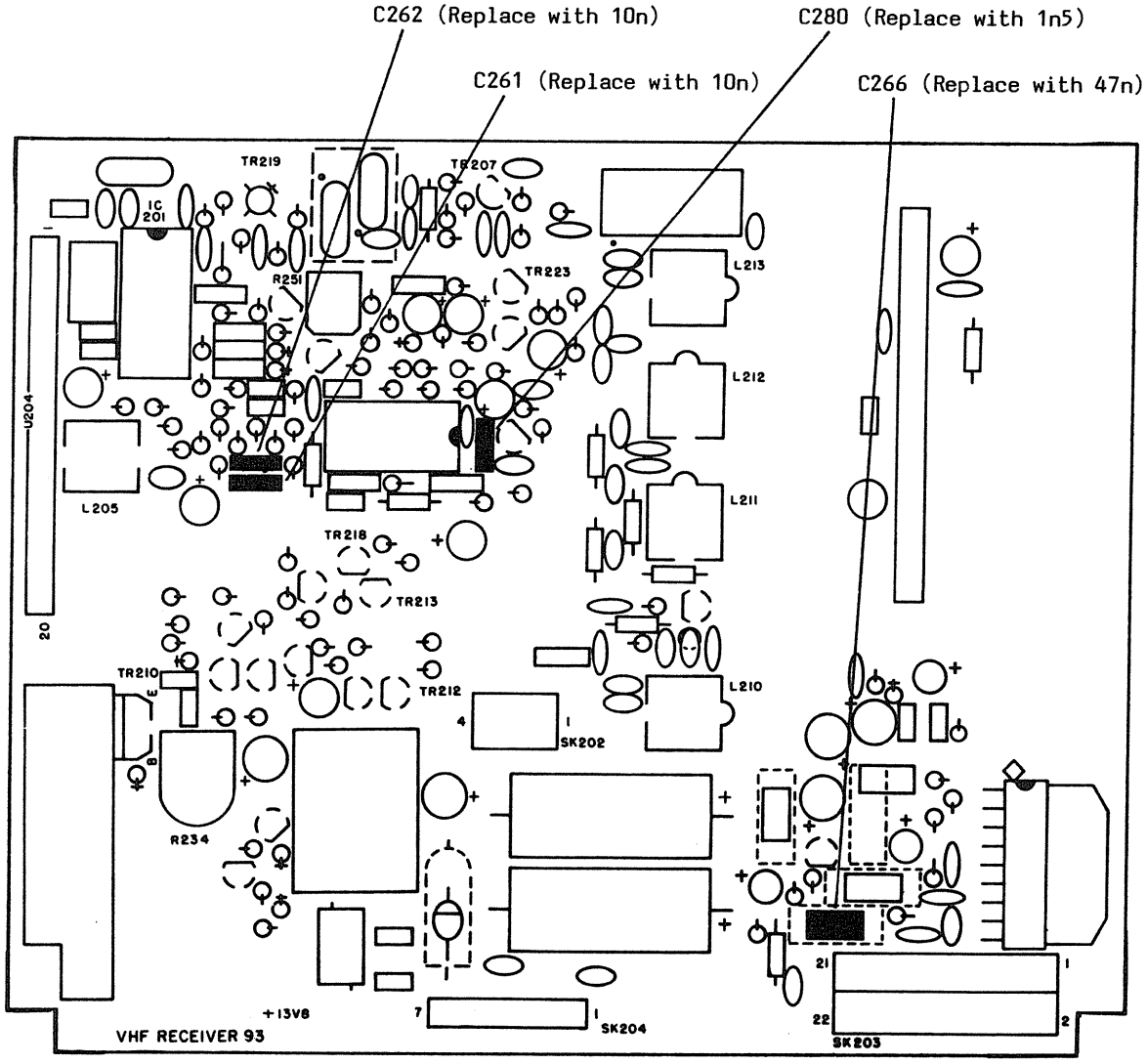
R271 10ohm resistor is to be removed and discarded

SECTION 5 300Hz TO 3000Hz AUDIO RESPONSE

5.1 GENERAL

This option provides for the audio response of the receiver to cover the full range of 300Hz to 3000Hz, +1dB,-3dB of 6dB/octave de-emphasised curve.

5.2 FITMENT

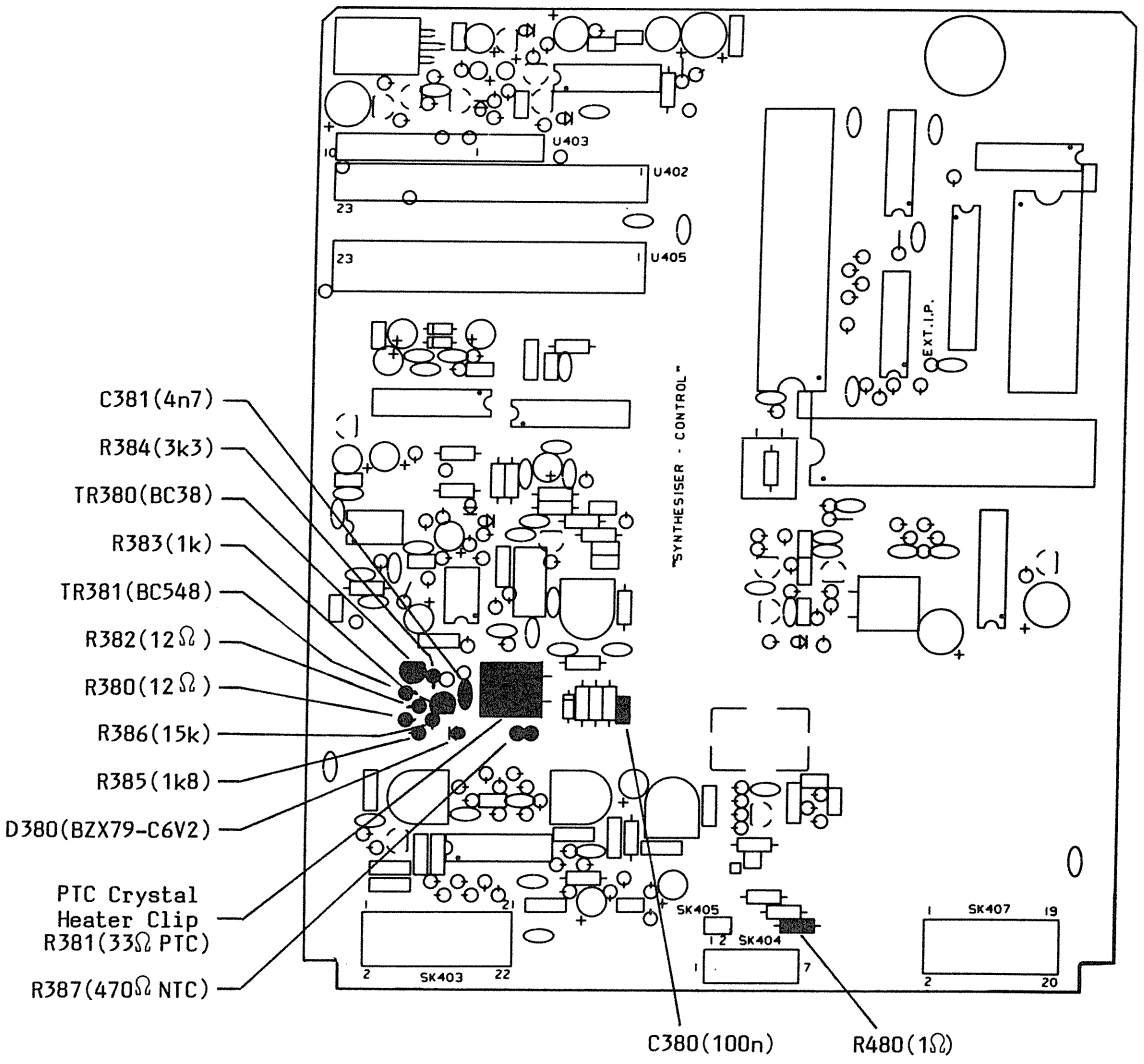


COMPONENTS

C261	Cap. Pest. STKD, 10n, 5%, 100V	2011 301 40251
C262	Cap. Pest. STKD, 10n, 5%, 100V	2011 301 40251
C266	Cap. Pest. STKD, 47n, 5%, 100V	2011 301 20501
C280	Cap. Pest. STKD, 1n5, 5%, 100V	2011 301 20028

SECTION 6 CRYSTAL HEATER KIT

6.1 FITMENT



COMPONENTS

R380	Res. Carb.F. 12E, 5%, ¼W	2120 101 46129
R381	Res. PTC 33E	2120 660 90006
R382	Res. Carb.F. 12E, 5%, ¼W	2120 101 46129
R383	Res. Carb.F. 1kO, 5%, ¼W	2120 101 46102
R384	Res. Carb.F. 3k3, 5%, ¼W	2120 101 46332
R385	Res. Carb.F. 1k8, 5%, ¼W	2120 101 46182
R386	Res. Carb.F. 15k, 5%, ¼W	2120 101 46153
R387	Thermistor NTC 47OE, 1%, 0.6W	2322 642 61471
R48O	Res. Carb.F. 1EO, 5%, ¼W	2120 101 46108

C380	Cap.Cer.D/M 100n, 20%, Z5U	2022 552 02482
C381	Cap.Cer.Pl. 4n7, 10%,	2222 630 08472
D380	Diode BZX79-C6V2	9331 177 40112
TR380	Transistor BC328	9331 491 90112
TR381	Transistor BC548	9331 976 40112

6.2 TEST PROCEDURE

a) With transceiver in Rx standby condition note the supply standby current.

b) Spray instant freeze onto the crystal heater and note that the supply current increases by approximately 300mA.

c) Stop spray and note that the supply current returns to the original value in less than 40 seconds.

SECTION 7 VOLTAGE CONTROLLED TEMPERATURE COMPENSATED CRYSTAL OSCILLATOR

7.1 FITMENT

REMOVE COMPONENTS:

Resistors
R323,324,325,328,329,
333,334,346,348,354,
355,356.

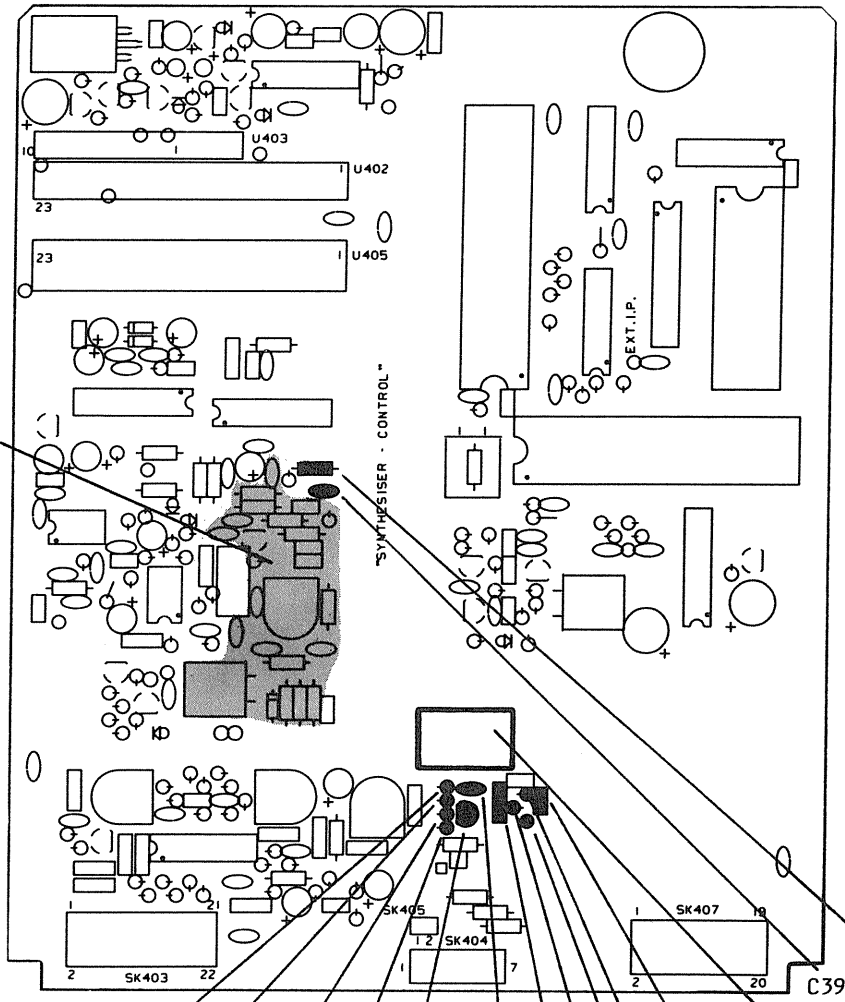
Capacitors
C320,321,325,328,330,
332,333,342,348.

Semiconductors
TR302,
D302.

Xtal
XL301.

or

Disable Reference Oscillator
by disconnecting R333 & R323.



FIT COMPONENTS: R394(10k)
R391(120k)
R392(33k)
R393(2k7)
TR390(BC558)
C391(470p)
C392(220n)
R395(120k)
R396(68k)
R397(150k)
C394(100n)

COMPONENTS

R391	Res. Carb.F. 120k, 5%, ¼W	2120 101 46124	C391	Cap.Cer.Pl. 470p,10%,100V	2222 630 08471
R392	Res. Carb.F. 33E, 5%, ¼W	2120 101 46339	C392	Cap.Cer Con/ML. 220n,10%,50V	2022 552 02379
R393	Res. Carb.F. 2k7, 5%, ¼W	2120 101 46272	C394	Cap.Cer.D/M 100n,20%,Z5U	2022 552 01751
R394	Res. Carb.F. 10k, 5%, ¼W	2120 101 46103	C395	Cap.Cer.Pl. 1n0,10%,100V	2222 630 08102
R395	Res. Carb.F. 120k, 5%, ¼W	2120 101 46124			
R396	Res. Carb.F. 68k, 5%, ¼W	2120 101 46683	TR390	Transistor BC558	9331 977 30112
R397	Res. Carb.F. 150k, 5%, ¼W	2120 101 46154			
R398	Res. Carb.F. 10k, 5%, ¼W	2120 101 46103	XL302	VCTCXO 10MHz	2722 171 08177